REFLECTION POSITIVITY FOR CHAMBER QED ON THE INFINITE LATTICE

Abstract. We consider the real curvature field of three-dimensional lattice QED with chamber fermions and all flux sectors included. At fixed positive charge, Wilson parameter in $(0, 1)$, positive hopping weights and arbitrary real mass parameter, the cubic finite-volume laws are tight. Every limit along arbitrary integer side lengths is invariant under the lattice symmetries and satisfies sharp Gaussian domination and reflection positivity at all three coordinate and six diagonal families of lattice mirrors. The reflection form is nonnegative on the full L^2 space generated by plaquettes in either open side. The proof compares cubic laws with reflected perturbations on adapted tori. Local operator estimates and finite layer elimination recover translation symmetry after the perturbation is removed; unequal slab folds and convex separation then give positivity for the original law.

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1. Introduction

We fix the lattice spacing and use unit lattice coordinates. The photon field is a real one-cochain on $\mathbb{Z}^3$. Write d for the lattice coboundary, orient a plaquette by the ordered pair (e_j, e_k), $j < k$, and let $\mathcal{P}_2$ contain the plaquettes with these orientations. The space in which the volume limit is taken is

$$\Omega = \{F \in \mathbb{R}^{\mathcal{P}_2} : dF = 0\},$$

with the product topology. Reversing a plaquette's orientation negates its field value. In particular, pullback by a lattice isometry acts on F as a two-cochain, including the orientation sign.

Here is the finite model. Let $W = \mathbb{Z}^3 \rtimes (O(3) \cap \mathrm{GL}(3, \mathbb{Z}))$ and $C_0 = \{0 < x_3 < x_2 < x_1 < 1/2\}$. Fix $v_0 \in C_0$, and let s_0, s_1, s_2, s_3 be the reflections in its walls $x_3 = 0$, $x_2 = x_3$, $x_1 = x_2$, $x_1 = 1/2$, respectively. The fermion sites are $\mathcal{X} = Wv_0$, and the four neighbors of $x = gv_0$ are $x^{(i)} = gs_iv_0$. Write $\pi(x)$ for the nearest point of $\mathbb{Z}^3$ to x. With $\gamma_j = \mathrm{diag}(\sigma_j, -\sigma_j)$, where σ_j are the Pauli matrices, put $\gamma_v = \sum_j v_j\gamma_j$. For charge q, Wilson parameter ρ, weights w_i, and mass parameter M, the chamber operator is

$$(\mathcal{D}_A\psi)(x) = M\psi(x) - \sum_{i=0}^{3} \frac{w_i}{2} \left(\rho - \gamma_{(x^{(i)}-x)/|x^{(i)}-x|}\right) e^{iqA(\pi(x),\pi(x^{(i)}))}\psi(x^{(i)}),$$

where $A(y,x) = -A(x,y)$ and $A(x,x) = 0$.

On the cubic torus $(\mathbb{Z}/n\mathbb{Z})^3$, the photons are periodic and $\mathcal{D}_A^{C(n)}$ acts on spinors satisfying $\psi(x + ne_j) = -\psi(x)$ for $j = 1, 2, 3$. Put $r_q = 2\pi/q$. For a bounded Borel function G on Ω, define

$$\mu_n(G) = \frac{1}{Z_{C(n)}[1]} \sum_{\substack{z \in C^2((\mathbb{Z}/n\mathbb{Z})^3;\mathbb{Z}) \\ dz=0}} \int_{[0,r_q)^{3n^3}} G(\widetilde{F})\, e^{-\|F\|^2/2} \det \mathcal{D}_A^{C(n)} \, \mathrm{d}A, \qquad F = dA + r_q z.$$

Here $\widetilde{F}$ is the periodic extension of F, the norm sums over the torus plaquettes, and $Z_{C(n)}[1]$ is the same expression without G. Thus every closed integer cochain is summed, with no restriction on its flux. The determinant is nonnegative and $0 < Z_{C(n)}[1] < \infty$; these facts are proved in propositions 2.2 and 2.3.

Theorem A (Reflection positivity in infinite volume)**.** Fix $q > 0$, $0 < \rho < 1$, $M \in \mathbb{R}$ and $w_0, \dots, w_3 > 0$. The family $(\mu_n)_{n \geq 4}$ is tight on Ω. Let $n_i \geq 4$ be any sequence of integers tending to infinity, and let ν be any subsequential weak limit of (μ_{n_i}). Then:

(i) For every finitely supported real two-cochain β,

$$\int_\Omega e^{\langle \beta, F \rangle} \, \mathrm{d}\nu(F) \leq e^{\|\beta\|^2/2}, \qquad \langle \beta, F \rangle = \sum_{\sigma \in \mathcal{P}_2} \beta_\sigma F_\sigma, \qquad \|\beta\|^2 = \sum_{\sigma \in \mathcal{P}_2} \beta_\sigma^2.$$

(ii) The law ν is invariant under lattice translations, signed coordinate permutations, and $F \mapsto -F$.

(iii) The law ν is invariant and reflection positive at every mirror

$$x_j = c \quad (c \in \tfrac{1}{2}\mathbb{Z}), \qquad x_j \pm x_k = c \quad (c \in \mathbb{Z},\ j < k).$$

More precisely, write a mirror as $\Pi = \{\ell(x) = c\}$ and let

$$\theta_\Pi x = x - \frac{2(\ell(x) - c)}{|\nabla \ell|^2} \nabla \ell.$$

Choose either open side H of Π, and let Σ_H be the sigma-algebra generated by the values F_σ for which every corner of σ lies in H. Then

$$\int_\Omega \overline{A(\theta_\Pi^* F)} A(F) \, \mathrm{d}\nu(F) \geq 0 \qquad (A \in L^2(\Sigma_H, \nu)).$$

■□

There are three coordinate and six diagonal mirror families in (iii). The support convention excludes plaquettes on or crossing the mirror. For example, reflection in $x_1 = 0$ changes the signs of F_{12} and F_{13} and preserves the sign of F_{23}. These signs occur both in invariance of the law and in its reflected form.

1.1. **The comparison argument.** The coordinate mirrors are compatible with every cubic torus. Their finite-volume reflected forms, together with a Gaussian bound, give tightness and coordinate reflection positivity of every weak limit. A diagonal mirror requires a different period lattice. The task is therefore to obtain diagonal positivity for the limit of the original cubic laws.

Reflected slab operators give a comparison with a diagonal torus of the same volume. Insert a real local trigonometric polynomial O in one cell, reflect it into all cells, and tilt the diagonal law by the exponential of t times their sum. The resulting law has reflection positivity at the cell walls. Its expectation of O bounds the cubic expectation, up to a pressure error divided by t. The pressure error tends to zero as the volume grows, so the volume limit must precede $t \downarrow 0$.

The perturbation occupies the entire torus. Its pointwise bound does not control local expectations uniformly in volume. We obtain that control by amplifying local insertions in two cell structures and applying a coordinate operator bound. Heat regularization then proves approximate reflection symmetry on two adjacent fermion layers. Fermion elimination and an imaginary contour shift

reduce each curvature character to these two layers, with an error of order t independent of the volume. After the ordered limits, the comparison law is invariant under coordinate link reflections and hence under translations by $2\mathbb{Z}^3$.

These translations allow a given observable to be placed inside a cell with any prescribed diagonal wall. For each real local trigonometric polynomial O we obtain a comparison law ν_O with positivity at that wall and $\nu(O) \le \nu_O(O)$. The comparison law may depend on O. The laws satisfying the Gaussian bound and positivity at the wall form a compact convex set, and separation places ν itself in that set.

The repeated reflections require divisible side lengths, whereas the theorem allows arbitrary integers n_i. An unequal-height slab estimate compares each original cube to a nearby divisible companion, whose side m_i satisfies $m_i/n_i \to 1$. The Hölder-weighted volume of the exceptional slabs is a vanishing fraction of the original volume, and a uniform pressure bound controls their contribution. All auxiliary states are constructed on the companions; the final separation argument still applies to the prescribed limit of the original cubes.

We give the finite-volume identities and limiting arguments needed for the fixed-lattice theorem, including the unequal-slab comparison for arbitrary cubic side lengths and the extension of the reflection form to the full open-side L^2 space. Repeated reflection and the chessboard method belong to the general theory of Fröhlich, Israel, Lieb, and Simon [Frö+78, Section 4]. Here the slab trace must also retain the photon flux sum and the fermion boundary character when the period lattice changes. The local operator and layer-elimination arguments establish the additional symmetry needed for that comparison. The theorem concerns fixed lattice parameters and subsequential volume limits; it asserts neither uniqueness nor a continuum limit.

2. The chamber regularization

The photon field is a real one-cochain on the cubic lattice. The fermion has one site in each chamber of the lattice mirrors. This choice has a specific consequence: a fermion bond crossing a mirror joins a site to its reflected site and is perpendicular to the mirror. The hopping term then admits a positive reflected expansion. We prove this expansion and the resulting finite-volume reflection positivity below. The construction also requires a nonzero partition function when the bare mass has undetermined sign. For this purpose we identify a nonzero coefficient of the fermion determinant, independent of the mass. The determinant sign and positive normalization are proved in propositions 2.2 and 2.3.

Throughout this section, $0 < \rho < 1$, $q > 0$, and the mass parameter M is real. The hopping weights $w_0, \dots, w_3$ are positive. All constants may depend on these fixed parameters unless a uniform bound is stated.

2.1. The photon field and the flux sum.

We orient each link (x, μ) from x to $x + e_\mu$ and each plaquette $(x; \mu\nu)$, $\mu < \nu$, by the ordered pair (e_μ, e_ν). The real k-cochains are denoted by C^k, and the integer cochains by $C^k(\mathbb{Z})$. The coboundary d and its adjoint d^* use these orientations. Thus

$$(dA)_{\mu\nu}(x) = A_\mu(x) + A_\nu(x + e_\mu) - A_\mu(x + e_\nu) - A_\nu(x). \tag{1}$$

An oriented link may also be written as a pair of its endpoints; $A(y, x) = -A(x, y)$ and $A(x, x) = 0$. Pullback by a lattice isometry includes the orientation of the image cell. In particular, pullback commutes with d. For a coordinate reflection, this gives a minus sign on the curvature components containing the normal direction and a plus sign on the tangential component. These are the two-form reflection signs used throughout.

Let $\Gamma \subset \mathbb{Z}^3$ be a rank-three subgroup with

$$\min\{|g|_\infty : g \in \Gamma \setminus \{0\}\} \ge 4.$$

We write $C^k(\Gamma)$ for the cochains on $\mathbb{Z}^3/\Gamma$ and identify them with their periodic extensions. The torus has $V = [\mathbb{Z}^3 : \Gamma]$ photon sites, $3V$ links and $3V$ plaquettes. The inner product on real cochains is the sum over positively oriented cells, and $\|\cdot\|$ denotes its norm.

A photon configuration is a class in

$$\mathrm{Conf}(\Gamma) = \{(A, z) : A \in C^1(\Gamma),\ z \in C^2(\Gamma;\mathbb{Z}),\ dz = 0\}/\sim, \qquad (A, z) \sim \left(A + \frac{2\pi}{q}k, z - dk\right), \quad (2)$$

where $k \in C^1(\Gamma;\mathbb{Z})$. Its curvature and link phases are

$$F = dA + \frac{2\pi}{q}z, \qquad u_b = \exp(iqA(b)). \quad (3)$$

Both are independent of the representative, and $dF = 0$. The integer cochain is required to be closed; there is no restriction on its flux. The curvature therefore remains a real, unbounded field.

Put $r_q = 2\pi/q$. Each class has a unique representative with $A(b) \in [0, r_q)$ at every link. The measure on the quotient is defined by

$$\int_{\mathrm{Conf}(\Gamma)} \Phi(A, z)\,\mathrm{d}(A, z) = \sum_{\substack{z \in C^2(\Gamma;\mathbb{Z}) \\ dz = 0}} \int_{[0,r_q)^{3V}} \Phi(A, z)\,\mathrm{d}A. \quad (4)$$

This formula holds for nonnegative measurable functions and for absolutely integrable complex functions. Any measurable fundamental domain may be used instead. Indeed, the changes of representative preserve Lebesgue measure in A and counting measure in z.

We will repeatedly use the finite number

$$C_q(\varepsilon) = \sup_{y \in \mathbb{R}} \sum_{j \in \mathbb{Z}} e^{-\varepsilon(y + r_q j)^2}, \qquad \varepsilon > 0. \quad (5)$$

It satisfies

$$C_q(\varepsilon) \le 2 + \frac{q}{2\pi}\sqrt{\frac{\pi}{\varepsilon}}.$$

To see this, retain the two terms nearest the origin and bound each remaining term by r_q^{-1} times the Gaussian integral over the adjacent interval of length r_q closer to the origin. Those intervals have disjoint interiors. Write $C_q = C_q(1/2)$. Dropping the constraint $dz = 0$ in the flux sum gives

$$0 < \int_{\mathrm{Conf}(\Gamma)} e^{-\|F\|^2/2}\,\mathrm{d}(A, z) \le (r_q C_q)^{3V}. \quad (6)$$

The lower bound follows already from $z = 0$; for the upper bound, sum each z_σ separately at fixed A.

The periodic extensions of the curvatures lie in

$$\Omega = \{F \in C^2 : dF = 0\},$$

with the topology of pointwise convergence on plaquettes. This is a closed subspace of a countable product of copies of $\mathbb{R}$. Local observables depend on finitely many plaquette values. For a finitely supported real two-cochain β, the character $\exp(i\langle\beta, F\rangle)$ is bounded and continuous on Ω.

2.2. **Chambers and the fermion graph.** The lattice mirrors are

$$\{x_\mu = c\}, \quad c \in \tfrac{1}{2}\mathbb{Z}, \qquad \{x_\mu \pm x_\nu = c\}, \quad c \in \mathbb{Z}, \quad \mu < \nu. \quad (7)$$

They form three coordinate families and six diagonal families. Let $\mathcal{L}$ be the corresponding linear forms ℓ and write $\ell(x) = n_\ell \cdot x$. The admissible positions are $\mathrm{Adm}(x_\mu) = \frac{1}{2}\mathbb{Z}$ and $\mathrm{Adm}(x_\mu \pm x_\nu) = \mathbb{Z}$. The reflection in a mirror $\Pi = (\ell, c)$ is

$$\theta_\Pi x = x - \frac{2(\ell(x) - c)}{|n_\ell|^2} n_\ell. \quad (8)$$

Its upward normal is $n_\ell/|n_\ell|$. Coordinate mirrors at integer positions are called site planes, and those at half-integer positions are called link planes.

The group generated by these reflections is

$$W = \mathbb{Z}^3 \rtimes O_{\mathrm{cub}}, \qquad O_{\mathrm{cub}} = O(3) \cap \mathrm{GL}(3,\mathbb{Z}).$$

Here O_{cub} consists of the 48 signed permutation matrices. A fundamental chamber is

$$C_0 = \{0 < x_3 < x_2 < x_1 < \tfrac{1}{2}\}.$$

Its wall reflections are

$$\begin{aligned} s_0(x) &= (x_1, x_2, -x_3), & s_1(x) &= (x_1, x_3, x_2),\\ s_2(x) &= (x_2, x_1, x_3), & s_3(x) &= (1 - x_1, x_2, x_3). \end{aligned}$$

Denote the walls by Π_i and their inward unit normals by

$$n_0 = e_3, \quad n_1 = (e_2 - e_3)/\sqrt{2}, \quad n_2 = (e_1 - e_2)/\sqrt{2}, \quad n_3 = -e_1. \tag{9}$$

Fix $v_0 = (v_1, v_2, v_3) \in C_0$. The fermion sites and their neighbors are

$$\mathcal{X} = W v_0, \qquad (g v_0)^{(i)} = g s_i v_0. \tag{10}$$

The unordered bond $\{x, x^{(i)}\}$ has type i. A site has the unique form $x = k + r v_0$, with $k \in \mathbb{Z}^3$ and $r \in O_{\mathrm{cub}}$; put $\pi(x) = k$ and $y(x) = r v_0$. Thus $\pi(x)$ is its nearest photon site.

Proposition 2.1. The chambers are precisely gC_0, $g \in W$, and W acts on them simply transitively. No fermion site lies on a mirror. Every bond crosses exactly one mirror, its chamber wall, and crosses it orthogonally at its midpoint. Consequently a bond crossing Π is $\{x, \theta_\Pi x\}$.

The projection π is W-equivariant. Bonds of types $0, 1, 2$ have the same photon site at both ends; a type-3 bond joins two nearest photon sites. Exactly eight fermion bonds lie over each photon link. If a mirror does not cut the link and its open segment lies on one side, all eight bonds lie on that side. The graph is connected and bipartite, and each site has one neighbor of each type. ■

Proof. Coordinate reflections and coordinate transpositions generate O_{cub}, and $\theta_{x_\mu,1/2}\theta_{x_\mu,0} = \tau_{e_\mu}$ gives the translations. Conversely, (8) is an affine signed permutation with integer translation, so the mirror group is W.

For a point on no mirror, subtract the unique nearest lattice point. The resulting three coordinates are nonzero, have distinct absolute values, and lie in $(-1/2, 1/2)$. There is exactly one signed permutation making them positive and decreasing. This places the point in exactly one translate of C_0. These translates are disjoint open convex sets, meet no mirror, and cover its complement. They are therefore the chambers, with the asserted free transitive action.

For the four bonds from v_0, the midpoint is, respectively,

$$(v_1, v_2, 0), \quad (v_1, (v_2 + v_3)/2, (v_2 + v_3)/2), \quad ((v_1 + v_2)/2, (v_1 + v_2)/2, v_3), \quad (1/2, v_2, v_3).$$

Each belongs to the relative interior of its wall and to no other mirror. Each half of the bond, except its midpoint, lies in one of the two adjacent chambers. Transport by $g \in W$ proves the bond assertion at every site.

In the coordinates $x = k + r v_0$, the first three neighbors are $k + r s_i v_0$, whereas the fourth is $k + r e_1 + r r_{e_1} v_0$, where r_{e_1} changes the sign of the first coordinate. This proves the statements about π and its equivariance. Over (k, μ) the lower ends are exactly the eight sites for which $y_\mu = v_1$: the remaining two absolute values can be ordered in two ways, with four choices of their signs. Their type-3 neighbors are the upper ends of these eight bonds.

The bond over $b = (k, \mu)$ has the segment $k + t e_\mu + y_\perp$, $v_1 \le t \le 1 - v_1$, where the other two coordinates of $y_\perp$ have absolute values v_2, v_3. For a diagonal height involving x_μ, its varying part is $t \pm y_\nu \in (0, 1)$, the same interval between integers as on the open photon link. For heights not involving x_μ, the changes are y_ν or $y_\nu \pm y_{\nu'}$, whose absolute values are smaller than $1/2$ or 1,

respectively. Away from a mirror containing b these preserve its side. The coordinate height x_μ gives the same side except at the unique link plane through the midpoint, which cuts b. This proves the assertion about photon links.

The generators s_0, s_1, s_2 generate the signed permutations, and $s_3 r_{e_1} = \tau_{e_1}$; conjugation gives all coordinate translations. Thus the four wall reflections generate W, proving connectivity. The two parts are the sites gv_0 with $\det r = 1$ and with $\det r = -1$, since each neighbor changes this determinant. □

The geometric location records reflection, while (k, r) labels the fermion sites by a photon site and a signed permutation. If $d_i = \operatorname{dist}(v_0, \Pi_i)$, then

$$x^{(i)} - x = -2d_i\, r n_i, \qquad \frac{x^{(i)} - x}{|x^{(i)} - x|} = -r n_i.$$

The directions, bond types and projected endpoints in these coordinates do not depend on v_0. Hence neither the operator below nor its determinant depends on v_0 after this identification.

2.3. **The chamber operator and the finite-volume law.** Let σ_μ be the Pauli matrices and

$$\gamma_\mu = \operatorname{diag}(\sigma_\mu, -\sigma_\mu), \qquad \gamma_v = \sum_\mu v_\mu \gamma_\mu.$$

They form a Hermitian Clifford triple. For a unit vector n, put $P_n^\pm = (1 \pm \gamma_n)/2$. Both projections have rank two, and

$$\gamma_n \gamma_v \gamma_n = -\gamma_{r_n v}, \qquad r_n v = v - 2(v \cdot n) n. \tag{11}$$

This follows by multiplying $\gamma_n \gamma_v + \gamma_v \gamma_n = 2n \cdot v$ by γ_n.

For an oriented bond (x, x') of type i define

$$U_A(x, x') = \exp(iqA(\pi(x), \pi(x'))), \qquad w(x, x') = \frac{w_i}{2}\left(\rho - \gamma_{(x'-x)/|x'-x|}\right). \tag{12}$$

The chamber operator is

$$(\mathcal{D}_A \psi)(x) = M\psi(x) - \sum_{i=0}^{3} w(x, x^{(i)}) U_A(x, x^{(i)}) \psi(x^{(i)}). \tag{13}$$

Its lattice mass is

$$m = M - \frac{\rho}{2} \sum_i w_i. \tag{14}$$

A fermion boundary condition is a character $\chi : \Gamma \to \{1, -1\}$. Write $\mathbb{T} = (\Gamma, \chi)$. Its spinors satisfy $\psi(x + g) = \chi(g)\psi(x)$ for $g \in \Gamma$. The restriction $\mathcal{D}_A^{\mathbb{T}}$ acts on a space of dimension $192V$, with the inner product obtained by summing over representatives of $\mathcal{X}/\Gamma$. For each site introduce generators $\psi_\alpha(x), \bar\psi_\alpha(x)$, $1 \le \alpha \le 4$, obeying the same boundary conditions. Their Grassmann algebra is denoted by $\mathrm{Gr}_{\mathbb{T}}$. Its Berezin integral is normalized by

$$\int_{\mathbb{T}} \prod_{x \in \mathcal{X}/\Gamma} \prod_{\alpha=1}^{4} \psi_\alpha(x) \bar\psi_\alpha(x) = 1. \tag{15}$$

Products of pairs are even, so their ordering is immaterial. With $S_{\mathbb{T}}(A) = \sum_x \bar\psi(x)(\mathcal{D}_A\psi)(x)$, this normalization gives $\int_{\mathbb{T}} e^{-S_{\mathbb{T}}(A)} = \det \mathcal{D}_A^{\mathbb{T}}$.

The unnormalized functional is

$$Z_{\mathbb{T}}[G] = \int_{\mathrm{Conf}(\Gamma)} e^{-\|F\|^2/2} \int_{\mathbb{T}} e^{-S_{\mathbb{T}}(A)} G(A, z)\, \mathrm{d}(A, z). \tag{16}$$

For a scalar G the inner integral is $G(A, z) \det \mathcal{D}_A^{\mathbb{T}}$. For a Grassmann-valued integrand, absolute convergence is understood coefficient by coefficient. Bounded measurable coefficients suffice: $e^{-S_{\mathbb{T}}(A)}$

has finitely many coefficients, each a polynomial in the link phases, and (6) supplies an integrable majorant.

Proposition 2.2. For every real A and M,

$$\det \mathcal{D}_A^{\mathbb{T}} \geq 0, \qquad \|\mathcal{D}_A^{\mathbb{T}}\| \leq |M| + \frac{1+\rho}{2} \sum_i w_i.$$

Moreover $\Re \mathcal{D}_A^{\mathbb{T}} = m + \frac{\rho}{2} \nabla_A^* \nabla_A$, where

$$(\nabla_A \psi)(x,i) = \sqrt{w_i/2}\, (\psi(x) - U_A(x, x^{(i)}) \psi(x^{(i)})).$$

The determinant is not identically zero as a function of the link phases, for every real mass parameter M. ∎

Proof. Write

$$H_A \psi(x) = \sum_i w_i U_A(x, x^{(i)}) \psi(x^{(i)}), \qquad J_A \psi(x) = \sum_i w_i \gamma_{(x^{(i)}-x)/|x^{(i)}-x|} U_A(x, x^{(i)}) \psi(x^{(i)}).$$

Reversing a bond conjugates its phase and reverses its direction. Reindexing by the involution $x \mapsto x^{(i)}$ therefore gives $H_A^* = H_A$ and $J_A^* = -J_A$. Hence

$$\mathcal{D}_A = M - \frac{\rho}{2} H_A + \frac{1}{2} J_A, \qquad \nabla_A^* \nabla_A = \sum_i w_i - H_A.$$

This proves the real-part identity. In the splitting $\mathbb{C}^4 = \mathbb{C}^2 \oplus \mathbb{C}^2$, the two blocks of H_A coincide, and the two blocks of J_A have opposite signs. It follows that

$$\mathcal{D}_A^{\mathbb{T}} = \operatorname{diag}(\check{\mathcal{D}}_A^{\mathbb{T}}, (\check{\mathcal{D}}_A^{\mathbb{T}})^*), \qquad \det \mathcal{D}_A^{\mathbb{T}} = |\det \check{\mathcal{D}}_A^{\mathbb{T}}|^2. \tag{17}$$

Each hopping matrix has norm $w_i(1+\rho)/2$, while the associated phase-weighted permutation of sites is unitary. Their sum gives the norm bound.

For the last assertion we view the determinant as a Laurent polynomial $P_{\mathbb{T}}$ in the independent variables u_b. Only type-3 bonds contain a variable, with u_b at the lower end and u_b^{-1} at the upper end. Fix a coordinate direction μ. The coefficient of

$$\prod_{b \| e_\mu} u_b^{32} \tag{18}$$

will be computed without using M.

Partition the sites modulo Γ into six layers according to the increasing values

$$y_\mu = -v_1, -v_2, -v_3, v_3, v_2, v_1.$$

Each layer has $8V$ sites. Each site has one upward neighbor, one downward neighbor, and two neighbors in its own layer. The upward map is a permutation of the sites and visits successive layers modulo six. Its bond types are

$$(i_1, i_2, i_3, i_4, i_5, i_6) = (2, 1, 0, 1, 2, 3). \tag{19}$$

For layers $1, \ldots, 5$ the upward phase is one; in layer six it is u_b for the link $(\pi(x), \mu)$. These facts follow by changing the smallest sign or exchanging successive absolute values in the neighbor formulas above. The sixth step changes v_1 to $-v_1$ and increases the photon coordinate by one.

There are eight lower and eight upper fermion ends over each photon link, each carrying four matrix rows. An exponent 32 at a link therefore requires every lower row to take the upward type-3 block and every upper row to avoid the reverse block. Imposing this at all links in direction μ forces every row in every layer to take its upward block. Indeed, layer-six rows fill all columns in layer one. A layer-one row cannot then use a column in its own layer, and the reverse block is excluded by the exponent condition; it must go upward to layer two. Inductively the columns in the current and

preceding layers are already occupied, so layers two through five must also go upward. The selected terms use no other link variables.

The remaining freedom is a permutation of the four spin indices at each site. The site permutation, repeated four times, has sign one. Summing over the spin permutations therefore gives the product of the 4×4 determinants of the selected hopping blocks. The boundary signs have fourth power one, and

$$\det w(x, x^{\uparrow}) = (w_{i_k}/2)^4 (1-\rho^2)^2$$

at a site in layer k. Consequently the coefficient of (18) is C_*^V, where

$$C_* = (w_0/2)^{32}(w_1/2)^{64}(w_2/2)^{64}(w_3/2)^{32}(1-\rho^2)^{96} > 0. \tag{20}$$

This coefficient is independent of M. In particular the determinant polynomial, and its restriction to the phase torus, are nonzero. □

The real-part estimate alone would require control of the sign of m to prove invertibility. The coefficient computation instead proves that the determinant has positive integral at every real M, which is all that is needed to define the law.

Proposition 2.3. The partition function satisfies $0 < Z_{\mathbb{T}}[1] < \infty$. More precisely,

$$r_q^{3V} e^{-6r_q^2 V} C_*^V \le Z_{\mathbb{T}}[1] \le (r_q C_q)^{3V} \left(|M| + \frac{1+\rho}{2} \sum_i w_i \right)^{192V}. \tag{21}$$

Thus $\mathbb{E}_{\mathbb{T}}[G] = Z_{\mathbb{T}}[G]/Z_{\mathbb{T}}[1]$ defines a probability law on scalar observables. Its curvature law $\mu_{\mathbb{T}}$ is the law of the periodic extension $\widetilde{F}$ on Ω. ■

Proof. For a nonnegative Laurent polynomial P on the phase torus, every Fourier coefficient satisfies

$$|\widehat{P}(\lambda)| = \left| \int P(u) \overline{u^\lambda} \, \mathrm{d}u \right| \le \int P(u) \, \mathrm{d}u,$$

where Haar measure is normalized. By the preceding proposition and (20),

$$\int_{[0,r_q)^{3V}} \det \mathcal{D}_A^{\mathbb{T}} \, \mathrm{d}A \ge r_q^{3V} C_*^V.$$

On this box $|(dA)_\sigma| \le 2r_q$ and $\|dA\|^2 \le 12 V r_q^2$. The $z = 0$ term in (16) gives the lower bound in (21). For the upper bound use $|\det \mathcal{D}_A^{\mathbb{T}}| \le \|\mathcal{D}_A^{\mathbb{T}}\|^{192V}$ and (6). The density $e^{-\|F\|^2/2} \det \mathcal{D}_A^{\mathbb{T}}$ is nonnegative. After division by its strictly positive integral it is a probability density, whose image under $(A, z) \mapsto \widetilde{F}$ is the asserted law. □

3. The reflected form on an adapted torus

The chamber bonds crossing a mirror join reflected endpoints. Their Clifford projections give positive fermion coefficients. We combine this expansion with a Fourier representation of the integer closedness constraints to obtain a positive form, including all flux sectors. The positive representation keeps the contributions from both reflection seams explicitly.

3.1. Symmetries and reflected Grassmann products.

The field strength transforms as a two-cochain, with the orientation of each plaquette included. For an antiperiodic cubic torus the law is invariant under W and under $F \mapsto -F$. We record the corresponding fermionic identities, since reflection positivity uses their signs.

Gauge covariance follows directly from

$$U_{A-d\lambda}(x, x') = e^{iq\lambda(\pi(x))} U_A(x, x') e^{-iq\lambda(\pi(x'))}.$$

For lattice covariance set $\gamma_5 = \left(\begin{smallmatrix} 0 & 1_2 \\ 1_2 & 0 \end{smallmatrix}\right)$. It anticommutes with the Clifford triple, and

$$(\gamma_n\gamma_5)\gamma_v(\gamma_n\gamma_5)^{-1} = \gamma_{r_n v}.$$

If the linear part of $g \in W$ is $r_{n_1}\cdots r_{n_k}$, the corresponding product of these spin matrices, followed by the permutation $x \mapsto g^{-1}x$, intertwines $\mathcal{D}^{\mathbb{T}}_A$ and $\mathcal{D}^{g\mathbb{T}}_{(g^{-1})^*A}$, where $g\mathbb{T} = (r\Gamma, \chi \circ r^{-1})$ and r is the linear part of g. Taking determinants and changing variables in (16) proves

$$\mu_{g\mathbb{T}} = ((g^{-1})^*)_*\mu_{\mathbb{T}}.$$

The configuration measure is preserved because pullback permutes cochain coordinates up to signs and maps one fundamental domain to another. All translations preserve $\mathbb{T}$, and all elements of W preserve an antiperiodic cubic torus, proving the stated invariance.

For charge conjugation let $C = \sigma_2 \oplus \sigma_2$. The Pauli identity $C\gamma_v^\top C^{-1} = -\gamma_v$, together with reversal of the oriented bond, gives

$$C(\mathcal{D}^{\mathbb{T}}_A)^\top C^{-1} = \mathcal{D}^{\mathbb{T}}_{-A}.$$

Thus the determinant is unchanged by $A \mapsto -A$. The substitution $(A, z) \mapsto (-A, -z)$ preserves the configuration measure and sends F to $-F$, proving charge symmetry of the curvature law.

At a mirror Π with upward normal n, the Grassmann reflection Θ_Π is antilinear and reverses products. In components it is

$$\Theta_\Pi\psi_\alpha(x) = -\sum_\beta \bar\psi_\beta(\theta_\Pi x)(\gamma_n)_{\beta\alpha}, \qquad \Theta_\Pi\bar\psi_\alpha(x) = -\sum_\beta (\gamma_n)_{\alpha\beta}\psi_\beta(\theta_\Pi x). \tag{22}$$

On scalar coefficient functions it acts as $\Phi(A, z) \mapsto \overline{\Phi(\theta_\Pi^* A, \theta_\Pi^* z)}$. The Clifford identities give $\Theta_\Pi^2 = 1$ and

$$\Theta_\Pi(\bar\psi(x)Y\psi(x')) = \bar\psi(\theta_\Pi x')\gamma_n Y^*\gamma_n\psi(\theta_\Pi x). \tag{23}$$

For a hopping term, $\gamma_n w(x, x')\gamma_n = w(\theta_\Pi x', \theta_\Pi x)$, by (11). Reflection therefore permutes the oriented terms of the action, including the real mass terms.

If Y is a finite set of sites, then

$$\int_{\theta_\Pi Y} \Theta_\Pi\xi = \overline{\int_Y \xi}. \tag{24}$$

Only the top-degree coefficient needs to be checked. At one site reflection maps $\prod_\alpha \psi_\alpha\bar\psi_\alpha$ to the same product at the reflected site, multiplied by $(\det\gamma_n)^2 = 1$. Products from different sites are even. This proves (24), and a measure-preserving change of photon variables gives

$$Z_{\theta_\Pi\mathbb{T}}[\Theta_\Pi G] = \overline{Z_{\mathbb{T}}[G]} \tag{25}$$

for bounded coefficient integrands, and for the integrable Gaussian-growth classes considered below.

On even integrands, compositions of these reflections define an action of W. The ambiguity of a reflection word acts only by the common sign on all fermion generators. To verify this, compose two words representing the same lattice isometry. The resulting even word has trivial geometric action, and its spin matrix is $\mathrm{diag}(u, u)$ with u a product of an even number of Pauli matrices σ_n. It commutes with all Pauli matrices, hence is scalar; its determinant is one, so $u = \pm 1_2$. Its action on an even monomial is consequently the identity. We write G^g for this action when folding even integrands. For a real scalar observable it is $G^g(F) = G(g^*F)$.

3.2. **The fermion crossing at a mirror.** Let a bond of type j cross Π, with lower endpoint x and upper endpoint $x' = \theta_\Pi x$. Its direction is the upward normal n. The two hopping matrices are

$$\begin{aligned} w(x, x') &= \tfrac{w_j}{2}\big((1+\rho)P_n^- - (1-\rho)P_n^+\big), \\ w(x', x) &= \tfrac{w_j}{2}\big((1+\rho)P_n^+ - (1-\rho)P_n^-\big). \end{aligned} \tag{26}$$

Choose orthonormal bases $e^{\pm}_{n,1}, e^{\pm}_{n,2}$ of the two projected spaces. For each $k = 1, 2$ introduce four crossing modes. Their coefficients and phases are displayed together:

	ζ	ω	u
1	$(e^{-}_{n,k})^{*}\psi(x')$	$\frac{w_j}{2}(1+\rho)$	$U_A(x,x')$
2	$(e^{+}_{n,k})^{*}\psi(x')$	$\frac{w_j}{2}(1-\rho)$	$U_A(x,x')$
3	$\bar{\psi}(x')e^{+}_{n,k}$	$\frac{w_j}{2}(1+\rho)$	$\overline{U_A(x,x')}$
4	$\bar{\psi}(x')e^{-}_{n,k}$	$\frac{w_j}{2}(1-\rho)$	$\overline{U_A(x,x')}$

(27)

All coefficients are positive in the range $0 < \rho < 1$.

Proposition 3.1. Let J be a finite collection of bonds crossing Π, and let $S^{\times}_{\Pi}(J)$ be their part of the action. Index all their crossing modes by a finite set $\mathcal{I}$. Then

$$-S^{\times}_{\Pi}(J) = \sum_{i\in\mathcal{I}} \omega_i u_i \Theta_{\Pi}(\zeta_i)\zeta_i. \tag{28}$$

Choose an order on $\mathcal{I}$ and, for $I \subset \mathcal{I}$, put $\zeta_I = \prod_{i\in I}\zeta_i$, $\omega_I = \prod_{i\in I}\omega_i$ and $u_I = \prod_{i\in I} u_i$. Then

$$e^{-S^{\times}_{\Pi}(J)} = \sum_{I\subset\mathcal{I}} \omega_I u_I \Theta_{\Pi}(\zeta_I)\zeta_I. \tag{29}$$

At site and diagonal mirrors all u_i are one. At a link plane, groups one and two have charge $+1$, groups three and four charge -1, and their phases are $e^{iqq(i)A(b)}$ at the cut photon link b. ■

Proof. The reflections of the four modes in (27) are, respectively,

$$\bar{\psi}(x)e^{-}_{n,k}, \quad -\bar{\psi}(x)e^{+}_{n,k}, \quad -(e^{+}_{n,k})^{*}\psi(x), \quad (e^{-}_{n,k})^{*}\psi(x).$$

Multiplying each reflected mode by the original mode and using anticommutation gives the four projected terms in (26), with the indicated phases. Summation over k uses $P^{\pm}_{n} = \sum_k e^{\pm}_{n,k}(e^{\pm}_{n,k})^{*}$ and gives the negative of the two bond terms. Summation over bonds proves (28).

Each $\Theta_{\Pi}(\zeta_i)\zeta_i$ is even and has square zero. These elements commute. Reversal of the order under Θ_{Π} gives

$$\prod_{i\in I}\Theta_{\Pi}(\zeta_i)\zeta_i = \Theta_{\Pi}(\zeta_I)\zeta_I,$$

with no remaining sign. Expanding the product of the finite exponentials proves (29). Only type-3 bonds carry a photon phase, and they cross link planes. Their projected endpoints are the oriented endpoints of the cut link, giving the last assertion. □

3.3. **Adapted tori and the second mirror.** An adapted torus for $\ell \in \mathcal{L}$ has the form

$$\Gamma = \Gamma_{\parallel} \oplus \mathbb{Z}v, \quad \Gamma_{\parallel} = \Gamma \cap \{\ell = 0\}, \quad v = \frac{p}{|n_\ell|^2} n_\ell \in \mathbb{Z}^3, \quad \chi(v) = -1, \tag{30}$$

where $p = \ell(v) > 0$ is the height period. For an admissible c set

$$c' = c + p/2, \quad \Pi = (\ell, c), \quad \Pi' = (\ell, c'), \quad U_+ = \{c < \ell < c'\}, \quad U_- = \{c - p/2 < \ell < c\}.$$

The two mirrors are admissible: for coordinate families $p \geq 4$, while for diagonal families p is even and $p \geq 8$. Their reflections satisfy $\theta_{\Pi'} = \tau_v \theta_{\Pi}$ and induce the same geometric reflection on the torus, exchanging its two halves. An antiperiodic coordinate torus is adapted to each coordinate family. The torus with periods $ae_1 - ae_2$, $be_1 + be_2$ and ce_3, antiperiodic on these periods, is adapted to the corresponding two diagonal families and the third coordinate family. Cubic images give adapted tori for all nine families.

Choose representatives $\mathcal{X}_+$ in U_+ and $\mathcal{X}_- = \theta_{\Pi}\mathcal{X}_+$ in U_-. They partition the fermion sites of the torus. Let S_+ contain the mass terms and oriented bonds entirely in U_+, and set $\theta = \theta_{\Pi}$, $\Theta = \Theta_{\Pi}$; the second boundary mirror induces the same geometric reflection on the torus. The coefficient functions of S_+ involve only links in the closed upper slab; the reflected part is ΘS_+.

The crossing modes at Π lie in $\mathrm{Gr}(\mathcal{X}_+)$. At Π' their upper endpoints are outside U_+; write $\check{\zeta}_{i'} = \Theta_{\Pi'}\zeta_{i'}$, so the checked modes lie in $\mathrm{Gr}(\mathcal{X}_+)$. The boundary condition gives

$$\Theta_{\Pi'} = \chi(v)^{\deg}\Theta_{\Pi} \quad \text{on } \mathrm{Gr}_{\mathbb{T}}\,. \tag{31}$$

In the crossing term at Π' we therefore have

$$\Theta_{\Pi'}(\zeta_{i'})\zeta_{i'} = -\chi(v)\Theta(\check{\zeta}_{i'})\check{\zeta}_{i'}.$$

The minus sign is from exchanging the two odd factors. Since $\chi(v) = -1$, it is canceled by antiperiodicity. Both boundary crossings thus have the same positive coefficients, and

$$e^{-S_{\mathbb{T}}} = e^{-S_+}\Theta(e^{-S_+})\sum_{I,I'}\omega_I\omega_{I'}u_Iu_{I'}\Theta(\zeta_I)\zeta_I\Theta(\check{\zeta}_{I'})\check{\zeta}_{I'}. \tag{32}$$

Here I and I' range over the finite crossing-mode sets at the two mirrors.

The reflected Berezin integral separates the two halves:

$$\int_{\mathbb{T}}\Theta(\xi')\xi = \overline{\int_{\mathcal{X}_+}\xi'}\int_{\mathcal{X}_+}\xi, \qquad \xi,\xi' \in \mathrm{Gr}(\mathcal{X}_+). \tag{33}$$

Indeed, the top monomial is the product of the two half-top monomials, and reflection maps one to the other by (24). We must still separate the photon flux constraints; these couple the halves at the cut cubes.

3.4. **The photon crossing and Gauss' law.** Let $\mathcal{M}$ be the two boundary mirrors, understood periodically. Classify a cell by its relative interior: it is upper or lower if that interior is in the respective half, lies on the mirrors if the whole cell does, and is cut if its relative interior meets a mirror without being contained in it. Write $B_+, B_-, B_0, B_\times$ for the four sets of links, $P_+, P_-, P_0, P_\times$ for the plaquettes, and $Q_+, Q_-, Q_\times$ for the cubes.

Reflection exchanges the upper and lower cells. It fixes a cell lying on the mirrors with orientation sign $+1$, and a cut cell with orientation sign -1. Each cell meets at most one boundary mirror: its height variation is at most $|n_\ell|^2$, smaller than $p/2$. To justify the orientation assertion, a point in the intersection of the relative interior with the mirror is fixed and lies in the relative interior of the reflected cell. Relative interiors of different cells are disjoint, so the cell is fixed. Reflection is the identity on its affine hull in the first case and a reflection of that affine hull in the cut case.

An upper cell has only upper faces or faces lying on the mirrors; a cut cell has only upper, lower or cut faces. Moreover a plaquette containing a cut link is cut, and a cube containing a cut plaquette is cut. An upper face can touch the boundary at a relative interior point only when its affine height is constant. A face of a cut cell cannot lie in the mirror: reflection in the supporting plane of a face would not preserve that cell. Finally, the height takes both signs on a cut face, hence on the relative interior of its coface. In particular, the incidences at a cut plaquette and a cut cube give

$$(dA)_\sigma = (d_+A)_\sigma - (d_+\theta^*A)_\sigma + (d_\times A)_\sigma, \qquad (dz)(Q) = (d_+z)(Q) - (d_+\theta^*z)(Q) + (d_\times z)(Q), \tag{34}$$

where d_+ and $d_\times$ retain the incidences from upper and cut faces, respectively. For example, $(d_+A)_\sigma = \sum_{b\in B_+}[\sigma : b]A(b)$. The minus sign in the lower term follows from orientation reversal of the cut cell and the identity $d\theta^* = \theta^*d$.

Use representatives

$$A = (A_+, A_-, A_0, a), \qquad z = (z_+, z_-, z_0, z_\times).$$

Choose the interval $[0, r_q)$ for upper and mirror links, its signed reflection for lower links, and $[-r_q/2, r_q/2)$ for cut links. This is a fundamental domain. Reflection maps the lower variables measure-preservingly to an independent copy of the upper variables, fixes (A_0, z_0), and changes the sign of a.

The upper and mirror photon weights are

$$W_+ = \prod_{\sigma\in P_+} e^{-F_\sigma^2/2} \prod_{Q\in Q_+} 1_{\{(dz)(Q)=0\}}, \qquad W_0 = \prod_{\sigma\in P_0} e^{-F_\sigma^2/2}. \tag{35}$$

The full Maxwell weight with the closedness constraint is $W_0W_+\Theta(W_+)$ times the Gaussian and constraint factors at the cut plaquettes and cubes. The half weights contain only upper and mirror variables. Their integrals are bounded uniformly in the mirror variables by

$$\int W_+ \,\mathrm{d}(A_+, z_+) \le r_q^{|B_+|} C_q^{|P_+|}, \qquad \int W_0 \,\mathrm{d}(A_0, z_0) \le r_q^{|B_0|} C_q^{|P_0|}.$$

The proof drops the cube constraints and uses (5) on each plaquette.

At a cut cube impose integer closedness by the Fourier multiplier

$$1_{\{(dz)(Q)=0\}} = \int_0^{2\pi} e^{i\alpha_Q (dz)(Q)} \frac{\mathrm{d}\alpha_Q}{2\pi}.$$

For $\alpha \in [0,2\pi)^{Q_\times}$ and $k \in \mathbb{Z}^{P_\times}$ define

$$\begin{aligned} \alpha'_\sigma &= \sum_{Q\in Q_\times} [Q:\sigma]\alpha_Q, \quad \beta_\sigma = \frac{q}{2\pi}(\alpha'_\sigma - 2\pi k_\sigma), \\ u_{\alpha,k} &= \exp\left(i\sum_Q \alpha_Q (d_+ z)(Q) - i\sum_\sigma \beta_\sigma (d_+ A)_\sigma \right). \end{aligned} \tag{36}$$

The function $u_{\alpha,k}$ depends only on the upper variables and has modulus one.

Poisson summation, in the normalization needed here, reads

$$\sum_{j\in\mathbb{Z}} e^{-(y+r_q j)^2/2 + ij\alpha} = \frac{q}{\sqrt{2\pi}} \sum_{k\in\mathbb{Z}} e^{-\beta_k^2/2 - i\beta_k y}, \qquad \beta_k = \frac{q}{2\pi}(\alpha - 2\pi k). \tag{37}$$

It follows by Fourier transforming the Gaussian under $t = y + r_q s$. Insert the cube multipliers and apply this formula at each cut plaquette. Splitting dA and dz by (34) gives the exact identity

$$\begin{aligned} &\sum_{z_\times} \prod_{\sigma\in P_\times} e^{-F_\sigma^2/2} \prod_{Q\in Q_\times} 1_{\{(dz)(Q)=0\}} \\ &= \int \prod_{Q\in Q_\times} \frac{\mathrm{d}\alpha_Q}{2\pi} \sum_{k\in\mathbb{Z}^{P_\times}} \prod_{\sigma\in P_\times} \frac{q}{\sqrt{2\pi}} e^{-\beta_\sigma^2/2}\, u_{\alpha,k}\Theta(u_{\alpha,k}) \exp\left(iq \sum_{b\in B_\times} a_b(d^*k)(b) \right). \end{aligned} \tag{38}$$

Here k is extended by zero outside $P_\times$. To obtain the final factor, observe that at a cut link b,

$$\sum_{\sigma\in P_\times} [\sigma:b]\alpha'_\sigma = \sum_{Q\in Q_\times} \alpha_Q \sum_\sigma [Q:\sigma][\sigma:b] = 0.$$

The restriction to cut plaquettes may be dropped because any plaquette containing a cut link is cut; the last sum is zero by $d^2 = 0$. Thus the part of β involving α' cancels from the cut-link factor.

All the weights before the two reflected factors on the right of (38) are positive. Their total integral is at most $C_q^{|P_\times|}$. Indeed, with $y = 0$ in (37),

$$\frac{q}{\sqrt{2\pi}} \sum_k e^{-\beta_k^2/2} \le \sum_j e^{-(r_q j)^2/2} \le C_q$$

at each cut plaquette. The original Gaussian sum also has an absolute bound $C_q^{|P_\times|}$, justifying insertion of the Fourier multipliers and the exchanges of their sums and integrals.

Cut links occur only at coordinate link planes. A link fixed by a mirror reflection but not contained in its plane has its endpoints exchanged; hence it is perpendicular to the mirror, whose normal is a coordinate vector. A diagonal mirror has cut plaquettes and cubes, but no cut links.

For crossing-mode subsets I, I' let $\mathfrak{q}(I \cup I')(b)$ be the sum of their charges at modes lying over b. The fermionic phases in (32) are

$$u_I u_{I'} = \exp\left(iq \sum_{b \in B_\times} a_b \mathfrak{q}(I \cup I')(b) \right).$$

Integration of the cut links over one period gives

$$\int_{[-r_q/2, r_q/2)^{B_\times}} e^{iq \sum_b a_b ((d^*k)(b) + \mathfrak{q}(I \cup I')(b))} \, \mathrm{d}a = r_q^{|B_\times|} 1_{\{d^*k + \mathfrak{q}(I \cup I') = 0 \text{ on } B_\times\}}. \tag{39}$$

The coefficients $(d^*k)(b) + \mathfrak{q}(I \cup I')(b)$ are integers. This is precisely Gauss' law at the cut links, with a nonnegative multiplier. When there are no cut links the condition is empty.

3.5. **The reflected form on an adapted torus.** Let $\mathcal{B}^+$ be the Grassmann-valued integrands whose generators are in $\mathcal{X}_+$ and whose bounded measurable coefficient functions depend only on the upper and mirror curvatures and link phases. For the later operator estimates we also use $\mathcal{G}^+$, with the same generator support and coefficient-variable dependence as $\mathcal{B}^+$, but with boundedness replaced by the following condition: for some $C < \infty$ and $\varepsilon > 0$, each coefficient Φ satisfies

$$W_+ |\Phi| \le C \prod_{\sigma \in P_+} e^{-\varepsilon F_\sigma^2}. \tag{40}$$

This condition is imposed on the upper and mirror variables and places no growth factor on mirror plaquettes. The bound is relevant only where the upper cube constraints hold. Such half-configurations can be extended to full closed configurations by reflecting their upper z, keeping z_0, and taking $z_\times = 0$; the cut-cube identity (34) proves closedness. Thus these coefficient functions have an unambiguous reading wherever $W_+ \neq 0$. In particular $\mathcal{B}^+ \subset \mathcal{G}^+$.

Proposition 3.2. For a torus adapted to ℓ and $G, G' \in \mathcal{G}^+$, the integral $Z_{\mathbb{T}}[\Theta(G')G]$ converges absolutely, coefficient by coefficient, and admits a representation

$$Z_{\mathbb{T}}[\Theta(G')G] = \int \overline{\widehat{G'}(s)} \widehat{G}(s) \, \mathrm{d}\mathfrak{m}(s), \tag{41}$$

where $\mathfrak{m}$ is a finite positive measure and $G \mapsto \widehat{G}$ is linear with bounded measurable values. This holds for even, odd and mixed-parity integrands. In particular $Z_{\mathbb{T}}[\Theta(G)G] \ge 0$.

For every bounded scalar observable Φ depending only on curvatures of plaquettes wholly contained in U_+, the curvature law satisfies

$$\int_\Omega \overline{\Phi(\theta^* F)} \Phi(F) \, \mathrm{d}\mu_{\mathbb{T}}(F) \ge 0. \tag{42}$$

It is invariant under this reflection. ∎

Proof. The measure space has coordinates

$$s = (A_0, z_0, \alpha, k, I, I'),$$

with mirror variables (A_0, z_0), cut-cube variables $\alpha \in [0, 2\pi)^{Q_\times}$, integers $k \in \mathbb{Z}^{P_\times}$, and the two finite mode subsets. Define

$$\mathrm{d}\mathfrak{m}(s) = W_0 \, \mathrm{d}(A_0, z_0) \prod_{Q \in Q_\times} \frac{\mathrm{d}\alpha_Q}{2\pi} \prod_{\sigma \in P_\times} \frac{q}{\sqrt{2\pi}} e^{-\beta_\sigma^2/2} \cdot \omega_I \omega_{I'} r_q^{|B_\times|} 1_{\{d^*k + \mathfrak{q}(I \cup I') = 0 \text{ on } B_\times\}}, \tag{43}$$

where z_0, k, I, I' carry counting measure. Put

$$\widehat{G}(s) = \int W_+ u_{\alpha,k} \int_{\mathcal{X}_+} G e^{-S_+} \zeta_I \check{\zeta}_{I'} \, \mathrm{d}(A_+, z_+). \tag{44}$$

The action S_+ has bounded coefficients depending only on upper and mirror links. Thus the integrand in this formula is a half-integrand of the stated kind.

We first prove the identity for bounded coefficients and definite parities. Inserting (32) into the reflected functional combines each term with the half factors

$$\xi = Ge^{-S_+}\zeta_I\check{\zeta}_{I'}, \qquad \xi' = G'e^{-S_+}\zeta_I\check{\zeta}_{I'}.$$

The sign from this rearrangement is $(-1)^{(|G|+|G'|)(|I|+|I'|)}$. It is one when the parities agree. When they disagree, one of ξ, ξ' is odd, so its half Berezin integral vanishes; the corresponding full integral also vanishes by (33). The sign consequently disappears from the integrated formula in all cases.

Now insert (38) and integrate the cut links. Formula (39) supplies the last factor in (43). At fixed s, the remaining factors are

$$\Theta(W_+u_{\alpha,k}\xi')W_+u_{\alpha,k}\xi.$$

The lower photon variables map measure-preservingly onto an independent set of upper variables. Combining this change of variables with (33) gives $\overline{\widehat{G'}(s)}\widehat{G}(s)$, proving (41) once the exchanges of integration have been justified.

For that justification let C bound all the finitely many half Berezin coefficients of $Ge^{-S_+}\zeta_I\check{\zeta}_{I'}$ and of the corresponding expression with G'. The scalar phase factors have modulus one. Before the cut-link integration, an absolute majorant therefore has total integral at most

$$C^2(r_qC_q)^{3V}\prod_{i \text{ at } \Pi}(1+\omega_i)\prod_{i' \text{ at } \Pi'}(1+\omega_{i'}). \tag{45}$$

The upper, lower and mirror variables are bounded by their Gaussian sums; the cut-link intervals contribute their lengths; the dual photon weights are bounded by $C_q^{|P_\times|}$; and the finite mode sums give the two products. This proves absolute integrability and all required uses of Fubini's theorem. The same bounds without the half factors show that $\mathfrak{m}$ is finite. It is positive by construction. Formula (44) is measurable and bounded uniformly in s by $Cr_q^{|B_+|}C_q^{|P_+|}$. Mixed parities follow by sesquilinearity.

For $G, G' \in \mathcal{G}^+$, use (40) with exponents ε and ε'. The algebraic identities above are unchanged. The absolute majorant now replaces the upper and lower Gaussian sums by $C_q(\varepsilon)^{|P_+|}$ and $C_q(\varepsilon')^{|P_-|}$, respectively, while the mirror and cut sums retain exponent $1/2$. These quantities are finite by (5). This proves coefficientwise integrability, the same representation, and the boundedness of $\widehat{G}$. Taking $G' = G$ proves positivity.

For a scalar Φ with this plaquette dependence, divide this positivity by $Z_{\mathbb{T}}[1] > 0$ and use the definition of the curvature law. Reflection invariance follows from (25), since the adapted torus is preserved by reflection. This proves (42) and completes the theorem. □

The same form remains positive after multiplication of the weight by $\Phi_+\Theta(\Phi_+)$, with Φ_+ bounded and scalar on the upper half: apply the theorem to $G\Phi_+$. This observation will be used for the reflected arrays of observables. It uses positivity of the joint reflected form before integrating out the fermion. The determinant's pointwise nonnegativity by itself does not imply reflection positivity.

4. Slab operators and finite-volume comparison

A cubic torus admits the coordinate reflection-positive representations of the preceding section. A diagonal mirror requires a different period lattice. To compare the two geometries, we express the functional as a trace of operators associated with slabs. Reflection gives the adjoint of a slab operator; repeated reflected copies give its Schatten norm. The resulting comparison changes the closing vector, and, when needed, the fermion character. Three such changes connect the cubic torus with the diagonal torus and with a second coordinate torus of the same volume. Their pressures differ by a quantity tending to zero. This is the finite-volume input for diagonal reflection positivity.

The slab kernel rules and trace are proved in proposition 4.1; the folds and pressure estimates are proved in propositions 4.2 to 4.4.

The same boundary construction will give the coordinate trace used in section 5 to prove Gaussian domination. There the real external two-cochain enters through unitary electric translations and magnetic multiplication operators; two choices of time direction remove its electric and magnetic parts.

All parameters remain fixed. We use the counting Berezin convention (15).

4.1. **Local observables and their lifts.** We need local observables that may contain fermion variables and whose coefficients grow with the curvature. Their definition must make sense in every flux sector. Let $B = y + [-r, r]^3$, and write $B^{(j)}$ for the oriented lattice j-cells contained in B. The fermion sites assigned to B are

$$\mathcal{X}(B) = \{x \in \mathcal{X} : \pi(x) + [-\tfrac{1}{2}, \tfrac{1}{2}]^3 \subset B\}.$$

This support condition includes the photon site at which the fermion transforms under gauge transformations.

Let $\mathcal{G}(B)$ consist of the Borel maps $G : \mathbb{R}^{B^{(1)}} \to \mathrm{Gr}(\mathcal{X}(B))$ with gauge-invariant values and with the following growth bound. If $G(\widehat{A}) = \sum_\Psi a_\Psi(\widehat{A})\Psi$, then, for some $C < \infty$ and $\varepsilon > 0$,

$$|a_\Psi(\widehat{A})| \le C \exp\left((\tfrac{1}{2} - \varepsilon) \sum_{\sigma \in B^{(2)}} (d\widehat{A})_\sigma^2 \right). \tag{46}$$

Here gauge invariance means

$$a_\Psi(\widehat{A} - d\lambda) e^{iq\langle \lambda, \mathfrak{q}(\Psi)\rangle} = a_\Psi(\widehat{A}) \tag{47}$$

for every real function λ on the photon sites of B; $\mathfrak{q}(\Psi)$ counts the ψ generators minus the $\bar\psi$ generators at fermion sites projecting to each photon site. We denote the even subspace by $\mathcal{G}^{\mathrm{ev}}(B)$.

A local lift of a configuration (A, z) on B is a real one-cochain $\widehat{A}$ on $B^{(1)}$ satisfying

$$d\widehat{A} = F \quad \text{on } B^{(2)}, \qquad e^{iq\widehat{A}(b)} = e^{iqA(b)} \quad (b \in B^{(1)}). \tag{48}$$

We evaluate G at this lift and then impose the fermion identifications of the torus or cylinder. We now verify that this procedure is Borel and independent of the lift.

Replace B by its integer box $\prod_\mu [a_\mu, b_\mu]$, where $a_\mu = \lceil y_\mu - r \rceil$ and $b_\mu = \lfloor y_\mu + r \rfloor$. If this box has no sites, use the unique empty lift. Otherwise its axial spanning tree consists of all links in direction 3, the links in direction 2 on $x_3 = a_3$, and the links in direction 1 on $x_2 = a_2$, $x_3 = a_3$. Prescribe a one-cochain on this tree. Its values on the remaining links are determined by

$$\begin{aligned} \widehat{A}_1(x + e_2) &= \widehat{A}_1(x) + \widehat{A}_2(x + e_1) - \widehat{A}_2(x) - F_{12}(x), && x_3 = a_3, \\ \widehat{A}_1(x + e_3) &= \widehat{A}_1(x) + \widehat{A}_3(x + e_1) - \widehat{A}_3(x) - F_{13}(x), \\ \widehat{A}_2(x + e_3) &= \widehat{A}_2(x) + \widehat{A}_3(x + e_2) - \widehat{A}_3(x) - F_{23}(x). \end{aligned} \tag{49}$$

First use the first line in increasing x_2, then the last two lines in increasing x_3. Each link outside the tree occurs as an unknown once. The construction is linear in F and the tree values. It gives the prescribed 13- and 23-curvatures and the 12-curvature on the bottom face. If $dF = 0$, induction over the cubes gives the remaining 12-curvatures: on each cube the two closed two-cochains F and $d\widehat{A}$ agree on five faces, hence on the sixth.

On the tree choose $\widehat{A}(b) = \arg_q(e^{iqA(b)})$, where $\arg_q$ takes values in $[0, r_q)$. This is a Borel choice. Subtract the recursion for A from (49). The tree differences lie in $r_q\mathbb{Z}$, and every subsequent difference is a signed sum of earlier ones and of a value of $r_q z$. Thus $\widehat{A} - A$ is integer-valued in units of r_q on every link. This proves (48), with Borel dependence on F and the tree phases.

Two local lifts differ by $r_q k$, where k is an integer closed one-cochain. Sum k along the tree paths from its bottom corner to define an integer function λ. Then $d\lambda = k$ on the tree. The

zero-curvature recursion shows that $k - d\lambda = 0$ on every link. Equation (47), applied with $-r_q\lambda$, gives $a_\Psi(\widehat{A} + r_q d\lambda) = a_\Psi(\widehat{A})$, since $\langle \lambda, \mathfrak{q}(\Psi)\rangle$ is integer. The value of G is therefore independent of the lift.

For an open set U, let $\mathcal{A}(U)$ be the linear span of the maps

$$\widehat{A} \longmapsto \Phi(d\widehat{A}) \prod_{j=1}^{r} G_j(\widehat{A}|_{B_j^{(1)}}), \tag{50}$$

where the boxes $B_j \subset U$ are pairwise disjoint, $G_j \in \mathcal{G}^{\mathrm{ev}}(B_j)$, and Φ is a bounded Borel function of finitely many curvature values at plaquettes contained in U. A box B embeds in a period group Γ when the sets $B + [-1,1]^3 + \gamma$, $\gamma \in \Gamma$, are pairwise disjoint. Boxes $B_1, \dots, B_r$ embed disjointly in Γ when each embeds and all sets $B_j + \gamma$, indexed by distinct pairs (j, γ), are pairwise disjoint. A product is read on the quotient only when its boxes embed disjointly.

It remains to check that equal maps (50) have equal values on configurations. This requires one lift for every box in a representation. On $\mathbb{Z}^3$, a closed integer two-cochain z has the integer primitive

$$\begin{aligned} k_1(x) &= 0, \\ k_2(x) &= \sum_{s=0}^{x_1-1} z_{12}(s, x_2, x_3), \\ k_3(x) &= \sum_{s=0}^{x_1-1} z_{13}(s, x_2, x_3) + \sum_{s=0}^{x_2-1} z_{23}(0, s, x_3). \end{aligned}$$

The sums are oriented for negative endpoints. The first two components of $dk = z$ follow by taking differences. In the third, $dz = 0$ changes the summand into a difference of consecutive z_{23} values, so the sum telescopes. Therefore $\widehat{A} = A + r_q k$ is a lift on the whole lattice. It need not be periodic, even when (A, z) is periodic.

The same conclusion holds on a closed slab. Choose a coordinate μ_0 absent from the normal n_ℓ. The slab cell complex is a planar strip times $\mathbb{Z}$ in direction μ_0. Let $\mu < \nu$ be the other coordinates, denoted by (a, r). Write $Q(a, r)$ for the unit square with lower corner (a, r) and I_r for the interval of a such that $[(a, r), (a, r+1)]$ lies in the strip. When I_r is nonempty choose $a_r \in I_r$. On the base sheet set the μ-link values of k to zero and put

$$k([(a, r), (a, r+1)] \times \{0\}) = \sum_{a_r \le s < a} z(Q(s, r) \times \{0\}).$$

All squares in this oriented sum lie in the strip because I_r is an interval. Set $k = 0$ on the μ_0-links. For a horizontal link b in direction $\nu' \in \{\mu, \nu\}$, extend by

$$k(b \times \{j\}) = k(b \times \{0\}) + s_{\nu'} \sum_{0 \le j' < j} z(b \times [j', j'+1]), \qquad s_{\nu'} = \begin{cases} -1, & \nu' < \mu_0, \\ +1, & \nu' > \mu_0. \end{cases}$$

The sums with reversed endpoints have the opposite sign. The resulting primitive has $dk = z$ on all vertical plaquettes and on the base sheet. The closed two-cochain $z - dk$ then has vanishing vertical components; closedness makes its horizontal component constant between sheets. It vanishes on the base sheet and hence everywhere. Again the primitive is taken on the lifted strip, not on its periodic quotient.

Evaluation at this common lift, followed by the quotient homomorphism on fermion generators, agrees with the separate local evaluations. It is linear in the map G, so it is independent of any representation of G as a sum of products. Moreover, different box plaquettes have different representatives on the quotient. Multiplying (46) by the Maxwell density therefore leaves

$$C \exp(-\varepsilon_G \|F\|^2), \qquad \varepsilon_G = \min(\tfrac{1}{2}, \varepsilon_1, \dots, \varepsilon_r) > 0. \tag{51}$$

The finite Grassmann expansion of the action has bounded coefficients for real A. The Gaussian sum (5) now proves absolute integrability, coefficient by coefficient. This also proves the required Borel measurability on the quotient.

4.2. **Boundary spaces and slab kernels.** Fix a mirror family ℓ and a cross-section $\Xi = (\Gamma_\parallel, \chi_\parallel)$, where $\Gamma_\parallel \subset \mathbb{Z}^3 \cap \{\ell = 0\}$ has rank two and systole at least four. Here and below, the systole of a period lattice Γ is $\min_{\gamma\in\Gamma\setminus\{0\}} |\gamma|_\infty$. Work on the cylinder $\mathbb{Z}^3/\Gamma_\parallel$. The admissible mirror positions are $c \in \frac{1}{2}|n_\ell|^2\mathbb{Z}$. Write $\mathcal{M}^j(c)$ for the j-cells contained in $\{\ell = c\}$, $\mathcal{X}^j(c)$ for those cut through their relative interiors by this plane, and $\mathcal{B}^j(c, c')$ for the j-cells contained in $\{c \le \ell \le c'\}$ and in neither bounding plane. In particular a slab cell may touch either boundary. All these sets are finite on a bounded slab.

For a chain of admissible mirrors, the cylinder cells partition into the cells in its mirrors, those cut by them, and those lying between consecutive mirrors. A cell cut by c has its height range $[c - |n_\ell|^2/2, c + |n_\ell|^2/2]$. Its non-cut faces lie on one side of the plane; the other faces are cut by the same plane. This partition applies also when two mirrors are consecutive. A fermion bond crosses at most one mirror of such a chain, and no fermion site lies in a mirror.

A closed-slab configuration $\varpi = (A, z)$ assigns real link values and integer plaquette values to all cells of the closed slab, with $dz = 0$ on its cubes. We integrate its interior values using $A(b) \in [0, r_q)$ and counting measure for z. Its interior photon weight is

$$W_{c,c'}(\varpi) = \prod_{\sigma\in\mathcal{B}^2(c,c')} e^{-F_\sigma^2/2} \prod_{Q\in\mathcal{B}^3(c,c')} \mathbf{1}[(dz)(Q) = 0]. \tag{52}$$

We use the larger class $\mathcal{I}_\Xi[c, c']$ of Borel Grassmann-valued functions of closed-slab configurations, with fermion support $c < \ell < c'$, invariant under $(A, z) \mapsto (A + r_q k, z - dk)$, whose coefficients obey

$$|a_\Psi(\varpi)| \le C_G \exp\left((\tfrac{1}{2} - \varepsilon_G) \sum_{\sigma\in\mathcal{B}^2(c,c')} F_\sigma^2 \right), \qquad 0 < \varepsilon_G \le \tfrac{1}{2}. \tag{53}$$

The local products supported in the open slab give elements of this class by the lift construction above. Products in consecutive slabs remain in the class, with the smaller of the two positive ε's. Reflection and translation also preserve the class.

At the mirror c, let $\mathcal{J}(c)$ be the ordered set of crossing modes from (27), taken modulo $\Gamma_\parallel$. Let $\mathcal{F}(c)$ be the exterior space with orthonormal basis e_I, $I \subset \mathcal{J}(c)$. Put $\omega_c e_I = \omega_I e_I$ and $U_\mathrm{f} e_I = (-1)^{|I|} e_I$. Translation of the modes includes both the character $\chi_\parallel$ and the sign required to reorder their product:

$$U_v e_I = s_v(I) e_{r_v(I)}, \qquad \tau_v \zeta_I^{(c)} = s_v(I) \zeta_{r_v(I)}^{(c+\ell(v))}.$$

The U_v are unitary, satisfy the translation law, and commute with parity. For an integer translation parallel to the normal, choose the orders compatibly so that $s_v(I) = 1$.

For a Grassmann element H supported in the open slab define

$$N_{c,c'}[H]_{I',I} = \int_{(c,c')} \zeta_I^{(c)} H \Theta_{c'}(\zeta_{I'}^{(c')}). \tag{54}$$

An operator Υ on $\mathcal{F}(c)$ is represented at the mirror by

$$\mathrm{Ins}_c[\Upsilon] = \sum_{I,J} \Upsilon_{IJ} \Theta_c(\zeta_J^{(c)}) \zeta_I^{(c)}.$$

The particular insertion $\mathrm{Ins}_c[\omega_c \,\mathrm{diag}(u_I(A))]$ is the crossing exponential (29). The ordering in these definitions gives

$$N_{c,c''}[H \,\mathrm{Ins}_{c'}[\Upsilon] H'] = N_{c',c''}[H'] \Upsilon N_{c,c'}[H]. \tag{55}$$

Indeed, an entry of the left side is a sum of products $[\zeta_I H \Theta_{c'} \zeta_J][\zeta_K H' \Theta_{c''} \zeta_{I''}]$ in disjoint sets of slab generators. Berezin factorization gives the entries on the right without moving any of these factors. Reflecting the integrand reverses their order and conjugates the Berezin coefficient; with $v = 2(c'-c)n_\ell/|n_\ell|^2$ this gives

$$N_{c',2c'-c}[\Theta_{c'} H] = U_v N_{c,c'}[H]^*. \tag{56}$$

We next include the photon crossing in the boundary data. The cut incidence maps d_c and d_c^+ are the restrictions of the coboundary to cut faces and to upper faces, as in (34). Introduce one angle ϑ_Q for each cut cube and one integer k_σ for each cut plaquette, and set

$$\begin{aligned} \varkappa_\sigma &= r_q^{-1}((d_c^* \vartheta)_\sigma - 2\pi k_\sigma), \\ \mathrm{d}\mu_c^\times &= \prod_{Q \in \mathcal{X}^3(c)} \frac{\mathrm{d}\vartheta_Q}{2\pi} \prod_{\sigma \in \mathcal{X}^2(c)} \frac{q}{\sqrt{2\pi}} e^{-\varkappa_\sigma^2/2}, \\ e_c((\vartheta,k);\varpi) &= e^{i\langle \vartheta, d_c^+ z\rangle - i\langle \varkappa, d_c^+ A\rangle}, \\ D_c(k,I) &= r_q^{|\mathcal{X}^1(c)|} \mathbf{1}[d_c^* k + \mathfrak{q}(I) = 0]. \end{aligned} \tag{57}$$

Here $\mathfrak{q}(I)(b)$ is the number of modes of groups $1,2$ over the cut link b, minus the number of groups $3,4$ over it. There are no cut links at diagonal mirrors. At a coordinate site plane there are no cut cells at all; its Fourier space is a point.

The one-mirror photon identity proved by the Fourier and Poisson calculation of the preceding section is

$$\begin{aligned} \sum_{z^\times} \int \mathrm{d}A^\times \prod_{\sigma \in \mathcal{X}^2(c)} e^{-F_\sigma^2/2} \prod_{Q \in \mathcal{X}^3(c)} \mathbf{1}[(dz)(Q) = 0]\, u_I(A^\times) \\ = \int \mathrm{d}\mu_c^\times\, D_c(k,I) e_c((\vartheta,k);\varpi_+) \overline{e_c((\vartheta,k);\theta_c^* \varpi_-)}. \end{aligned} \tag{58}$$

The upper and lower data on the right have the same values on the mirror. Their upper and reflected lower faces provide the non-cut faces on the left. Cube Fourier multipliers enforce closedness, Poisson summation gives the factors $e^{-\varkappa^2/2}$, and compact cut-link integration gives $D_c(k,I)$. This identity is used in the direction shown when constructing a kernel and in the opposite direction when gluing two kernels.

Mirror variables are $\xi = (\mathfrak{m}, \vartheta, k, I)$, with $\mathfrak{m} = (A_0, z_0) \in [0, r_q)^{\mathcal{M}^1(c)} \times \mathbb{Z}^{\mathcal{M}^2(c)}$. On their space Y_c put

$$\mathrm{d}\mu_c(\xi) = \mathrm{d}A_0 \prod_{\sigma \in \mathcal{M}^2(c)} e^{-F_\sigma(\mathfrak{m})^2/2} \mathrm{d}\mu_c^\times(\vartheta,k) D_c(k,I), \qquad \mathcal{H}(c) = L^2(Y_c, \mu_c). \tag{59}$$

The integer and mode variables carry counting measure. This is a finite positive measure. Summing the mirror plaquette Gaussians gives $C_q^{|\mathcal{M}^2(c)|}$, while Poisson summation at fixed angles gives

$$\frac{q}{\sqrt{2\pi}} \sum_{k_\sigma} e^{-\varkappa_\sigma^2/2} \le \sum_{j \in \mathbb{Z}} e^{-r_q^2 j^2/2}.$$

The remaining factors are bounded by $r_q^{|\mathcal{M}^1(c)|+|\mathcal{X}^1(c)|} 2^{|\mathcal{J}(c)|}$. The induced signed translations $V_v : \mathcal{H}(c) \to \mathcal{H}(c + \ell(v))$ are unitary and satisfy the group law. The parity operator multiplies the component I by $(-1)^{|I|}$.

Let $S_{(c,c')}(A)$ be the fermion action containing precisely the mass and bond terms with fermion endpoints in the open slab. For $G \in \mathcal{I}_\Xi[c,c']$, define the kernel

$$\begin{aligned} \mathfrak{S}[G](\xi',\xi) = \sqrt{\omega_{I'}\omega_I} \int \mathrm{d}\varpi^\circ\, W_{c,c'}(\varpi) e_c((\vartheta,k);\varpi) \overline{e_{c'}((\vartheta',k');\theta_{c'}^* \varpi)} \\ \cdot N_{c,c'}[G(\varpi) e^{-S_{(c,c')}(A)}]_{I',I}, \qquad \varpi = \mathfrak{m} \vee \varpi^\circ \vee \mathfrak{m}'. \end{aligned} \tag{60}$$

Set the whole integrand to zero where $W_{c,c'}(\varpi) = 0$; $G(\varpi)$ is evaluated only on the closed configurations in its domain. The associated integral operator maps $\mathcal{H}(c)$ to $\mathcal{H}(c')$. The square roots of the mode weights belong in the kernel; the measure (59) contains none of these weights. This places one full crossing weight at a glued mirror.

Proposition 4.1. Every $\mathfrak{S}[G]$ is Hilbert–Schmidt and depends linearly on G. If G has constants C_G, ε_G in (53), then

$$\begin{aligned} |\mathfrak{S}[G](\xi', \xi)| &\le C_{c,c'} C_G C_q(\varepsilon_G)^{|\mathcal{B}^2(c,c')|}, \\ \|\mathfrak{S}[G]\|_2 &\le \sqrt{\mu_c(Y_c)\mu_{c'}(Y_{c'})}\, C_{c,c'} C_G C_q(\varepsilon_G)^{|\mathcal{B}^2(c,c')|}. \end{aligned} \tag{61}$$

The constant depends on the fixed cross-section, mirror positions, q, M, ρ, w_i, and chamber base point. For consecutive slabs,

$$\begin{aligned} \mathfrak{S}[G']\mathfrak{S}[G] &= \mathfrak{S}[GG'], \\ \mathfrak{S}[\tau_v G] &= V_v \mathfrak{S}[G] V_v^{-1}, \\ \mathfrak{S}[\Theta_{c'} G] &= V_{2(c'-c)n_\ell/|n_\ell|^2} \mathfrak{S}[G]^*. \end{aligned} \tag{62}$$

If $\mathbb{T} = (\Gamma, \chi)$ has cross-section Ξ, $g \in \Gamma$ is a closing vector with minimal positive height $p = \ell(g)$, and $c_0 < \cdots < c_r = c_0 + p$ are admissible, $r \ge 2$, and $G_j \in \mathcal{I}_\Xi[c_{j-1}, c_j]$ for every j, then

$$Z_{\mathbb{T}}[G_1 \cdots G_r] = \operatorname{Tr}_{\mathcal{H}(c_0)} \left(V_g^{-1} U_{\mathrm{f}}^{(1+\chi(g))/2} \mathfrak{S}[G_r] \cdots \mathfrak{S}[G_1] \right). \tag{63}$$

These identities apply to arbitrary parities. For homogeneous G, $U_{\mathrm{f}} \mathfrak{S}[G] = (-1)^{\deg G} \mathfrak{S}[G] U_{\mathrm{f}}$. ∎

Proof. Each coefficient of the finite Grassmann polynomial $e^{-S_{(c,c')}(A)}$ is bounded uniformly in real A. Each coefficient in the matrix in (60) is consequently bounded by a fixed multiple of the majorant for G. The phases have modulus one. After multiplication by the photon weight, the absolute integrand is bounded by

$$C C_G \prod_{\sigma \in \mathcal{B}^2(c,c')} e^{-\varepsilon_G F_\sigma^2}.$$

Drop the cube constraints and sum the integers separately. Each sum is at most $C_q(\varepsilon_G)$; each compact link integral contributes r_q. The mode factors and finite matrix dimension enter $C_{c,c'}$. This proves the pointwise bound in (61). The boundary measures are finite, so the kernel is in $L^2(Y_{c'} \times Y_c)$, with the stated norm. These bounds also justify all the gluing integrals below before any cancellation.

For composition, multiply the two kernels and integrate over the common boundary. Their two square roots give $\omega_{I'}$ for the middle mode. The two phases and its boundary measure are precisely the right side of (58). Applying that identity restores the photon variables on the cut cells, their Maxwell weights, their cube constraints, and the phase $u_{I'}$ of the crossing modes. Sum the middle modes by (55), with $\Upsilon = \omega_{c'} \operatorname{diag}(u_{I'})$. This restores the crossing fermion action. The plaquettes in the middle mirror come from its boundary measure, and the cell partition restores the full weight on the combined slab. No cube lies in a mirror, so every closedness constraint is present once. This is the kernel of $\mathfrak{S}[GG']$.

Translation acts by a signed relabeling of the modes and an orientation-preserving relabeling of cells, proving covariance. For reflection use (56); the two boundary phases exchange their roles and the interior weight is preserved. Reflection may reverse a compact integration interval, but integer shift invariance makes any interval of length r_q equivalent to $[0, r_q)$. The reflected upper boundary is the translate of the original lower one by $2(c' - c)n_\ell/|n_\ell|^2$. This gives the last identity of (62). Parity follows because a Berezin integral of odd total degree vanishes.

For the trace, composition reduces the claim to two slabs. A product of two Hilbert–Schmidt kernels is trace class, and its trace is the absolutely convergent integral of their cyclic product. The

closing operator identifies the upper mirror data with the translate by g of the lower data. Its contribution to this cyclic kernel is

$$(-\chi(g))^{|I_0|}s_g(I_0).$$

Here s_g is the mode relabeling sign; $(-\chi(g))^{|I_0|}$ is the remaining Grassmann closing sign. It is 1 for antiperiodic closure and the fermion parity for periodic closure. To see this sign directly, translate the lower crossing product to the upper boundary. Each of its $|I_0|$ generators contributes $\chi(g)$, while moving it across the adjacent upper product contributes $(-1)^{|I_0|}$ in every term of even total degree. Their product is $(-\chi(g))^{|I_0|}$.

Apply (58) at the two mirrors. For fixed photon variables the remaining mode sum is

$$\operatorname{tr}_{\mathcal{F}(c_0)}(U_g^{-1}U_{\mathrm{f}}^{(1+\chi(g))/2}N_2\Upsilon_1N_1\Upsilon_0), \qquad \Upsilon_j=\omega_{c_j}\operatorname{diag}(u_I(A)).$$

Berezin composition and closure replace this trace by the full torus Berezin integral. The two open-slab actions and the two crossing actions include each mass and hopping term once. The photon variables likewise include every torus link and plaquette once, and the cut and interior cube constraints impose $dz=0$ everywhere. Their measure is therefore the measure in (16). This proves (63) for even total degree. With odd total degree both sides vanish, the operator under the trace anticommuting with parity. Finally, expand nonhomogeneous entries into their even and odd parts. □

4.3. **Reflected folds and Schatten estimates.** For an operator Y between Hilbert spaces, write $\|Y\|_s=(\operatorname{Tr}(Y^*Y)^{s/2})^{1/s}$. We use Schatten's inequality in the following form; see [Pon25, slides 55–57]. If $1\le s_j<\infty$, $Y_j:H_{j-1}\to H_j$ belongs to the Schatten class of exponent s_j, $H_r=H_0$, and $\sum_j s_j^{-1}=1$, then the cyclic product is trace class and

$$|\operatorname{Tr}(Y_r\cdots Y_1)|\le\prod_{j=1}^{r}\|Y_j\|_{s_j}. \tag{64}$$

The statement for different spaces follows from the usual inequality on their orthogonal direct sum: extend each Y_j by zero on all other summands. The nonzero block of the product is the displayed cyclic product, and the singular values of each extension are those of Y_j. A unitary closing identification can be absorbed into Y_r without changing its Schatten norm.

Let $\mathbb{T}$ have height period p in the family ℓ. Its straightened torus $\mathbb{T}^0$ has the same cross-section and the closing vector

$$g^0=\frac{p}{|n_\ell|^2}n_\ell, \qquad \Gamma^0=\Gamma_\parallel\oplus\mathbb{Z}g^0, \qquad \chi^0(g^0)=-1. \tag{65}$$

We use this torus only when $g^0\in\mathbb{Z}^3$ and its lattice has systole at least four. Its volume is the volume of $\mathbb{T}$: both lattices have the same covolume in the mirror and the same normal height period.

Divide one period into $2k$ slabs of equal height h', with all $c_j=c_0+jh'$ admissible and $p=2kh'$. For an integrand G in one of these slabs, form its fold by reflecting it successively in the bounding planes, taking all $2k$ copies. This is an integrand on $\mathbb{T}^0$. Its fermion boxes are disjoint: they lie strictly between different consecutive mirrors; within each slab the cross-section embedding separates their translates.

Proposition 4.2. For integrands $G_j\in\mathcal{I}_\Xi[c_{j-1},c_j]$,

$$Z_{\mathbb{T}^0}[\operatorname{fold}_\ell G_j]=\|\mathfrak{S}[G_j]\|_{2k}^{2k},$$
$$|Z_{\mathbb{T}}[G_1\cdots G_{2k}]|\le\prod_{j=1}^{2k}Z_{\mathbb{T}^0}[\operatorname{fold}_\ell G_j]^{1/(2k)}. \tag{66}$$

In particular every folded factor on the right is nonnegative. ■

Proof. Fix j and write $a = c_{j-1}$. Put $Y = \mathfrak{S}[G_j] : \mathcal{H}(a) \to \mathcal{H}(a+h')$, and form the fold on $[a, a+p]$, with initial boundary a in the trace formula. Let $v = 2h'n_\ell/|n_\ell|^2$, which is integer because both mirror positions are admissible. The first reflected pair has product $V_v Y^*Y$ by (62). Its translate by sv has product $V_{(s+1)v}Y^*YV_{sv}^{-1}$. Multiplying all k pairs telescopes to

$$V_{kv}(Y^*Y)^k.$$

Since $kv = g^0$ and $\chi^0(g^0) = -1$, the closing translation in (63) cancels V_{g^0} and introduces no parity. The trace is $\mathrm{Tr}_{\mathcal{H}(a)}(Y^*Y)^k = \|Y\|_{2k}^{2k}$. This calculation takes place at the starting boundary of each slab and proves the first identity for every G_j.

For the original torus, the operator $V_g^{-1}U_{\mathrm{f}}^{(1+\chi(g))/2}$ is unitary. Apply (64) to its trace formula, with every exponent equal to $2k$. The first identity identifies the resulting norms with folded partition functions. The Hilbert–Schmidt estimate ensures all operators belong to the required Schatten classes. □

We next apply this inequality to cells. This is the repeated-reflection chessboard argument of [Frö+78, Section 4], with the period lattice and its fermion character carried through each fold. A frame is a triple of the mirror families with mutually orthogonal normals n_1, n_2, n_3. The two frames needed below are (x_1, x_2, x_3) and $(x_1 - x_2, x_1 + x_2, x_3)$. Choose admissible offsets o_a and an integer height width $h > 0$. The open cells are

$$Q_\kappa = \{x : o_a + h\kappa_a < \ell^a(x) < o_a + h(\kappa_a + 1),\ a = 1, 2, 3\}, \qquad |Q| = \frac{h^3}{|n_1||n_2||n_3|}. \tag{67}$$

The reflections in the planes $\ell^a = o_a + hj$ generate a product of three infinite dihedral groups. Exactly one element of this group carries a given open cell to any specified open cell. Its action on even Grassmann-valued functions is generated by the reflections Θ and translations used above. Denote the resulting copy of $G \in \mathcal{A}(Q_\kappa)$ in $Q_{\kappa'}$ by $G^{[\kappa']}$.

A rectangular torus in this frame has periods $\gamma_a = p_a n_a/|n_a|^2 \in \mathbb{Z}^3$, with $\chi(\gamma_a) = -1$. If each $p_a \in 2h\mathbb{Z}$, the copies fit its period lattice. Let $N = |\mathbb{T}|/|Q|$ be the number of cells, and define

$$\mathrm{Fold}_{\mathbb{T}} G = \prod_{\kappa' \bmod \Gamma} G^{[\kappa']}, \qquad \|G\|_{\mathbb{T}} = Z_{\mathbb{T}}[\mathrm{Fold}_{\mathbb{T}} G]^{1/N}. \tag{68}$$

This is an unnormalized seminorm. Its defining partition function is nonnegative, by the first identity of (66) applied to one layer of the folded array. The value is independent of the chosen representatives of the cells.

Proposition 4.3. If a rectangular torus $\mathbb{T}$ carries the cells (67), then

$$\left| Z_{\mathbb{T}} \left[\prod_\kappa G_\kappa \right] \right| \le \prod_\kappa \|G_\kappa\|_{\mathbb{T}}, \qquad G_\kappa \in \mathcal{A}(Q_\kappa). \tag{69}$$

Let $C(a, b, c)$ have coordinate periods ae_1, be_2, ce_3, all with character -1. Let $e_- = e_1 - e_2$, $e_+ = e_1 + e_2$, and let $D(a, b, c)$ have periods ae_-, be_+, ce_3, all with character -1. Let $D'(a, b, c)$ have the same periods and characters, except for $\chi(be_+) = +1$. For the diagonal frame and $n \in 2h\mathbb{Z}$, $n \ge 8$, the cells have representatives

$$0 \le \kappa_1 < n/h, \qquad 0 \le \kappa_2 < 2n/h, \qquad 0 \le \kappa_3 < n/h.$$

They represent the cells of $C(n, n, n)$, $D'(n/2, n, n)$, and $D(n/2, n, n)$, and

$$\left| Z_{\mathbb{T}} \left[\prod_\kappa G_\kappa \right] \right| \le \prod_\kappa \|G_\kappa\|_{D(n/2,n,n)}, \quad \mathbb{T} = C(n, n, n) \text{ or } D'(n/2, n, n). \tag{70}$$

For coordinate cells of width h, $a, c \in 2h\mathbb{Z}$, $a, c \geq 4$, one has

$$\left| Z_{D(a,2a,c)} \left[\prod_{\kappa} G_\kappa \right] \right| \leq \prod_{\kappa} \|G_\kappa\|_{C(a,4a,c)}. \tag{71}$$

■

Proof. Group the cell integrands into layers perpendicular to the first normal, and apply (66). Its right side consists of the arrays obtained by retaining one layer and reflecting it throughout the straightened torus. Next apply the same inequality to each such array in direction two, and then in direction three. The reflected array after these three steps is the full fold of the integrand from one original cell. The three exponents multiply to

$$\frac{h}{p_1}\frac{h}{p_2}\frac{h}{p_3} = \frac{1}{N}.$$

Thus the last product consists of one factor $\|G_\kappa\|_{\mathbb{T}_3}$ for each original cell, where $\mathbb{T}_3$ is the final rectangular torus.

We verify that all three arrays have consistent periods and that the same cells represent them. If Γ_0 is the initial lattice, then straightening successively in the frame gives

$$\begin{aligned} \Gamma_1 &= (\Gamma_0 \cap \{\ell^1 = 0\}) \oplus \mathbb{Z}\gamma_1, \\ \Gamma_2 &= (\Gamma_0 \cap \{\ell^1 = \ell^2 = 0\}) \oplus \mathbb{Z}\gamma_1 \oplus \mathbb{Z}\gamma_2, \\ \Gamma_3 &= \mathbb{Z}\gamma_1 \oplus \mathbb{Z}\gamma_2 \oplus \mathbb{Z}\gamma_3, \qquad \gamma_a = p_a n_a / |n_a|^2. \end{aligned}$$

The kernel in the second line is a rank-one group along n_3; its positive generator is γ_3, by the definition of the third height period. Characters already fixed at γ_1, γ_2 are retained in subsequent cross-sections, so all three final characters are -1. The brick $0 \leq \kappa_a < p_a/h$ represents Γ_3. It represents Γ_2 as well; then its section $\kappa_2 = 0$ represents the second cross-section, and its section $\kappa_1 = 0$ represents the first one. Restoring the normal coordinate in each case gives representatives for the preceding lattice. These groups all have the same covolume.

We also check the stronger box embedding condition used in the local classes. If $B = y + [-r, r]^3$ lies in an open cell, then $2|n_a|^2 r < h$ for every a, by taking the range of ℓ^a on B. For a nonzero period γ, at least one height is nonzero and $|\ell^a(\gamma)| \geq 2h$. Consequently $|\gamma|_\infty > 4r$, and the systole also gives $|\gamma|_\infty \geq 4$. These inequalities imply $|\gamma|_\infty > 2r + 2$: use the second if $r < 1$, the first if $r \geq 1$. Thus translates of $B + [-1, 1]^3$ are disjoint. Boxes in different cell orbits cannot meet after period translations, since the open cells themselves are disjoint. It follows that every layer belongs to its slab class. This justifies the three applications.

For a rectangular torus, every straightening leaves the torus unchanged, which proves (69). For a cubic torus with even side n, the diagonal operations are

$$C(n,n,n) \xrightarrow{\ell_-} D'(n/2,n,n) \xrightarrow{\ell_+} D(n/2,n,n), \qquad D(n/2,n,n) \xrightarrow{x_1} C(n/2,2n,n). \tag{72}$$

For the first operation the cross-section is generated by ne_+, ne_3, with characters $+1, -1$; its height period is n, and straightening replaces the closing vector ne_1 by $(n/2)e_-$. For the second, the cross-section is generated by $(n/2)e_-, ne_3$, both with character -1. Its closing vector ne_+ already points along the normal, and the operation changes its character from $+1$ to -1. For the third, the cross-section is generated by $2ne_2, ne_3$, both with character -1; the height period is $n/2$ and the straight closing vector is $(n/2)e_1$. The lattices have systole at least four.

The first two diagonal operations, followed by the unchanged x_3 operation, have height periods $(n, 2n, n)$. Hence they give (70); starting at D' instead, use the order (ℓ_+, x_3, ℓ_-) to obtain the same final fold. Likewise, starting at $D(a, 2a, c)$ and using coordinate directions gives periods $(a, 4a, c)$ and ends at $C(a, 4a, c)$, proving (71). □

4.4. **Pressure and the comparison error.** Write $C(n) = C(n,n,n)$ and

$$f(n_1,n_2,n_3) = \frac{1}{n_1 n_2 n_3} \log Z_{C(n_1,n_2,n_3)}[1].$$

The mass and charge in this function are held fixed. The following error will compare local expectations in the cubic and diagonal geometries.

Proposition 4.4. The pressure f is nonincreasing in each side, and

$$f(n_1,n_2,n_3) \longrightarrow f_\infty := \inf f \in \mathbb{R} \quad \text{as } \min_\mu n_\mu \longrightarrow \infty.$$

For every even $n \geq 8$,

$$Z_{C(n)}[1] \leq Z_{D'(n/2,n,n)}[1] \leq Z_{D(n/2,n,n)}[1] \leq Z_{C(n/2,2n,n)}[1]. \tag{73}$$

With $\Delta(n) = f(n/2,2n,n) - f(n,n,n)$,

$$0 \leq \Delta(n) \leq f(n/2,n/2,n/2) - f_\infty \longrightarrow 0. \tag{74}$$

The logarithms of any two partition functions in (73) differ in absolute value by at most $n^3\Delta(n)$. ■

Proof. Fix the cross-section with sides n_2, n_3 and let $Y = \mathfrak{S}[1] : \mathcal{H}(0) \to \mathcal{H}(1/2)$ be the coordinate half-step. The first identity of (66), applied to $2n_1$ half-slabs, gives

$$Z_{C(n_1,n_2,n_3)}[1] = \operatorname{Tr}(Y^*Y)^{n_1}. \tag{75}$$

The positive trace-class operator Y^*Y is independent of n_1 and is nonzero, since the partition function is positive. Consequently

$$f(n_1,n_2,n_3) = \frac{1}{n_2 n_3} \log \|Y^*Y\|_{n_1}.$$

Schatten norms decrease as their exponent increases, so f decreases in n_1. Permuting the time direction proves the other two claims. The same trace identity shows separate convexity of $\log Z$ in the side lengths: Hölder's inequality gives $\operatorname{Tr} A^{(1-t)s+ts'} \leq (\operatorname{Tr} A^s)^{1-t}(\operatorname{Tr} A^{s'})^t$ for $A = Y^*Y$.

The lower partition bound (21) gives the volume-independent floor

$$f \geq \log C_* + 3\log r_q - 6r_q^2.$$

Together with the upper bound in the same equation, it makes f_∞ finite. Monotonicity now gives convergence in any manner with all sides tending to infinity: choose a finite triple at which $f < f_\infty + \delta$ and then exceed its three sides.

Apply (66) with constant integrands to each of the three operations (72). A fold of the constant is the constant, so $Z_{\mathbb{T}}[1] \leq Z_{\mathbb{T}^0}[1]$. This proves the chain. Each torus in it has volume n^3, so its first and last logarithms differ by $n^3\Delta(n)$. Side monotonicity gives $f(n/2,2n,n) \leq f(n/2,n/2,n/2)$, while $f(n,n,n) \geq f_\infty$, proving (74). All intermediate logarithms lie between the endpoints, which proves the last assertion. □

Thus the estimate concerns the logarithm per unit volume. It does not give a bounded ratio of partition functions: the retained bound is $e^{n^3\Delta(n)}$, with $\Delta(n) \to 0$. The local comparison divides the logarithmic error bound $n^3\Delta(n)$ by the number of copies of the observable and by the perturbation strength.

5. Gaussian domination on coordinate tori

The compactness estimate comes from the coordinate trace. We first retain the real slice connections and all magnetic fluxes in that trace, then remove the external curvature field by two applications of Schatten Hölder in different time directions. The all-flux trace (86) gives the Gaussian bound in proposition 5.1.

5.1. **Coordinate slices and the all-flux trace.** For $s > 0$ write

$$g_s(x) = (2\pi s)^{-1/2} e^{-x^2/(2s)}.$$

Tensor powers of g_s denote product densities. Thus $e^{-a|E|^2}$ has convolution kernel g_{2a} in each real link variable.

For Gaussian domination we use a slice representation with real spatial connections. Take direction 1 as time on $C_{\mathbf{n}} = C(n_1, n_2, n_3)$, $\mathbf{n} = (n_1, n_2, n_3)$, $n_\mu \geq 4$, and let $\Sigma = \mathbb{Z}/n_2\mathbb{Z} \times \mathbb{Z}/n_3\mathbb{Z}$. Its links and plaquettes are $\Sigma^{(1)}$ and $\Sigma^{(2)}$, its coboundary is d_s, and $v = |\Sigma| = n_2 n_3$. A connection has spatial slices A_j and temporal links a_j, and its curvature has the parts

$$F_j^{\mathrm{el}} = A_{j+1} - A_j - d_s a_j + r_q z_j^{\mathrm{el}}, \qquad F_j^{\mathrm{mag}} = d_s A_j + r_q z_j^{\mathrm{mag}}. \tag{76}$$

Closedness reads $z_{j+1}^{\mathrm{mag}} - z_j^{\mathrm{mag}} = d_s z_j^{\mathrm{el}}$.

We first unfold the integer variables. Put $k_0 = 0$, $k_{j+1} = k_j + z_j^{\mathrm{el}}$. Closedness is equivalent to

$$z_j^{\mathrm{mag}} = z_0^{\mathrm{mag}} + d_s k_j, \qquad d_s k_{n_1} = 0.$$

The integer shift with spatial values k_j and zero temporal values removes all electric integers except the last one, k_{n_1}, and changes every spatial slice to $A_j + r_q k_j$. Summing k_j while integrating A_j over its compact box unfolds the intermediate slices to all of $\mathbb{R}^{\Sigma^{(1)}}$. The last integer remains as a winding vector

$$g = r_q k_{n_1} \in \Gamma_s, \qquad \Gamma_s = r_q(\mathbb{Z}^{\Sigma^{(1)}} \cap \ker d_s).$$

Fix a plaquette σ_0 of Σ and write $k = \sum_\sigma z_0^{\mathrm{mag}}(\sigma)$. Then $z_0^{\mathrm{mag}} - k\mathbf{1}_{\sigma_0} = d_s k'$ for an integer one-cochain k'. Integer exactness follows from the dual graph whose vertices are the plaquettes. A link of Σ is a dual edge directed toward the plaquette where its coefficient in d_s is positive. The dual graph is connected. Choose a spanning tree rooted at σ_0. Unit signed flow along the path from the root to a vertex σ has divergence $\delta_\sigma - \delta_{\sigma_0}$. Summing these paths with the integer coefficients of a zero-sum two-cochain gives precisely that cochain as its divergence, hence as $d_s k'$. Two choices of k' differ by an integer element of $\ker d_s$.

Shift every slice by $r_q k'$. The magnetic integers now equal $k\mathbf{1}_{\sigma_0}$ in every slice. Summation over k' modulo closed integer cochains replaces the compact box for A_0 by a fundamental domain of Γ_s. We arrive at paths

$$A_0 \in \mathbb{R}^{\Sigma^{(1)}}/\Gamma_s, \qquad A_j \in \mathbb{R}^{\Sigma^{(1)}} \ (0 < j < n_1), \qquad A_{n_1} = A_0 + g, \quad g \in \Gamma_s,$$

with $a_j \in [0, r_q)^\Sigma$ and flux $k \in \mathbb{Z}$. Their curvatures are

$$F_j^{\mathrm{el}} = A_{j+1} - A_j - d_s a_j, \qquad F_j^{\mathrm{mag}} = d_s A_j + r_q k \mathbf{1}_{\sigma_0}. \tag{77}$$

Thus integration over the original configuration quotient equals summation over k, g and integration over these paths. For a nonnegative integrand the identity follows by Tonelli's theorem. For an absolutely integrable complex integrand it follows by applying that identity to its modulus and then using Fubini's theorem. Each change above preserves link phases and curvature, so it applies to the determinant and to all curvature observables used here.

There are two finite Fock spaces in this representation: $\mathcal{F}'$ at a coordinate site plane and $\mathcal{F}$ at a link plane. The matrix of the open half-slab gives a map $N(A) : \mathcal{F}' \to \mathcal{F}$, and we include both crossing weights by writing

$$\widehat{N}(A) = \omega^{1/2} N(A) (\omega')^{1/2}.$$

Let $u(a)$ be the diagonal unitary on $\mathcal{F}$ with entries $e^{iq\langle a, \mathfrak{q}(I)\rangle}$. The fermion trace is

$$\det \mathcal{D}_A^{C_{\mathbf{n}}} = \mathrm{tr}_{\mathcal{F}'} \prod_{j=n_1-1,\dots,0} (\widehat{N}(A_{j+1})^* u(a_j) \widehat{N}(A_j)). \tag{78}$$

This formula also holds on an unfolded path, since $A_{n_1} - A_0 \in \Gamma_s$ leaves $\widehat{N}$ unchanged.

To derive (78), split the torus at each site and link plane. The half-slab above a link plane is the reflected half-slab below it, hence its matrix is $N(A_{j+1})^*$. The crossing at a site plane has no link phase and gives ω'. At a link plane it gives $\omega^{1/2}u(a_j)\omega^{1/2}$. Compose the matrices and close antiperiodically; the closing parity is the identity. Cyclicity moves half of the final ω' to the front, giving (78). Every entry of $\widehat{N}(A)$ is a polynomial in the link phases: the half-slab action is quadratic in finitely many generators, its exponential terminates, and Berezin integration extracts a coefficient. Consequently

$$C_N := \sup_{A\in\mathbb{R}^{\Sigma^{(1)}}} \|\widehat{N}(A)\| < \infty, \qquad \widehat{N}(A + r_q k') = \widehat{N}(A) \quad (k' \in \mathbb{Z}^{\Sigma^{(1)}}). \tag{79}$$

On real slice connections the two Hilbert spaces are

$$H^s = \bigoplus_{k\in\mathbb{Z}} L^2(\mathbb{R}^{\Sigma^{(1)}}/\Gamma_s; \mathcal{F}'), \qquad H^l = \bigoplus_{k\in\mathbb{Z}} L^2(\mathbb{R}^{\Sigma^{(1)}}/\Gamma_s; \mathcal{F}).$$

All quotient measures are induced by Lebesgue measure. Define commuting self-adjoint electric operators E_b by $e^{-i\langle x,E\rangle}\Phi(A) = \Phi(A - x)$. On H^l let

$$U_G(a)\Phi(A) = u(a)\Phi(A - d_s a), \qquad P_G = r_q^{-v}\int_{[0,r_q)^\Sigma} U_G(a)\,\mathrm{d}a, \qquad K = r_q^v e^{-|E|^2/2}P_G. \tag{80}$$

The integer mode charges make U_G a representation of the compact gauge group $\mathbb{R}^\Sigma/r_q\mathbb{Z}^\Sigma$. Its average P_G is the orthogonal projection onto invariant vectors. It commutes with every function of E, because $u(a)$ acts on the Fock index and all spatial translations commute. Hence

$$K^{1/2} = r_q^{v/2}e^{-|E|^2/4}P_G. \tag{81}$$

For a real slice two-cochain b, define the multiplication operator $M_b : H^s \to H^l$ by

$$(M_b\Phi)_k(A) = e^{-\|d_sA + r_qk\mathbf{1}_{\sigma_0} + b\|^2/4}\widehat{N}(A)\Phi_k(A), \qquad X_b = K^{1/2}M_b. \tag{82}$$

The magnetic weight is bounded and $\widehat{N}$ is bounded, so M_b is bounded. We must prove that X_b is Hilbert–Schmidt; the real quotient is not compact and there are infinitely many flux sectors.

The periodized Gaussian

$$\widetilde{g}_{1/2}([x]) = \sum_{\gamma\in\Gamma_s}\prod_{b\in\Sigma^{(1)}} \pi^{-1/2}e^{-(x_b+\gamma_b)^2}$$

has integral one on the quotient and is bounded. Indeed $\Gamma_s \subset r_q\mathbb{Z}^{\Sigma^{(1)}}$, so the series is dominated by the product of one-dimensional Gaussian sums on that larger lattice. It is therefore in $L^1 \cap L^\infty \subset L^2$. The kernel of $K^{1/2}$ is

$$K^{1/2}(A', A) = r_q^{-v/2}\int_{[0,r_q)^\Sigma} \widetilde{g}_{1/2}([A' - A - d_s a])u(a)\,\mathrm{d}a.$$

Its operator norm is bounded by the nonnegative function

$$\widehat{g}([x]) = r_q^{-v/2}\int_{[0,r_q)^\Sigma} \widetilde{g}_{1/2}([x - d_s a])\,\mathrm{d}a,$$

which has finite L^1 and L^∞ norms and hence finite L^2 norm. Integrating the square of the kernel of X_b over A' now gives

$$\|X_b\|_2^2 \le (\dim\mathcal{F}')C_N^2\|\widehat{g}\|_2^2\sum_{k\in\mathbb{Z}}\int_{\mathbb{R}^{\Sigma^{(1)}}/\Gamma_s} e^{-\|d_sA + r_qk\mathbf{1}_{\sigma_0} + b\|^2/2}\,\mathrm{d}A. \tag{83}$$

To bound this last sum, split the quotient as

$$(\ker d_s/\Gamma_s) \times (\ker d_s)^\perp.$$

The first factor is compact: $\ker d_s$ is the kernel of an integer matrix, hence has an integer basis, and Γ_s has full rank in it. On the second factor d_s is injective, so $\|d_s A\|^2$ is positive definite. Moreover $\ker d_s^*$ consists of constant plaquette functions. This follows from the two equations

$$(d_s^* b)(y,2) = b(y) - b(y-e_3), \qquad (d_s^* b)(y,3) = b(y-e_2) - b(y).$$

Thus $\operatorname{ran} d_s$ is the orthogonal complement of the constant two-cochain. The orthogonal constant component of $r_q k \mathbf{1}_{\sigma_0} + b$ has squared norm $v^{-1}(r_q k + \sum_\sigma b_\sigma)^2$. Translate the variable in $(\ker d_s)^\perp$ to remove the remaining component. We obtain

$$\sum_k \int e^{-\|d_s A + r_q k \mathbf{1}_{\sigma_0} + b\|^2/2} \, \mathrm{d}A = C_s \sum_{k\in\mathbb{Z}} e^{-(r_q k + \sum_\sigma b_\sigma)^2/(2v)} < \infty, \tag{84}$$

where $C_s = \operatorname{vol}(\ker d_s/\Gamma_s) \int_{(\ker d_s)^\perp} e^{-\|d_s A\|^2/2} \, \mathrm{d}A < \infty$. This constant is independent of k and b. Equations (83)–(84) prove the Hilbert–Schmidt property for every real magnetic perturbation. The bounds here depend on the fixed cross-section; their purpose is to justify the trace argument.

For a real external two-cochain β on $C_{\mathbf{n}}$, put

$$Z_{C_{\mathbf{n}}}(\beta) = \int_{\mathrm{Conf}(C_{\mathbf{n}})} e^{-\|F+\beta\|^2/2} \det \mathcal{D}_A^{C_{\mathbf{n}}} \, \mathrm{d}(A,z). \tag{85}$$

Let β_j^{el} and β_j^{mag} be its electric and magnetic slices. Then

$$Z_{C_{\mathbf{n}}}(\beta) = (2\pi)^{n_1 n_2 n_3} \operatorname{Tr}_{H^s} \prod_{j=n_1-1,\dots,0} (X_{j+1}^* U_j X_j), \quad X_j = X_{\beta_j^{\mathrm{mag}}}, \quad U_j = e^{i\langle \beta_j^{\mathrm{el}}, E\rangle}, \quad X_{n_1} = X_0. \tag{86}$$

Here is the conversion from the unfolded path integral. Write $g_t(x) = (2\pi t)^{-1/2} \exp(-x^2/(2t))$ for $t > 0$. The electric Maxwell factors are the kernels

$$e^{-|A_{j+1} - A_j - d_s a_j + \beta_j^{\mathrm{el}}|^2/2} = (2\pi)^v g_1^{\otimes 2v}(A_{j+1} - A_j - d_s a_j + \beta_j^{\mathrm{el}}).$$

After summing the winding vector they are periodized on Γ_s. Gaussian convolution with a translated kernel is $e^{i\langle \beta_j^{\mathrm{el}}, E\rangle} e^{-|E|^2/2}$. Integration over a_j includes the matrix $u(a_j)$ from (78); it is exactly the gauge average in (80). The magnetic Gaussian in slice j splits into the two quarter-Gaussians in M_j and M_j^*. The determinant half-slab matrices supply their other factors. Thus the cyclic path integral is the trace on H^l of

$$\prod_{j=n_1-1,\dots,0} (K^{1/2} U_j K^{1/2} M_j M_j^*).$$

Each time slice contributed $(2\pi)^v$, giving the constant in (86). Move the final M_0^* to the front and regroup the products as $X_{j+1}^* U_j X_j$. The Hilbert–Schmidt property above and the absolute Gaussian path majorant justify this cyclicity and every interchange. In particular the external-field integral is finite; alternatively this follows directly by absorbing a linear factor into a smaller Gaussian.

We spell out the path and kernel identification used here. On a fundamental domain write every intermediate real slice as $A_j = A_j^\flat + g_j$, $g_j \in \Gamma_s$, and set $g_0 = 0$ and $g_{n_1} = A_{n_1} - A_0$. Each unperiodized electric kernel then has the argument $A_{j+1}^\flat - A_j^\flat + g_{j+1} - g_j - d_s a_j$. The map $(g_1, \dots, g_{n_1}) \mapsto (g_{j+1} - g_j)_{0\le j<n_1}$ is a bijection of $\Gamma_s^{n_1}$. Summing these differences independently is therefore precisely periodization of every cyclic kernel. The magnetic factors and $\widehat{N}$ are unchanged by the shifts g_j. The finite Fock trace is changed from $\mathcal{F}'$ to $\mathcal{F}$ by moving its last $\widehat{N}(A_0)^*$ to the front; periodicity identifies it with $\widehat{N}(A_{n_1})^*$.

For absolute convergence replace every matrix factor by its norm, and bound each shifted electric Gaussian by a fixed constant times $e^{-|A'-A|^2/4}$ and each shifted magnetic factor by a constant times $e^{-\|F^{\mathrm{mag}}\|^2/4}$. This defines nonnegative scalar kernels. Their periodized convolutions are bounded and in L^2, and multiplication by the magnetic Gaussian is square integrable after summing fluxes by (84). Hence each scalar majorant operator is Hilbert–Schmidt. The integral of their cyclic product

is finite: group the first factor against the product of the others and use the Hilbert–Schmidt Cauchy–Schwarz inequality. Tonelli first gives this conclusion for the nonnegative majorants, and Fubini then permits the kernel expansions and periodization changes for the original integrand. This proves the trace identity without a formal interchange of the infinite flux sum and the operator trace.

5.2. **Gaussian domination.**

Proposition 5.1. For every coordinate torus with sides at least four and every real two-cochain β,

$$0 < Z_{C_{\mathbf{n}}}(\beta) \le Z_{C_{\mathbf{n}}}(0). \tag{87}$$

Consequently its normalized curvature law obeys

$$\mathbb{E}_{C_{\mathbf{n}}} e^{\langle \beta, F\rangle} \le e^{\|\beta\|^2/2}. \tag{88}$$

For real two-cochains β_i and complex numbers z_i,

$$\left|\mathbb{E}_{C_{\mathbf{n}}} e^{\sum_i z_i \langle \beta_i, F\rangle}\right| \le \exp\left(\tfrac{1}{2}\left\|\sum_i (\operatorname{Re} z_i)\beta_i\right\|^2\right). \tag{89}$$

The constants are independent of the volume, the charge, and the mass. ■

Proof. The determinant is nonnegative and positive on a set of positive configuration measure, since $Z_{C_{\mathbf{n}}}(0) > 0$. The shifted photon density is strictly positive everywhere, proving the strict lower bound in (87).

In (86) move X_0^* through the trace to obtain a trace on H^l:

$$(2\pi)^{-n_1 n_2 n_3} Z_{C_{\mathbf{n}}}(\beta) = \operatorname{Tr}_{H^l}(U_{n_1-1} X_{n_1-1} X_{n_1-1}^* \cdots U_0 X_0 X_0^*).$$

Every U_j is unitary. Apply Schatten's inequality with all exponents equal to n_1; the trace on the left is nonnegative. Since $X_j X_j^*$ and $X_j^* X_j$ have the same nonzero eigenvalues, this gives

$$Z_{C_{\mathbf{n}}}(\beta) \le \prod_{j=0}^{n_1-1} \left((2\pi)^{n_1 n_2 n_3} \operatorname{Tr}(X_j^* X_j)^{n_1}\right)^{1/n_1} = \prod_{j=0}^{n_1-1} Z_{C_{\mathbf{n}}}(\beta^{(j)})^{1/n_1}. \tag{90}$$

Here $\beta^{(j)}$ has zero electric part and the same magnetic part β_j^{mag} in every time slice. The equality uses the trace formula again, with the same X_j in each slice and all unitary factors equal to the identity.

Each $\beta^{(j)}$ is supported only on 23-plaquettes. Interchange directions 1 and 2. The torus becomes $C(n_2, n_1, n_3)$ and the perturbation is supported only on 13-plaquettes, so its magnetic part for the new time direction is zero. The reduction (90) on this torus bounds its partition function by the unperturbed one. Lattice isometry covariance of the determinant and photon measure identifies these partition functions with $Z_{C_{\mathbf{n}}}(\beta^{(j)})$ and $Z_{C_{\mathbf{n}}}(0)$. Substitution in (90) proves the upper bound in (87).

Replacing β by $-\beta$ and completing the square gives (88). For complex parameters, take the absolute value inside the probability integral and apply that bound to $\sum_i (\operatorname{Re} z_i)\beta_i$, proving (89). □

5.3. **Coordinate limits and the reflection form.**

The coordinate part of Theorem A. The determinant and normalization results make μ_n a probability law on Ω for every integer $n \ge 4$. If a finitely supported cochain β embeds in the torus without overlap, its periodization β_n has

$$\langle \beta_n, F\rangle = \langle \beta, \widetilde{F}\rangle, \qquad \|\beta_n\| = \|\beta\|.$$

Proposition 5.1 therefore gives the claimed Gaussian bound for this test under μ_n. Every fixed test embeds once n is large. A single plaquette embeds for every $n \geq 4$: its box of radius $1/2$, enlarged by $[-1,1]^3$, has side three. Applying the bound with $\beta = \pm R 1_\sigma$ and then Markov's inequality gives

$$\mu_n\{|F_\sigma| > R\} \leq 2e^{-R^2/2}, \qquad R > 0,$$

uniformly in $n \geq 4$.

Enumerate the plaquettes as $\sigma_1, \sigma_2, \ldots$. Given $0 < \varepsilon < 1$, put

$$R_j = \sqrt{2\log(2^{j+1}/\varepsilon)}, \qquad K_\varepsilon = \Omega \cap \prod_{j\geq 1}[-R_j, R_j].$$

The set Ω is closed in the countable product, so K_ε is compact. The union bound gives $\mu_n(\Omega \setminus K_\varepsilon) \leq \varepsilon$ for all $n \geq 4$. Thus the family is tight. Since Ω is Polish, Prokhorov's theorem gives subsequential weak limits; see [Sor24, Theorem 1.1].

Let $\mu_{n_i} \to \nu$ along such a subsequence, and fix β. For all sufficiently large i, the support of β embeds. For $K > 0$ the function $\min\{e^{\langle \beta, F\rangle}, K\}$ is bounded and continuous. Weak convergence followed by monotone convergence in K gives (i). Each μ_{n_i} is invariant under W and charge conjugation by the finite-volume symmetry identities. The cubic period lattice and its character, equal to -1 on each coordinate period, are preserved by every signed coordinate permutation. Translations preserve them as well. Their pullbacks and $F \mapsto -F$ are continuous on Ω, so these symmetries pass to ν. This proves (ii) and reflection invariance at every mirror.

Fix a coordinate mirror $\Pi = \{x_j = c\}$, $c \in \frac{1}{2}\mathbb{Z}$, and a bounded continuous function G of finitely many plaquettes wholly in $\{x_j > c\}$. Choose $h_G > 0$ so that every corner of these plaquettes satisfies $c < x_j < c + h_G$. For $n_i > 2h_G$, the support lies in the open positive half-period slab

$$\{c < x_j < c + n_i/2\}.$$

The cube is adapted to this mirror, with closing vector $n_i e_j$ and character -1. Its other seam is at $c + n_i/2$, which belongs to $\frac{1}{2}\mathbb{Z}$ for both even and odd n_i. Thus both seams are admissible, including when one is a site plane and the other a link plane. The adapted positivity theorem applies and gives

$$\mu_{n_i}\big(\overline{G(\theta_\Pi^* F)}\, G(F)\big) \geq 0.$$

The integrand is bounded and continuous, so the inequality passes to ν.

We give the extension to the whole side, which will also be used for the diagonal mirrors. Suppose a law λ is invariant under θ_Π and its reflected form is nonnegative on bounded continuous local tests on one open side. On each finite-coordinate marginal, compactly supported continuous functions are dense in L^2. Reflection invariance and Cauchy–Schwarz give

$$\big|\lambda\big((\Theta A)A - (\Theta B)B\big)\big| \leq \|A - B\|_{L^2(\lambda)}\big(\|A\|_{L^2(\lambda)} + \|B\|_{L^2(\lambda)}\big), \qquad \Theta A = \overline{A \circ \theta_\Pi^*}.$$

Hence positivity extends to all finite-coordinate L^2 tests. Enumerate the plaquettes in the chosen open side, and let $\mathcal{F}_m$ be generated by the first m. For $A \in L^2(\Sigma_H, \lambda)$, the conditional expectations $\lambda(A \mid \mathcal{F}_m)$ converge to A in L^2, because Σ_H is generated by $\bigcup_m \mathcal{F}_m$. The same inequality extends positivity to A. Reflection carries one open side to the other, giving the assertion on either side. Applied to $\lambda = \nu$, this proves the coordinate part of (iii). The diagonal part remains to be proved by comparison. □

Optimality of the Gaussian constant. Choose a nonzero finitely supported integer two-cochain z_0 with $dz_0 = 0$; for example, take $z_0 = dk$ with k equal to one on a single oriented link and zero on the other links. Put $\beta = r_q z_0$. For all sufficiently large n, these cochains embed in the cubic torus.

We use the same symbols for their periodizations, which preserve $dz_0 = 0$, the source norm, and the pairing with the periodically extended field. Completing the square gives

$$\sum_{dz=0} \int e^{-\|dA+r_q z\|^2/2+\langle \beta, dA+r_q z\rangle} \det \mathcal{D}_A^{C(n)} \, \mathrm{d}A$$
$$= e^{\|\beta\|^2/2} \sum_{dz=0} \int e^{-\|dA+r_q(z-z_0)\|^2/2} \det \mathcal{D}_A^{C(n)} \, \mathrm{d}A = e^{\|\beta\|^2/2} Z_{C(n)}[1].$$

Here the sum runs over all closed integer two-cochains and the integrals have the same domain as in the definition of μ_n. The last equality reindexes this entire sum; the determinant depends only on A. Thus $\mu_n(e^{\langle \beta, F\rangle}) = e^{\|\beta\|^2/2}$. Put $X(F) = e^{\langle \beta, F\rangle}$. The same identity for 2β gives $\mu_n(X^2) = e^{2\|\beta\|^2}$ for large n, and (i) gives $\nu(X^2) \le e^{2\|\beta\|^2}$. For either of these laws λ,

$$0 \le \lambda(X) - \lambda(X \wedge K) \le K^{-1} \lambda(X^2) \le K^{-1} e^{2\|\beta\|^2}.$$

The function $X \wedge K$ is bounded and continuous. First passing to the weak limit at fixed K, then letting $K \to \infty$, gives

$$\nu(e^{\langle \beta, F\rangle}) = e^{\|\beta\|^2/2} \qquad (\beta = r_q z_0).$$

Since $\beta \neq 0$, no constant smaller than one can replace the constant in (i) at the fixed parameters of the theorem.

6. Reflected perturbations and pressure comparison

6.1. **The finite-volume fold inequality.** We give the local form of the comparison that will be used to obtain diagonal positivity. Fix diagonal cells of even height width h, and let O be a real trigonometric polynomial in finitely many curvature values at plaquettes contained in the base cell Q_0. Thus O is bounded and depends only on this open cell. Let $n \ge 8$ be divisible by $4h$, and abbreviate $D_n = D(n/2, n, n)$. Its cell count, and that of the cubic torus, is

$$N_n = \frac{n^3}{|Q|} = \frac{2n^3}{h^3}.$$

The translations between cells form the integer lattice

$$\Lambda_h = \frac{h}{2}\mathbb{Z}e_- + \frac{h}{2}\mathbb{Z}e_+ + h\mathbb{Z}e_3.$$

On $C(n)$ form the translated copies, and on D_n the reflected copies:

$$\widetilde{O}_n(F) = \sum_{v \in \Lambda_h/n\mathbb{Z}^3} O(\tau_v^* F), \qquad O_n^{\mathrm{fold}}(F) = \sum_{\kappa \bmod \Gamma_{D_n}} O(g_\kappa^* F). \tag{91}$$

Each sum has N_n terms. The first sum is read on cubic-periodic curvatures, the second on D_n-periodic curvatures. Here g_κ is the reflected map from the base cell to Q_κ.

Let $\mu_{C(n)}$ and μ_{D_n} be the normalized curvature laws. For $t > 0$ define the tilted diagonal law

$$\mu_{D_n}^t(\Phi) = \frac{\mu_{D_n}(e^{tO_n^{\mathrm{fold}}}\Phi)}{\mu_{D_n}(e^{tO_n^{\mathrm{fold}}})}. \tag{92}$$

The folded array is invariant under the group generated by the cell walls: that group permutes the cells of D_n and their reflected copies. Thus the tilt retains these symmetries.

Proposition 6.1. For the preceding data and every $t > 0$,

$$\mu_{C(n)}(O) \le \mu_{D_n}^t(O) + \frac{|Q|\Delta(n)}{t}. \tag{93}$$

The error is the fixed-parameter pressure error of (74). ∎

Proof. Jensen's inequality and translation invariance on $C(n)$ give

$$\log \frac{Z_{C(n)}[e^{t\widetilde{O}_n}]}{Z_{C(n)}[1]} \geq t\mu_{C(n)}(\widetilde{O}_n) = tN_n\mu_{C(n)}(O). \tag{94}$$

The diagonal chessboard inequality gives

$$Z_{C(n)}[e^{t\widetilde{O}_n}] \leq \|e^{tO}\|_{D_n}^{N_n} = Z_{D_n}[e^{tO_n^{\mathrm{fold}}}]. \tag{95}$$

To check that its N_n factors have the same seminorm, translate the base observable by $v \in \Lambda_h$. Translation permutes the cells and conjugates their reflected maps, so the fold of the translated observable is the translate of the original full fold. The law on D_n is translation invariant, hence its partition function and the seminorm are unchanged. All observables here are bounded scalar functions supported in distinct open cells, so the chessboard inequality applies directly.

Set $L(t) = \log \mu_{D_n}(e^{tO_n^{\mathrm{fold}}})$. Boundedness permits differentiation, giving $L'(t) = \mu_{D_n}^t(O_n^{\mathrm{fold}})$. Convexity and $L(0) = 0$ give $L(t) \leq tL'(t)$. Wall-group invariance of the tilted law makes the expectation of every reflected copy equal to that of O; hence

$$\log \frac{Z_{D_n}[e^{tO_n^{\mathrm{fold}}}]}{Z_{D_n}[1]} \leq tN_n\mu_{D_n}^t(O). \tag{96}$$

Combining (94)–(96) leaves, in addition, the normalization difference

$$\log Z_{D_n}[1] - \log Z_{C(n)}[1] \leq n^3\Delta(n).$$

Division by tN_n proves (93). □

For fixed cell geometry and $t > 0$, the pressure error tends to zero along admissible sides. To remove the tilt we need uniform local bounds and enough translation symmetry to place an arbitrary observable next to a prescribed diagonal wall. We prove the bounds in section 7. The layer reduction in section 8 then recovers coordinate link reflections and hence even translations in each comparison state. These are the symmetries used in the final convex separation.

Fix $q > 0$, $0 < \rho < 1$, positive weights w_i and a real mass M. All volume limits in this section keep these parameters fixed. For the comparison construction we use sides $m_i \to \infty$ such that eight divides all sufficiently large sides and the set

$$\mathcal{D} = \{m \in \mathbb{N} : m \mid m_i \text{ for all sufficiently large } i\}$$

is infinite. The set $\mathcal{D}$ is closed under divisors and least common multiples. The comparison sequence may be chosen so that every positive integer divides all sufficiently large sides; lemma 10.5 constructs such sides below any prescribed original sequence.

6.2. **The tilted law at its walls.** Fix diagonal cells with offsets (c_0, c_+, c_3), even width h, and $c_0, c_+ \in \mathbb{Z}$, $c_3 \in \mathbb{Z} + \frac{1}{2}$, and $4h \in \mathcal{D}$. Let O be the real trigonometric polynomial used in the finite fold comparison. Its support is contained in a box of radius r_O, and we require $h > 8r_O + 8$. Such widths can be chosen because $\mathcal{D}$ contains arbitrarily large multiples of eight. For large divisible sides, put $D_i = D(m_i/2, m_i, m_i)$ and $\mu_i^t = \mu_{D_i}^t$, as in (92). Write

$$O(F) = \sum_\beta a_\beta e^{i\langle\beta,F\rangle}, \qquad |O|_1 = \sum_\beta |a_\beta|.$$

The sum is over the distinct exponents in the chosen finite expansion. In particular $|O| \leq |O|_1$. Let $V_i = O_{m_i}^{\mathrm{fold}}$ and $N_i = m_i^3/|Q|$. For an admissible Grassmann integrand we use the same notation for the normalized functional $\mu_i^t(G) = Z_{D_i}[e^{tV_i}G]/Z_{D_i}[e^{tV_i}]$. On bosonic integrands it agrees with integration against the tilted probability law. The normalizing denominator satisfies

$$Z_{D_i}[e^{tV_i}] \geq e^{-tN_i|O|_1}Z_{D_i}[1] > 0, \qquad t \geq 0. \tag{97}$$

This uses only the positive bosonic measure; it makes no positivity assertion about a Grassmann integrand.

At a wall $\Pi = (\ell^a, o_a + jh)$, the adapted torus has height period p_a, with $(p_1, p_2, p_3) = (m_i, 2m_i, m_i)$. The number p_a/h is even. Split the cells into the two half-slabs adjacent to this wall, and define

$$\Phi_+(F) = \exp\left(t \sum_{Q_\kappa \subset \{o_a + jh < \ell^a < o_a + jh + p_a/2\}} O(g_\kappa^* F) \right),$$

with representatives in one half-period. Reflection in the wall maps these cells onto the other half-period modulo the normal closing vector. Because the reflected copies of O form one wall-invariant array, on periodic curvatures

$$e^{tV_i} = \Phi_+ \Theta_\Pi \Phi_+. \tag{98}$$

The function Φ_+ is bounded, real, and supported strictly within the upper half-slab. The support lies there because each plaquette of a copy of O is contained in its open cell.

Apply the adapted reflected representation to $\Phi_+ G$ and $\Phi_+ G'$. For local even integrands G, G' in the upper half-slab,

$$\mu_i^t(\Theta_\Pi G'\, G) = \frac{\mathrm{Tr}(\mathfrak{S}[\Phi_+ G']^* \mathfrak{S}[\Phi_+ G])}{Z_{D_i}[e^{tV_i}]}. \tag{99}$$

The numerator is the Hilbert–Schmidt inner product of the two slab operators. Thus, on D_i, the tilted functional is reflection positive at each cell wall for local even integrands supported in the corresponding open positive half-period slab. Wall reflection also gives $\mu_i^t(\Theta_\Pi G) = \overline{\mu_i^t(G)}$: the unperturbed functional is reflection covariant, the scalar weight e^{tV_i} is invariant, and its denominator is real and positive.

The bosonic law is invariant under the full group generated by the cell walls. Indeed, that group preserves D_i and permutes the reflected copies in V_i, while the unperturbed bosonic law is invariant under its lattice isometries. In particular the tilt is invariant under translations by he_-, he_+ and $2he_3$. These are the periods of the reflected cell array, rather than all lattice translations.

We use the Polish closed-cochain space Ω with its product topology. The pullback by a lattice reflection or translation is continuous, so invariance passes to weak limits. Positivity passes for each bounded continuous local positive-side test once its support lies in the finite positive half-period slab for all large volumes. Positivity against bounded Borel local tests follows once reflection invariance has passed: on their finite-dimensional marginal approximate the test in L^2 by compactly supported continuous functions. The reflected-form difference is at most

$$\|\Phi - \Phi_k\|_2 (\|\Phi\|_2 + \|\Phi_k\|_2),$$

because reflection preserves the L^2 norm. The Gaussian domination bound also passes to weak limits, by truncating $e^{\langle \beta, F \rangle}$ and then using monotone convergence. The conditional-expectation approximation in section 5.3 then extends positivity to the whole open-side L^2 space. These facts will be applied after obtaining tightness of the tilted laws; their extensive weight alone gives no uniform moment estimate.

7. Local operators and amplification

The reflected perturbation contains a number of terms proportional to the volume. To control one local insertion uniformly, we repeat it in large diagonal cells and then fold those cells in coordinate directions. The coordinate trace bounds each repeated operator by its local norm. Taking the root corresponding to the number of copies leaves an error that tends to one when volume tends to infinity before the perturbation is removed. The local norm, insertion bound and amplification estimate are proved in propositions 7.1 to 7.3.

The estimate to be proved is

$$|\mu_i^t(G)| \le C \gamma_{t,i}(k), \qquad \gamma_{t,i}(k) = \exp\bigl(2t|O|_1 k^3 + |Q| k^3 \Delta(m_i)\bigr).$$

Here C is the coordinate bound of the local insertion G, defined in (110); for the insertion G_X of (107) it is $\|X\|$. The integer k depends on the fixed support radius, cell width and side sequence, but not on t, i or the location of the support. We choose it before taking limits. The pressure error $\Delta(m_i)$ tends to zero, so $\lim_{t\downarrow 0}\lim_{i\to\infty}\gamma_{t,i}(k)=1$. Proposition 7.3 gives the precise assertion and proof. We first construct the local norm and the insertions to which it applies.

7.1. **Local operators at a coordinate link plane.** An insertion must carry an operator norm that is uniform in the cross-section and in electric characters. Fix a finite set S of links of $\mathbb{Z}^2$ and a nonempty finite set Λ of sites containing their endpoints. Let $\mathcal{F}_\Lambda$ be the Fock space of crossing modes over Λ, and put

$$H_S = L^2(\mathbb{R}^S)\otimes\mathcal{F}_\Lambda, \qquad E_b = -i\partial_{A(b)}.$$

The diagonal unitary $u(a)$ on $\mathcal{F}_\Lambda$ has integer mode charges. An operator Υ has charge $z\in\mathbb{Z}^\Lambda$ when $u(a)\Upsilon u(a)^{-1}=e^{iq\langle a,z\rangle}\Upsilon$.

Let $\mathfrak{A}(S)$ be the span of the functions $e^{-Q(\xi_{S'})+\langle\lambda,\xi_{S'}\rangle}$, where $S'\subset S$, Q is real and positive definite on $\mathbb{R}^{S'}$, and $\lambda\in\mathbb{C}^{S'}$. The empty subset gives the constant one. These functions are bounded and uniformly continuous. Their span is closed under products, conjugation and real translations: products add the positive quadratic forms on the union of their supports, while translation completes the same quadratic exponent.

For $\delta\in\mathbb{R}^S$, extended by zero to the full slice, require $d_s^*\delta = qz(\delta)$ with $z(\delta)\in\mathbb{Z}^\Lambda$. The local operator algebra consists of the finite sums

$$X=\sum_\delta e^{i\langle\delta,A\rangle}\Upsilon_\delta(E_S), \qquad \Upsilon_\delta\in\mathfrak{A}(S)\otimes\mathrm{End}_{z(\delta)}(\mathcal{F}_\Lambda). \tag{100}$$

We denote it by $\mathfrak{W}(S,\Lambda)$. The Weyl identity gives

$$\begin{aligned}(e^{i\delta A}\Upsilon(E_S))(e^{i\delta' A}\Upsilon'(E_S)) &= e^{i(\delta+\delta')A}\Upsilon(E_S+\delta')\Upsilon'(E_S),\\ (e^{i\delta A}\Upsilon(E_S))^* &= e^{-i\delta A}\Upsilon^*(E_S-\delta).\end{aligned} \tag{101}$$

Here δA abbreviates the real pairing. These formulas make $\mathfrak{W}(S,\Lambda)$ a unital $*$-algebra. The normal form is unique. In Fourier variables its terms send a function supported near ξ_0 to functions supported near $\xi_0+\delta$; small enough neighborhoods separate the finitely many distinct exponents. If the sum vanishes as an operator, every continuous symbol vanishes on each such neighborhood and hence everywhere.

Every X commutes with the local gauge transformations $\Phi(A)\mapsto u(a)\Phi(A-d_sa)$. The phase change $e^{-i\langle\delta,d_sa\rangle}$ cancels the Fock charge phase. The total charge of $z(\delta)$ is zero, since it is the divergence of a finitely supported cochain. Each Fock term is therefore even. For $\beta\in\mathbb{R}^S$, define

$$\alpha_\beta X = e^{i\langle\beta,A\rangle}Xe^{-i\langle\beta,A\rangle}=\sum_\delta e^{i\langle\delta,A\rangle}\Upsilon_\delta(E_S-\beta). \tag{102}$$

This is an isometric $*$-automorphism on H_S.

For a rectangular cross-section Ξ containing the smallest integer rectangle spanned by Λ, the same formulas define X_Ξ on the real slice quotient space H^l of section 5.1. The Fock operators are localized into the full crossing Fock space by ι. The localization is the tensor inclusion for the exterior space, with the reordering sign and cross-section character attached to each representative; it is a unital injective $*$-homomorphism. The phase $e^{i\langle\delta,A\rangle}$ is well defined on the quotient. Indeed, on the support hull an integer closed one-cochain is an integer coboundary, by the box Poincare argument. Thus for $\gamma\in\Gamma_s$, $\langle\delta,\gamma\rangle\in2\pi\mathbb{Z}$, using $d_s^*\delta=qz(\delta)$. The local Weyl formulas therefore remain valid on the quotient. Uniqueness of the normal form makes the representation independent of its presentation. It commutes with P_G and with functions of the electric variables outside S.

Define the norm needed for characters by

$$\|X\|_{\#} = \sup_{\Xi,\beta} \|(\alpha_\beta X)_\Xi\|. \tag{103}$$

The support hull must embed in the cross-section; the supremum is over such cross-sections and all $\beta \in \mathbb{R}^S$. The electric spectrum of a quotient can be smaller than $\mathbb{R}^S$, so its unshifted operator norm need not equal the local norm.

Proposition 7.1. For $X \in \mathfrak{W}(S,\Lambda)$,

$$\|X\|_{\#} = \|X\|_{\mathcal{L}(H_S)}.$$

The norm is unchanged by enlarging S, Λ. In particular it is a C^*-norm, $\|1\|_{\#} = 1$, and every quotient realization satisfies $\|(\alpha_\beta X)_\Xi\| \le \|X\|_{\#}$. ■

Proof. First remove the Fock localization. Its tensor extension is an injective $*$-homomorphism and hence isometric; we may work on $L^2(\mathbb{R}^{\Sigma^{(1)}}/\Gamma_s) \otimes \mathcal{F}_\Lambda$ with the original finite Fock operators. Call this realization X^0_Ξ. Introduce an additional variable $z \in \mathbb{R}^S$, with electric operators $E' = -i\partial_z$, on

$$L^2(\mathbb{R}^S_z) \otimes L^2(\mathbb{R}^{\Sigma^{(1)}}/\Gamma_s) \otimes \mathcal{F}_\Lambda.$$

The operator $X(z, E')$ has norm $\|X\|$, since it is X tensored with the identity. Choose a Borel representative $A^\flat$ of the quotient and use the two unitaries

$$(U_1\Phi)(z,[A]) = \Phi(z + A^\flat|_S, [A]), \qquad (U_2\Phi)(z,[A]) = \Phi(z, [A - z]),$$

where z is extended by zero outside S. The first conjugation multiplies $e^{i\delta z}$ by $e^{i\delta A}$. The second replaces E' by $E' + E_S$ and cancels the factor $e^{i\delta z}$. Hence

$$U_2U_1X(z,E')(U_2U_1)^{-1} = \sum_\delta e^{i\delta A}\Upsilon_\delta(E_S + E').$$

Fourier transform in z. This becomes a direct integral of $T_p = (\alpha_{-p}X)^0_\Xi$, $p \in \mathbb{R}^S$. Uniform continuity of the symbols makes $p \mapsto T_p$ norm continuous. Its direct-integral norm is therefore $\sup_p \|T_p\|$: the reverse inequality follows by using a unit vector nearly maximizing T_{p_0} on a small ball about p_0. It follows that

$$\|X\| = \sup_p \|(\alpha_{-p}X)^0_\Xi\| = \sup_p \|(\alpha_{-p}X)_\Xi\|.$$

This holds for each admissible cross-section and proves the asserted supremum identity. Enlarging S tensors with an identity on the additional L^2 factor; enlarging Λ uses the same isometric Fock inclusion. Both preserve the norm. □

7.2. **Insertions and a coordinate trace bound.** Let $c = j + \frac{1}{2}$. Write Ins_c for the crossing insertion map, and ω for its diagonal mode weights. For an even matrix kernel $k(A', A)$ on $\mathbb{R}^S$, set

$$\begin{aligned} G[k](\widehat{A}) = {} & \mathrm{Ins}_c\Big[\omega^{1/2} k(\widehat{A}_{j+1}|_S, (\widehat{A}_j + d_s\widehat{a}_j)|_S) u(\widehat{a}_j)\omega^{1/2}\Big] \\ & \cdot \prod_{b\in S} g_1(F^{\mathrm{el}}_j(b))^{-1} \exp(S^\times_c(\Lambda)). \end{aligned} \tag{104}$$

Here $S^\times_c(\Lambda)$ is the part of the fermion action crossing the plane over Λ. The two factors on the second line remove the original electric and fermion crossing weights on the support of k. Thus $G[k]$ inserts k in their place. Its definition uses the infinite lattice and is independent of a torus.

We require

$$\begin{aligned} u(a)^{-1}k(A' + d_sa, A + d_sa)u(a) &= k(A', A), \\ \|k(A', A)\| &\le C\exp(-\epsilon|A' - A|^2), \qquad \epsilon > 0. \end{aligned} \tag{105}$$

Choose a box containing the two copies of S, the temporal links over Λ, and the unit cubes centered at (j, y) and $(j + 1, y)$, $y \in \Lambda$. In what follows we use the cube $B^\#(c, S, \Lambda)$ obtained by taking twice

the largest half-width of the smallest rectangular box containing these sets, with the same center. This choice is covariant under coordinate translations and reflections. The insertion reads only these links and the crossing fields, so it is supported in $B^{\#}$.

The insertion is an even, gauge invariant local integrand. Indeed, an even kernel has entries only between occupation vectors of the same parity, so every monomial $\Theta_c\zeta_J\,\zeta_I$ in its insertion is even. If the gauge parameter is λ, its upper and lower slice values change the matrix in (104) to

$$u(\lambda_{j+1})^{-1}k(\widehat{A}_{j+1},\widehat{A}_j+d_s\widehat{a}_j)u(\widehat{a}_j)u(\lambda_j).$$

The matrix entry acquires precisely the inverse of the phase of $\Theta_c\zeta_J\,\zeta_I$. The electric curvature and the crossing action are gauge invariant. Finally, the kernel estimate and $g_1(x)^{-1}=\sqrt{2\pi}e^{x^2/2}$ bound every Grassmann coefficient by

$$C'\exp\left((\tfrac12-\epsilon')\sum_{b\in S}F_j^{\mathrm{el}}(b)^2\right),\qquad \epsilon'=\min(\epsilon,\tfrac12). \tag{106}$$

The other factors have bounded coefficients. This is the local integrand bound of section 4.1, since its electric plaquettes lie in the box.

For $X\in\mathfrak{W}(S,\Lambda)$ let $k_X^{(s)}$ be the continuous kernel of $e^{-s|E_S|^2}Xe^{-s|E_S|^2}$, with $s=\frac14$ or $\frac18$, and write

$$G_X=G[k_X^{(1/4)}],\qquad G'_X=G[k_X^{(1/8)}]. \tag{107}$$

Both kernels satisfy (105). To see their decay, consider one normal-form term $e^{i\delta A}\Upsilon(E)$ with $\Upsilon(\xi)=e^{-Q(\xi)+\lambda\xi}\Upsilon_0$. Identify Q with its symmetric matrix, so $Q(\xi)=\langle\xi,Q\xi\rangle$, extending both Q and λ by zero to the other links. The brackets in the following exponent are the complex-bilinear extension of the real Euclidean pairing. With $Q_s=4sI+2Q$ and $\lambda_s=\lambda-2s\delta$, its kernel is

$$\begin{aligned}k_X^{(s)}(A',A)=e^{i\delta A'}(2\pi)^{-|S|/2}(\det Q_s)^{-1/2}e^{-s|\delta|^2}\\ \cdot\exp\left(\tfrac12\langle\lambda_s+i(A'-A),Q_s^{-1}(\lambda_s+i(A'-A))\rangle\right)\Upsilon_0.\end{aligned} \tag{108}$$

The real part of this exponent has a strictly negative quadratic form in $A'-A$, with a fixed linear term. It gives the required Gaussian decay after decreasing its exponent. Gauge covariance follows by conjugating the regularized operator by the gauge unitary; equality of their continuous kernels gives the first identity in (105). Finite sums preserve these properties. The identity operator has kernel $\prod_{b\in S}g_1(A'(b)-A(b))$ when $s=\frac14$, so $G_1=1$; unused links and modes cancel in the same way.

Reflection acts on these insertions by adjoints:

$$\Theta_cG_X=G_{X^*},\qquad \Theta_cG'_X=G'_{X^*}. \tag{109}$$

The kernel is changed to $k(A,A')^*$. At the same time the two slices exchange, $\widehat{a}_j$ changes sign, and covariance moves $u(\widehat{a}_j)$ to the required side of the kernel. This gives (109), including its field signs. We also need coordinate translations and spatial reflections to preserve the operator norm, since each coordinate fold contains such images.

For a spatial reflection θ in the second or third coordinate, the spin matrix exchanges the two eigenspaces of γ_1. Its unitary matrix v on crossing modes preserves their weights and reverses their charges. On exterior products the reordering sign is $r_I=(-1)^{|I|(|I|-1)/2}$; hence the Fock unitary is $U=(\bigwedge v)r$. The anticommutation of the two field reflection maps contributes the sign $(-1)^{\deg}$, and moving the two mode products past one another contributes $(-1)^{|I||J|}$. For the parity-matched entries of an even kernel these signs leave exactly r_Ir_J. Consequently

$$\Theta_\theta\,\mathrm{Ins}_c[\Upsilon(\widehat{A})]=\mathrm{Ins}_c[U\overline{\Upsilon(\theta^*\widehat{A})}U^*].$$

The antiunitary $(J\Phi)(A)=U\overline{\Phi(\theta_s^*A)}$ gives this kernel transformation. It sends an oriented electric operator to the negative of the reflected electric operator, and sends $e^{i\delta A}$ to $e^{-i\langle\theta_s^*\delta,A\rangle}$. Thus it

preserves the charge constraint, the local algebra and the heat factors $e^{-s|E|^2}$. Translations act by the corresponding unitary relabeling. Together with adjoints at the link plane, these facts show that each translate or signed coordinate reflection of G_X is an insertion G_{X^g} with $\|X^g\| = \|X\|$. The same holds for G'_X. The box $B^\#$ transforms with the insertion.

An even local integrand G in a box B satisfies the coordinate bound $C \geq 0$ if, on every rectangular coordinate torus $T = (\Gamma, \chi) = C(n_1, n_2, n_3)$ with $n_\mu \geq 4$, for every nonempty finite family of images $g_a B$ under lattice translations and signed coordinate reflections that embeds disjointly in Γ, and every $\beta \in C^2(\Gamma)$,

$$\left| Z_T \left[\varphi_\beta \prod_{a=1}^{r} G^{g_a} \right] \right| \leq C^r Z_T[1], \qquad \varphi_\beta(F) = e^{i\langle \beta, F \rangle}, \tag{110}$$

provided that, in each coordinate direction μ, there is a plaquette σ orthogonal to e_μ such that none of the boxes or their period translates contains any plaquette $\sigma + ke_\mu$, $k \in \mathbb{Z}$. This condition places the flux representative used in the slice trace on plaquettes read by none of the insertions. When the coordinate cell structure cuts T and its walls are at half-integers, boxes contained in its open cells satisfy the free-column condition. Choose a plaquette orthogonal to the chosen direction that meets a wall normal to another direction. Period translations permute the open cells, while every plaquette in this column meets that wall. Thus none of these plaquettes is contained in a box or any of its period translates.

Proposition 7.2. The insertion G_X satisfies (110) with $C = \|X\|$. The insertion G'_X satisfies it with $C = 2^{|S|/2}\|X\|$. A bounded bosonic integrand satisfies it with $C = \|G\|_\infty$. These bounds survive box enlargement, multiplication by a curvature character, and transport by a signed permutation of the coordinates. ∎

Proof. We give the trace calculation, including the support condition. Choose the time direction to be the first coordinate. By the covariance just proved, the images of the insertions have the same operator norms and are at link planes $j + \frac{1}{2}$. The free column lets us choose the magnetic flux plaquette σ_0 so that no plaquette in its column is contained in an insertion box. At a fixed step j their spatial supports and modes are disjoint. Let S_j be the union of the supports of the primed insertions at that step.

The unfolded path variables are genuine local lifts on each $[j, j+1] \times \mathrm{hull}(\Lambda_a)$. On electric plaquettes their coboundary is $A_{j+1} - A_j - d_s a_j$. On magnetic plaquettes it is $d_s A_j$; the flux correction is supported at σ_0, which is not a plaquette of the local hull. Two local lifts with these curvatures and phases differ by an integer exact cochain $\frac{2\pi}{q} d\lambda$. Gauge invariance therefore allows the insertion to be read directly at the path variables. This is why the free-column hypothesis is needed.

At each crossing, the factors $e^{S_c^\times(\Lambda_a)}$ remove the crossing weights on the local modes. The remaining crossing factors, together with their insertions, form

$$\mathrm{Ins}_c \left[\omega^{1/2} \left(\prod_a \iota(k_a(A_{j+1}, A_j + d_s a_j)) \right) u(a_j) \omega^{1/2} \right].$$

The even localized matrices commute because their mode sets are disjoint. The factors g_1^{-1} similarly remove the electric Gaussian densities on the local links. The slice determinant factorization of section 5.1 now gives the path product with these modified crossing kernels. Its Gaussian majorant is integrable: every modified kernel has strict decay in its electric difference, all remaining links retain g_1, and the magnetic multiplier has its Gaussian weight. Hence unfolding, periodization and the path trace are valid for this product.

Separate the curvature character into electric factors $e^{i\langle \beta_j, F_j^{\mathrm{el}} \rangle}$ and magnetic unitaries u_j. Multiplying a regularized kernel by $e^{i\langle \beta_j, A' - A \rangle}$ changes the corresponding operator to

$$e^{-s|E-\beta_j|^2} (\alpha_{\beta_j} X) e^{-s|E-\beta_j|^2}.$$

Fourier transformation of the disjoint link kernels therefore gives

$$Z_T\left[\varphi_\beta \prod_a G_a\right] = (2\pi)^{|T|}\,\mathrm{Tr}_{H^l} \prod_{j=n_1-1,\dots,0} \left(K^{1/2}_{\beta_j,S_j} B_j K^{1/2}_{\beta_j,S_j} M_0 u_j M_0^*\right), \tag{111}$$

where

$$B_j = \prod_{a:j_a=j} (\alpha_{\beta_j} X_a)_\Xi, \qquad K_{\beta,S} = r_q^v e^{-|E-\beta|^2/2+|E-\beta|_S^2/4} P_G.$$

Here $r_q^v = (2\pi/q)^{|\Sigma|}$ is the gauge-volume factor in (80); the primed kernels change the electric Gaussian variance only on S. Both the kernels and the local operators commute with P_G. In particular $\|B_j\| \le \prod_{a:j_a=j} \|X_a\|$ by the norm identity above.

Set $T_j = K^{1/2}_{\beta_j,S_j} M_0$. These are Hilbert–Schmidt by the same quotient Gaussian calculation as (83). Move the first square root cyclically to the end of the trace. The product then has alternating factors $B_j T_j u_j T_{j-1}^*$. Schatten Hölder, with $2n_1$ equal exponents, bounds its absolute trace by

$$\prod_j \|B_j\| \prod_j \|T_j\|_{2n_1}^2.$$

The slice trace applied to $(T_j^* T_j)^{n_1}$ gives exactly

$$\mathrm{Tr}(T_j^* T_j)^{n_1} = (2\pi)^{-|T|} Z_T \left[\prod_{l=0}^{n_1-1} e^{i\langle \beta_j, F_l^{\mathrm{el}}\rangle} \prod_{l=0}^{n_1-1} \prod_{b\in S_j} \sqrt{2} e^{-F_l^{\mathrm{el}}(b)^2/2}\right]. \tag{112}$$

The second product changes g_1 into $g_{1/2}$ on those links. The positive bosonic law bounds the absolute value by $2^{|S_j|n_1/2}(2\pi)^{-|T|} Z_T[1]$. Consequently

$$\prod_j \|T_j\|_{2n_1}^2 \le 2^{\frac{1}{2}\sum_j |S_j|} (2\pi)^{-|T|} Z_T[1].$$

Together with (111) this proves the two operator bounds. For insertions in another time direction use an orientation-preserving coordinate permutation. The scalar bound follows directly from the positive law. Multiplication by a character merely changes β, and enlargement restricts the allowed placements. A signed permutation transports the torus, the free columns and the placements; conjugation of a character still has modulus one. These observations prove the remaining assertions. □

7.3. **Amplification in two cell structures.** A diagonal fold exchanges the first and second time directions. Its copies of a coordinate insertion therefore cannot all be put in one coordinate trace. We will place these copies in different coordinate cells and apply a second chessboard estimate. Each resulting coordinate fold contains copies of only one insertion, all with the same time direction, so (110) applies.

The placement requires two scales depending on the support radius r, but not its center or the tilt. Choose $h'' \in h\mathbb{Z}$ and $k \in \mathbb{N}$ such that

$$h'' \ge 2r + 6h, \qquad kh \ge 16h'', \qquad kh \in h''\mathbb{Z}, \qquad 4h'', 2kh \in \mathcal{D}. \tag{113}$$

These choices exist: take $4h''$ to be the least common multiple of $4h$ and an element of $\mathcal{D}$ exceeding $4(2r+6h)$, then take $2kh$ to be the least common multiple of $4h''$ and an element exceeding $32h''$. Fix choices depending only on h, r and the side sequence. For all sufficiently large i, the needed divisibility conditions then hold simultaneously.

Here is the geometry of the placement. Let $B = y + [-r,r]^3$. Choose coarse diagonal offsets $c'_\pm \in c_\pm + h\mathbb{Z}$ (with $c_- = c_0$) and $c'_3 \in c_3 + h\mathbb{Z}$ so that

$$\ell_-(y) - c'_- \in [2h'', 2h''+h), \quad \ell_+(y) - c'_+ \in [4h'', 4h''+h), \quad y_3 - c'_3 \in [h'', h''+h). \tag{114}$$

The box lies strictly in the coarse cell Q_0' of width kh. Every element of its wall group has the form

$$g = \tau_{kh(z_-e_-+z_+e_++2z_3e_3)}\theta_-^{a_-}\theta_+^{a_+}\theta_3^{a_3}, \qquad z \in \mathbb{Z}^3, \quad a \in \{0,1\}^3,$$

where the reflections are at the lower walls of Q_0'. From (114) one obtains

$$gy = y + h''\kappa(g) + s(a), \qquad |s(a)|_\infty < 2h, \tag{115}$$

with

$$\kappa(g) = (-2a_- - 4a_+,\, 2a_- - 4a_+,\, -2a_3) + \frac{kh}{h''}(z_- + z_+,\, z_+ - z_-,\, 2z_3).$$

The eight first vectors are distinct, and their pairwise differences have size at most six. Since $kh/h'' \geq 16$, equality of two indices first forces equality of their a's and then of their z's. Thus $g \mapsto \kappa(g)$ is injective.

Choose half-integer coordinate offsets o_μ in

$$I_\mu = (y_\mu + r + 2h - h'',\; y_\mu - r - 2h).$$

These intervals have length at least $2h$. Equation (115) places every gB strictly inside coordinate cell $\kappa(g)$ of width h''. We can also choose the offsets so that all these coordinate walls avoid every reflected copy of $\operatorname{supp} O$. Under the original diagonal wall group, the centers of the copies of a box containing O have at most four residue classes modulo h in either of the first two coordinates, and at most two in the third. Its radius r_O therefore forbids at most $4(2r_O+1) = 8r_O + 4$ half-integer classes. Since $h > 8r_O + 8$, an allowed class exists. Each I_μ contains a representative, and $h'' \in h\mathbb{Z}$ keeps every wall in the allowed class.

For large i, the adapted torus D_i carries the original and coarse diagonal structures; its period lattice lies in the translation subgroup of the coarse wall group. The coordinate structure of width h'' cuts D_i and is carried by $T_i = C(m_i/2, 2m_i, m_i)$. Formula (115) also shows that coarse diagonal cells distinct modulo the periods have their boxes in coordinate cells distinct modulo those periods. Indeed a period translation changes κ by that translation divided by h''; injectivity then gives the converse. Thus the counts are

$$N_i' = \frac{m_i^3}{|Q|k^3}, \qquad N_i'' = \frac{m_i^3}{(h'')^3}. \tag{116}$$

The first is the number of coarse diagonal cells in D_i; the second is the number of coordinate cells in either torus. The shortest periods are at least $m_i/2 \geq kh \geq 16h''$, so the boxes, with the margins used for local lifts, embed. Folding one such box in the coordinate structure gives a disjoint placement with free columns, since every image stays inside its open cell.

Proposition 7.3. Put $\Delta_i = \Delta(m_i)$, with Δ defined in (74). It tends to zero as $i \to \infty$, with the charge, mass and weights fixed. If G satisfies the coordinate bound C in a box of radius r, then, for k as above and all sufficiently large i,

$$|\mu_i^t(G)| \leq C\gamma_{t,i}(k), \qquad \gamma_{t,i}(k) = \exp(2t|O|_1k^3 + |Q|k^3\Delta_i), \qquad t \geq 0. \tag{117}$$

The threshold and k depend only on the side sequence, h and r. In particular they are uniform in the location of G. ∎

Proof. Each coarse diagonal cell contains k^3 original cells. Put $P = \exp(t\sum_{Q_\kappa \subset Q_0'} O^{[\kappa]})$. The tilt e^{tV_i} is exactly the product of the reflected copies of P over the coarse cells: their walls are original walls, so each reflection carries the original array to the same array. Apply the adapted chessboard estimate to the integrands PG in one coarse cell and P in the others. The fold of P is e^{tV_i}. After dividing its powers out, this gives

$$|\mu_i^t(G)|^{N_i'} \leq \frac{Z_{D_i}[e^{tV_i}\prod_{a=1}^{N_i'} G^{g_a}]}{Z_{D_i}[e^{tV_i}]}. \tag{118}$$

The right side is nonnegative by the reflected fold form. This step uses reflection positivity of even integrands and does not require G itself to be a positive scalar function.

Expand e^{tV_i} in curvature characters. The coefficient sum is at most $\exp(tN_i|O|_1)$: the character algebra is closed under multiplication, and the exponential series is absolutely summable in its coefficient norm. Every plaquette in a character of this expansion lies strictly within a coordinate cell, because the chosen walls avoid the support of every copy of O. For each character term, collect its factors by coordinate cells. Each cell then contains a character alone or a character times one G^{g_a}; precisely N_i' cells have the latter form. Period translations can be used to place the integrands in a chosen coordinate brick without changing their value on D_i.

Use the chessboard estimate from the adapted to the coordinate frame, (71). A cell with a character alone has fold bounded by $Z_{T_i}[1]$. A cell with a copy of G has a fold consisting of a character and its coordinate images; the geometry above supplies the free columns and disjoint embedding, so its fold is bounded by $C^{N_i''}Z_{T_i}[1]$. Taking the chessboard roots and multiplying the cell contributions gives, for every character term,

$$\left| Z_{D_i}\left[\varphi_\beta \prod_{a=1}^{N_i'} G^{g_a}\right]\right| \le C^{N_i'} Z_{T_i}[1].$$

The coefficient sum therefore bounds the numerator of (118) by $e^{tN_i|O|_1}C^{N_i'}Z_{T_i}[1]$. The denominator bound (97) contributes a second $e^{tN_i|O|_1}$. Finally, the pressure comparison gives $Z_{T_i}[1]/Z_{D_i}[1] \le e^{m_i^3\Delta_i}$. The root of order N_i' gives (117). □

We also need exponential moments, for which a coordinate operator bound is unnecessary. Repeat (118) with $G = e^{\langle\beta,F\rangle}$ supported in B. It is an admissible local integrand, since $\langle\beta,F\rangle \le \|\beta\|^2 + \frac{1}{4}\sum_{\operatorname{supp}\beta} F_\sigma^2$. Its diagonal copies have disjoint supports. Remove the tilt from the numerator by its pointwise bound, and apply the second chessboard estimate to their coordinate cell integrands. The fold of one exponential has N_i'' disjoint copies of β, and hence squared norm $N_i''\|\beta\|^2$. Gaussian domination on T_i bounds its chessboard root by $e^{\|\beta\|^2/2}Z_{T_i}[1]^{1/N_i''}$. There are N_i' nonempty cells. The same denominator and pressure bounds therefore yield

$$\mu_i^t(e^{\langle\beta,F\rangle}) \le e^{\|\beta\|^2/2}\gamma_{t,i}(k). \tag{119}$$

For each fixed t this proves tightness on Ω. Indeed, Markov's inequality applied to $\beta = \pm s1_\sigma$ gives a uniform coordinate tail $2e^{-sR+s^2/2}\gamma_{t,i}(k(h,1))$. Choose radii R_σ with these tails summable over plaquettes; the intersection of Ω with their product of compact intervals is compact. The finitely many initial laws are tight individually. Any weak limit ν_t consequently satisfies

$$\nu_t(e^{\langle\beta,F\rangle}) \le e^{\|\beta\|^2/2+2t|O|_1k^3}. \tag{120}$$

This uses lower semicontinuity of the exponential integral, as explained after (99).

8. Recovering coordinate reflection symmetry

The local operator estimate first gives reflection symmetry for observables on the two layers adjacent to a coordinate link plane. To use this estimate for an arbitrary curvature character, we remove its outer fermion fields by exact integration identities and its outer photon exponents by a finite contour shift. Only finitely many terms of the reflected perturbation change in each shift; their contribution is of order its strength, uniformly in volume. The two-layer regularization in section 8.1, the exact field identities in section 8.2, and the contour shift in section 8.3 give the reduction in proposition 8.2.

8.1. **The two adjacent fermion layers.** The remaining comparison argument uses gauge invariant polynomials

$$P(\widehat{A}) = \sum_a b_a e^{i\langle \delta_a, \widehat{A}\rangle} \Psi_a, \qquad d^*\delta_a = q\mathfrak{q}(\Psi_a),$$

where the cochains have finite support and the Ψ_a are ordered Grassmann monomials. Denote their class by $\mathcal{P}$. The charge condition is exactly gauge invariance. It also makes each term even: the total divergence is zero, so its numbers of ψ and $\bar\psi$ factors agree. The expansion is unique after equal pairs (δ_a, Ψ_a) have been collected, because the monomials and the distinct characters are linearly independent. Each such polynomial is an admissible local integrand with bounded coefficients.

Write $v_0 = (v_1, v_2, v_3)$, with $0 < v_3 < v_2 < v_1 < 1/2$, and set

$$(s_1, \dots, s_6) = (-v_1, -v_2, -v_3, v_3, v_2, v_1), \qquad \mathcal{X}_{j,a} = \{x \in \mathcal{X} : x_1 = j + s_a\}.$$

These are the six thin layers of the thick layer centered at $x_1 = j$. At a coordinate link plane $c = j + \frac{1}{2}$, a two-layer polynomial reads photon links in the planes j and $j+1$ and between them, and fermion fields only at the two ends of the bonds crossing c. These are the thin layers $\mathcal{X}_{j,6}$ and $\mathcal{X}_{j+1,1}$; the other thin layers of the two neighboring thick layers do not occur. Call this subclass $\mathcal{P}_1(c)$.

Two estimates connect this class to coordinate reflection symmetry. First, for a monomial $Y \in \mathcal{P}_1(c)$, heat regularization constructs a bounded operator $X_{Y,\epsilon}$ and an error $\xi_Y(\epsilon) \to 0$ such that

$$|\mu_i^t(\Theta_c Y) - \overline{\mu_i^t(Y)}| \le 4\sqrt{\gamma_{t,i}(k)^2 - 1}\, \|X_{Y,\epsilon}\| + C_Y \gamma_{t,i}(k) \xi_Y(\epsilon).$$

The fixed support scale k is independent of ϵ. The operator norm can grow exponentially in $1/\epsilon$; thus the volume limit must precede $t \downarrow 0$, and only then may $\epsilon \downarrow 0$. The construction and the monomial form of this estimate appear in (132); finite sums give the symmetry assertion for every two-layer polynomial. Second, proposition 8.2 assigns to every $P \in \mathcal{P}$ a $Y_P \in \mathcal{P}_1(c)$, independent of the torus and tilt, with

$$|\mu_i^t(P^g) - \mu_i^t((Y_P)^g)| \le C_{P,g} t \qquad (0 \le t \le 1,\ i \ge i_0(P,g))$$

for each fixed lattice isometry g. Its proof removes the outer fermion layers and then shifts the photon contour. This estimate extends the two-layer symmetry to the curvature characters used to identify the limiting law.

For a monomial $Y = e^{i\delta\widehat{A}}\Psi$, write its exponent as $(\delta_0, \delta_1, \delta_\perp)$ on the lower spatial, upper spatial and temporal links. If $\mathfrak{q}_\pm$ are the lower and upper charge densities, gauge invariance gives

$$d_s^*\delta_0 - \delta_\perp = q\mathfrak{q}_-, \qquad d_s^*\delta_1 + \delta_\perp = q\mathfrak{q}_+. \tag{121}$$

Let $S = \operatorname{supp}\delta_0 \cup \operatorname{supp}\delta_1$ and choose a nonempty Λ containing their endpoints, the support of $\delta_\perp$ and all sites over which the fields of Ψ occur.

Every such Y is a kernel insertion, but the operator between the usual quarter-heat factors need not be bounded. We give the different regularization that permits approximation by bounded operators. The crossing modes and their reflected modes form a basis of the two-layer generators. Since all their weights are positive when $0 < \rho < 1$, there is a unique finite matrix Υ_Y with

$$\Psi e^{-S_c^\times(\Lambda)}\Big|_{\widehat{a}_j = 0} = \operatorname{Ins}_c[\omega^{1/2}\Upsilon_Y\omega^{1/2}]. \tag{122}$$

To obtain it, expand the left side in the basis $\Theta_c\zeta_J\zeta_I$ and multiply the resulting matrix by $\omega^{-1/2}$ on both sides. Its nonzero entries have $\mathfrak{q}(I) - \mathfrak{q}(J) = \mathfrak{q}_- + \mathfrak{q}_+$. Thus Υ_Y has charge $q^{-1}d_s^*(\delta_0 + \delta_1)$ and is even.

Set

$$k_Y^0(A', A) = e^{i\delta_1 A'} \prod_{b \in S} g_1(A'(b) - A(b)) e^{i\delta_0 A} \Upsilon_Y. \tag{123}$$

It is the continuous kernel of $e^{i(\delta_0+\delta_1)A}e^{-|E_S+\delta_0|^2/2}\Upsilon_Y$. We also write k^0_Y for this integral operator. The charge identity and unchanged Gaussian difference prove its covariance. Moreover

$$Y = G[k^0_Y]. \tag{124}$$

Indeed, the Gaussian factors in the insertion cancel. Applying the automorphism that multiplies the lower fields by their charge phases to (122) replaces its matrix by $\omega^{1/2}\Upsilon_Y u(\widehat{a}_j)\omega^{1/2}$ and its left side by $e^{-iq\langle \mathfrak{q}_-,\widehat{a}_j\rangle}\Psi e^{-S_c^\times(\Lambda)}$. The remaining temporal exponent in $G[k^0_Y]$ is $d_s^*\delta_0 - q\mathfrak{q}_- = \delta_\perp$ by (121), proving the identity.

Put $\widetilde{E} = E_S + \delta_0$, $D = \delta_1 - \delta_0$ and $a = (|\delta_0|^2 + |\delta_1|^2)/8$. Define the bounded operator

$$X'_Y = e^{i(\delta_0+\delta_1)A}e^{-\frac{1}{4}|\widetilde{E}|^2+\frac{1}{4}\langle \widetilde{E},D\rangle + a}\Upsilon_Y. \tag{125}$$

It belongs to $\mathfrak{W}(S,\Lambda)$, and completion of the quadratic exponent gives

$$e^{-|E_S|^2/8}X'_Y e^{-|E_S|^2/8} = k^0_Y, \qquad \|X'_Y\| = \|\Upsilon_Y\|e^{|D|^2/16+a}.$$

For the first identity move the left heat factor through the phase using the Weyl rule. Its resulting squares are $\frac{1}{8}|\widetilde{E}+\delta_1|^2 + \frac{1}{8}|\widetilde{E}-\delta_0|^2$; their linear and constant terms cancel those in (125). Therefore $Y = G'_{X'_Y}$ and it satisfies the coordinate bound $2^{|S|/2}\|X'_Y\|$. This bound will also control the approximation error uniformly in the torus.

For $\epsilon > 0$ let $P_\epsilon = e^{-\epsilon|E_S|^2}$ and

$$X_{Y,\epsilon} = e^{i(\delta_0+\delta_1)A}\exp\left(-\epsilon|\widetilde{E}+\delta_1|^2 - \epsilon|\widetilde{E}-\delta_0|^2 + \tfrac{1}{2}\langle \widetilde{E},D\rangle + 2a\right)\Upsilon_Y. \tag{126}$$

The quadratic part is strictly negative, so this too is in $\mathfrak{W}(S,\Lambda)$. The same Weyl calculation gives

$$\begin{aligned} e^{-|E_S|^2/4}X_{Y,\epsilon}e^{-|E_S|^2/4} &= P_\epsilon k^0_Y P_\epsilon, \\ Y - G_{X_{Y,\epsilon}} &= G'_{X'_Y - P_\epsilon X'_Y P_\epsilon}. \end{aligned} \tag{127}$$

The second equality follows from the first and the primed representation, since their continuous kernels agree. Its coordinate bound is $2^{|S|/2}\xi_Y(\epsilon)$, where

$$\begin{aligned} \xi_Y(\epsilon) = \|\Upsilon_Y\| \sup_{x\in\mathbb{R}^S} & e^{-|x|^2/4+\langle x,D\rangle/4+a} \\ & \cdot (1 - e^{-\epsilon|x+\delta_1|^2-\epsilon|x-\delta_0|^2}) \le C_Y\epsilon. \end{aligned} \tag{128}$$

The last inequality uses $1-e^{-u} \le u$ and the boundedness of a Gaussian times a quadratic polynomial. By contrast,

$$\|X_{Y,\epsilon}\| = \|\Upsilon_Y\|\exp\left(\frac{(1/2-2\epsilon)^2|D|^2}{8\epsilon} + (1/4-\epsilon)(|\delta_0|^2+|\delta_1|^2)\right). \tag{129}$$

It can grow exponentially in $1/\epsilon$. This forces the order of the limits in the next result.

Reflection changes $(\delta_0,\delta_1,\delta_\perp)$ to $(-\delta_1,-\delta_0,\delta_\perp)$ and Υ_Y to Υ_Y^*. The latter follows by reflecting (122); $\mathrm{Ins}_c[\omega]$ is even and invariant. Taking adjoints in the two heat identities then gives $X'_{\Theta_c Y} = (X'_Y)^*$ and $X_{\Theta_c Y,\epsilon} = X^*_{Y,\epsilon}$; the heat operator is injective with dense range, so these identities determine the bounded middle operators uniquely. All constructions extend linearly to two-layer polynomials. Coordinate permutations give the same assertions at every coordinate link plane.

Proposition 8.1. For every $Y \in \mathcal{P}_1(c)$ at a coordinate link plane,

$$\lim_{t\downarrow 0}\limsup_{i\to\infty}|\mu_i^t(\Theta_c Y) - \overline{\mu_i^t(Y)}| = 0. \tag{130}$$

The assertion concerns the finite-volume functionals on fermion polynomials; it does not presume that they converge to fermion laws. ∎

Proof. For a fixed support, $o_i^t(X) = \mu_i^t(G_X)$ is a unital linear functional on the local operator algebra with $|o_i^t(X)| \le \gamma\|X\|$, where $\gamma = \gamma_{t,i}(k)$. A unital functional this close in norm to one is nearly Hermitian. If $X_0 = X_0^*$ and $\|X_0\| \le 1$, write $o_i^t(X_0) = u + iv$. For each real s,

$$(v+s)^2 \le |o_i^t(X_0+is)|^2 \le \gamma^2\|X_0+is\|^2 \le \gamma^2(1+s^2).$$

For $\gamma > 1$ take $s = v/(\gamma^2-1)$; this gives $|v| \le \sqrt{\gamma^2-1}$. For $\gamma = 1$ the same conclusion follows by letting s tend to either infinity. Decomposing a general X into its real and imaginary self-adjoint parts yields

$$|o_i^t(X^*) - \overline{o_i^t(X)}| \le 4\sqrt{\gamma^2-1}\|X\|. \tag{131}$$

Use (109) to interpret this as reflection covariance of the insertion expectation.

Apply (131) to $X_{Y,\epsilon}$ and use (127) for Y and its reflection. The amplification theorem controls the two errors and gives

$$\begin{aligned}|\mu_i^t(\Theta_c Y) - \overline{\mu_i^t(Y)}| \le 4\sqrt{\gamma^2-1}\|X_{Y,\epsilon}\| \\ + 2\cdot 2^{|S|/2}\gamma\xi_Y(\epsilon).\end{aligned} \tag{132}$$

The support box and its coarse scales do not depend on ϵ. First take $i \to \infty$, and then $t \downarrow 0$ with ϵ fixed. Since $\Delta_i \to 0$, the first term vanishes and the remaining upper bound is $2\cdot 2^{|S|/2}\xi_Y(\epsilon)$. Let $\epsilon \downarrow 0$ using (128). Finite sums and coordinate transport give the stated result. □

8.2. **Removing fields from a thin layer.** Consider the six thin layers $\mathcal{X}_{j,a}$ defined above. Each site has one neighbor in the next thin layer, one in the preceding one and two in its own layer. The forward matching bonds have types $2,1,0,1,2,3$ in succession. This follows directly from the chamber reflections: type zero reverses the coordinate of modulus v_3, types one and two interchange the adjacent moduli, and type three reverses the coordinate of modulus v_1 and shifts its nearest photon site by the corresponding unit vector. The bonds of types zero, one and two carry no photon phase. Type three bonds within thin layers $2,\dots,5$ carry spatial phases in plane j, while those between layer six and the next layer one carry temporal phases between j and $j+1$. These facts control which planes the recursion can read.

For a photon site y, let $\mathcal{P}^{\pm}(y)$ consist of the same exponential monomials as $\mathcal{P}$, with $d^*\delta = q(\mathfrak{q}(\Psi) \mp 1_y)$. They transform with gauge phase $e^{\pm iq\lambda(y)}$ and are odd. Removing a $\bar\psi(x)$ from an invariant monomial leaves an element of $\mathcal{P}^+(\pi(x))$; removing a $\psi(x)$ leaves one in $\mathcal{P}^-(\pi(x))$. Multiplication by a hopping phase transports the specified charge to the other endpoint. These assertions are the identities

$$d^*1_{(y,y')} = 1_{y'} - 1_y, \qquad U_A(x,x') = e^{iq\langle 1_{(\pi(x),\pi(x'))},A\rangle}.$$

They show that every term of the following replacements stays in $\mathcal{P}$.

Berezin integration by parts, with left derivatives, gives

$$\begin{aligned}Z_T[\Phi(\bar\psi\mathcal{D}_A)_\alpha(x')G] &= -Z_T[\Phi\partial_{\psi_\alpha(x')}G],\\ Z_T[\Phi(\mathcal{D}_A\psi)_\alpha(x')G] &= Z_T[\Phi\partial_{\bar\psi_\alpha(x')}G].\end{aligned} \tag{133}$$

The appropriate charge of G makes each integrand invariant. These identities first hold at fixed complex A: the derivatives of $e^{-S_T(A)}$ are respectively $(\bar\psi\mathcal{D}_A)_\alpha e^{-S_T(A)}$ and $-(\mathcal{D}_A\psi)_\alpha e^{-S_T(A)}$, and the Berezin integral annihilates a derivative. Integrating against any bounded bosonic factor $\Phi(F)$ then proves (133). The enlarged local box must embed so that all generators involved are distinct torus generators and the phases can be read on the local lift.

Let x' be the neighbor of x in the next thin layer. Its matching hopping matrix $w(x,x')$ has eigenvalues $\frac{w_i}{2}(\rho \mp 1)$ and is invertible for our fixed $0 < \rho < 1$. Solve the equation at x' for the field

at x. With the replaced field moved to the left of the monomial, the two rules are

$$\begin{aligned}\bar\psi(x) &\longmapsto \left(M\bar\psi(x') - \sum_{x''\sim x',\, x''\neq x} \bar\psi(x'') w(x'',x') U_A(x'',x') + \partial_{\psi(x')} \right) w(x,x')^{-1} U_A(x',x),\\ \psi(x) &\longmapsto U_A(x,x') w(x',x)^{-1} \left(M\psi(x') - \sum_{x''\sim x',\, x''\neq x} w(x',x'') U_A(x',x'') \psi(x'') - \partial_{\bar\psi(x')} \right).\end{aligned} \tag{134}$$

The derivatives act on the monomial to their right. Multiplying by the inverse matching matrix and its inverse phase in (133) proves equality of the expectations before and after either replacement, for every bosonic factor Φ. This equality is independent of a choice of time direction or a symmetry of the torus.

For a thin layer $\mathbb{L}$ define a linear map $R^+_{\mathbb{L}}$ by recursion on the number of its fields in an ordered monomial. Select the first such field, move it to the front with its Grassmann sign, and apply (134) through its matching neighbor in $\mathbb{L}^+$. That neighbor has only one neighbor in $\mathbb{L}$, the field just removed. All other fields created by the rule lie in $\mathbb{L}^+$ or $\mathbb{L}^{++}$; its derivatives remove fields in $\mathbb{L}^+$. Thus the number in $\mathbb{L}$ falls by exactly one at every recursive call. The map is well-defined, preserves $\mathcal{P}$, removes all fields in $\mathbb{L}$, and creates no new fields outside its next two layers. A downward map $R^-_{\mathbb{L}}$ is obtained using the preceding matching neighbor.

If the initial polynomial is read in a box B, the entire one-layer recursion is read in $B + [-1,1]^3$. The fields being removed stay in the original box, and their next neighbors and the other neighbors of those sites have photon projections equal or nearest to the original projection: among two successive different bonds at most one has type three. Hence these sites and every added phase are in the same enlarged box, irrespective of the number of recursive calls. Induction therefore gives

$$Z_T[\Phi P] = Z_T[\Phi R^\pm_{\mathbb{L}} P] \tag{135}$$

whenever this enlarged box embeds. The new photon links are only those of type three bonds with an endpoint in $\mathbb{L}^\pm$. The recursion commutes with multiplication by a curvature character, since it differentiates only fields and its choices depend only on the monomial.

We first use this recursion to bound an arbitrary polynomial; here its photon links need not yet be reduced. Move its fields upwards through all thin layers below $\mathcal{X}_{j,6}$ and downwards through those above $\mathcal{X}_{j+1,1}$. The last downward replacement can reach $\mathcal{X}_{j,6}$, and cannot reach a lower layer. We obtain, on every sufficiently large torus and with every bounded bosonic factor, an equality

$$Z_T[\Phi P] = \sum_a b_a Z_T[\Phi \varphi_{\beta_a} Y_a], \qquad Y_a \in \mathcal{P}_1(c). \tag{136}$$

Here the Y_a can be chosen with integer exponents qj_a. Indeed each remaining monomial has integer charge of total zero on the two adjacent photon planes. In their connected rectangular support graph choose paths from a fixed vertex to each charged vertex; the charge-weighted sum of these paths is an integer cochain j_a with the required divergence. Then $u_a = \delta_a - qj_a$ is divergence free in the same integer box and can be written $d^*\beta_a$ there.

The filling must stay inside the support box for the polynomial bound below. In $\prod_\mu [a_\mu, a'_\mu]$, set electric components $\beta^{\mathrm{el}}_j = -\sum_{l=a_1}^{j} (u_a)_l$ for $a_1 \le j < a'_1$. Subtracting $d^*\beta$ removes all spatial links below the top plane; divergence zero then removes the temporal links. On that top plane a second partial-sum sweep in the second coordinate removes the third-direction links below its top edge. Divergence zero removes the second-direction links and leaves only a terminal segment in direction three. Its divergence equation and vanishing boundary value make it zero. The two sweeps use only plaquettes of the box. This proves the asserted filling and thus the splitting $\delta_a = d^*\beta_a + qj_a$ in (136).

Apply the coordinate bound to each Y_a, absorb the extra character and use amplification. For any fixed box B_0 there are constants C_P, k_P and a volume threshold with

$$\sup_{\operatorname{supp}\beta\subset B_0} |\mu_i^t(\varphi_\beta P)| \le C_P \gamma_{t,i}(k_P), \qquad t \ge 0. \tag{137}$$

One may take $C_P = \sum_a |b_a| 2^{|S_a|/2} \|X'_{Y_a}\|$, where the factor $2^{|S_a|/2}\|X'_{Y_a}\|$ is one for $Y_a = 1$. Such a term therefore contributes $|b_a|$. All support boxes in the finite normal form and B_0 fit in one fixed box; its radius determines k_P. The size of β does not enter the bound. Reflected polynomials have the same kind of bound, since the normal form identity can be reflected on the reflected torus and the coordinate bounds are preserved by reflection.

8.3. **The contour shift and an outer plane.** A fermion recursion removes fields from a plane but leaves its photon exponent. An imaginary shift of links in the next plane removes that exponent together with the outgoing temporal one. The all-flux measure is essential for this shift: summation over exact integer fluxes unfolds the compact link intervals to real lines on which the contour can move.

Let u be a real cochain supported on finitely many links S of one coordinate plane, and let $P \in \mathcal{P}$. Choose a box B reading P and containing $b + [-1,1]^3$ for every $b \in S$. Put $P^{(u)}(\widehat{A}) = P(\widehat{A} - iu)$ by its entire exponential expansion. Define

$$\begin{aligned} Q_u = \sum_{b\in S} \sum_{(x,x') \text{ over } b} & \big[(e^{qu_b}-1)\bar\psi(x) w(x,x') U_A(x,x')\psi(x') \\ & + (e^{-qu_b}-1)\bar\psi(x') w(x',x) U_A(x',x)\psi(x)\big], \qquad V_u = e^{Q_u}. \end{aligned} \tag{138}$$

The bonds are oriented with b. Each term is invariant and even, and the exponential is a finite polynomial by nilpotence. It reads only those links of S with $u_b \neq 0$ and their overlying bonds. For a plane normal to e_1, those fields are in thin layers $2,\dots,5$ of its thick layer. The change in the action is $S_T(A - iu) - S_T(A) = -Q_u$, so $e^{-S_T(A-iu)} = e^{-S_T(A)} V_u$.

If $\Phi(F)$ is entire and bounded on each tube of bounded imaginary curvature, then on every torus in which B embeds we have

$$Z_T[\Phi P] = e^{\|du\|^2/2} Z_T[\Phi(F - idu)\varphi_{du}(F) V_u P^{(u)}]. \tag{139}$$

The norm here is the lattice norm, equal to the torus norm because its support embeds. We supply the contour justification. For each b choose its electric plaquette leaving the plane in the normal direction. Its only side in S is b, with coefficient ± 1. Consequently d is injective on cochains supported in S, and the action $z \mapsto z + dk$, $k \in \mathbb{Z}^S$, on closed integer fluxes is free. Choose orbit representatives z_0. The compact A_S intervals and the sum over k become the integral over $\mathbb{R}^S$, since $(A, z_0 + dk)$ represents the same configuration as $(A + \frac{2\pi}{q}k, z_0)$.

For fixed z_0 and the other real links, choose an integer one-cochain k' with $dk' = z_0$ in the local box. Then $A + \frac{2\pi}{q}k'$ is a local lift affine in the complex variables $\zeta = A_S$. The Berezin coefficients of $e^{-S_T(A)}$ are polynomials in the link phases, so the full integrand extends to an entire function $f(\zeta)$. On the strip $|\Im\zeta_b| \le \|u\|_\infty$ the selected plaquettes give

$$|f(\zeta)| \le C \prod_{b\in S} e^{-(\Re\zeta_b - a_b)^2/2}. \tag{140}$$

Indeed each of their real curvatures is $\pm\Re\zeta_b + a'_b$, all imaginary curvatures are bounded in this finite strip, the factor Φ is bounded there, and each remaining exponential phase is bounded there. The constant can be chosen independent of the remaining real links and the orbit representative; only the real Gaussian centers vary.

Move one variable at a time along the rectangle with real endpoints $\pm R$ and imaginary displacement $-iu_b$. The vertical sides tend to zero by (140). Fubini then gives $\int f(\xi)\,\mathrm{d}\xi = \int f(\xi - iu)\,\mathrm{d}\xi$. At the

shifted arguments the Maxwell exponent, using the complex-bilinear cochain pairing, becomes

$$-\tfrac{1}{2}\langle F-idu, F-idu\rangle = -\tfrac{1}{2}\|F\|^2 + i\langle du, F\rangle + \tfrac{1}{2}\|du\|^2.$$

The change in the fermion action is the polynomial V_u, and the observable becomes $P^{(u)}$. Both the initial integrand and the resulting real integrand are absolutely integrable by their local bounds. Hence we may sum the identities over z_0 and integrate the remaining links, which proves (139).

We now eliminate the lowest plane j read by P. This means that its fields have photon projection height at least j and its links have no endpoint below plane j. Apply the six upward thin-layer maps of this thick layer to obtain $P^{\uparrow} = \sum_a b_a e^{i\delta'_a \widehat{A}}\Psi_a$. It reads no fields of thick layer j and no links below it; the only new links are spatial links in plane j and temporal links from j to $j+1$. At each term, gauge invariance in plane j therefore says

$$(\delta'_a)^{\perp}_j = d^*_s(\delta'_a)_j. \tag{141}$$

For this term let u_a be supported on plane $j+1$ with $(u_a)_{j+1} = (\delta'_a)_j$. Define

$$P_a = b_a e^{\|du_a\|^2/2 + \langle \delta'_a, u_a\rangle} e^{i\langle \delta'_a + d^* du_a, \widehat{A}\rangle} V_{u_a}\Psi_a. \tag{142}$$

Then (135) and (139) give the exact relation

$$Z_T[\Phi P] = \sum_a Z_T[\Phi(F - idu_a)P_a]. \tag{143}$$

The construction is on the infinite lattice; the same relation holds for every Φ entire in the plaquette variables and bounded on each tube $|\Im F_\sigma| \le r$ for all σ, $r > 0$, and every torus containing the fixed enlarged support box.

Here is the cancellation of the eliminated exponent. Write $v = (\delta'_a)_j$. From the slice formulas $(du)^{\mathrm{el}}_l = u_{l+1} - u_l - d_s u^{\perp}_l$ and $(d^*\beta)_l = \beta^{\mathrm{el}}_{l-1} - \beta^{\mathrm{el}}_l + d^*_s\beta^{\mathrm{mag}}_l$, the only nonzero components of $d^* du_a$ are

$$\begin{aligned} (d^*du_a)_j &= -v, & (d^*du_a)^{\perp}_j &= -d^*_s v,\\ (d^*du_a)_{j+1} &= (2 + d^*_s d_s)v, & (d^*du_a)^{\perp}_{j+1} &= d^*_s v,\\ (d^*du_a)_{j+2} &= -v. \end{aligned} \tag{144}$$

Thus the spatial exponent in plane j vanishes, and (141) cancels its outgoing temporal exponent as well. The polynomial V_{u_a} reads only thick layer $j+1$. Hence P_a has lowest plane at least $j+1$ and highest plane at most the maximum of $j+2$ and the previous highest plane. The six field recursions enlarge the box by six, and the shifted links with their plaquettes require two more. The whole step is read in $B + [-8,8]^3$. Elimination from the highest plane is obtained by the downward maps and a shift in plane $j-1$. The corresponding incoming temporal exponent is $-d^*_s(\delta'_a)_j$, and the same two components cancel. The highest plane falls by one, while the lowest can fall only to $j-2$.

The source of error in the tilted law is precisely the shifted bosonic factor. For a finite cochain η, only finitely many copies in the array V_i meet its support. Their change is the fixed trigonometric polynomial

$$H_\eta(F) = \sum_{\kappa,b} a_b (e^{\langle (g_\kappa^{-1})^*\beta_b, \eta\rangle} - 1)\varphi_{(g_\kappa^{-1})^*\beta_b}(F), \tag{145}$$

where $O = \sum_b a_b \varphi_{\beta_b}$ and only contributing cells occur. At most $|\operatorname{supp}\eta|$ cells contribute; each lies in the support box of η enlarged by h. Its coefficient norm has the bound

$$|H_\eta|_1 \le C(\eta) := |\operatorname{supp}\eta|\,|O|_1 (e^{\max_b \|\beta_b\|_1 \|\eta\|_\infty} - 1). \tag{146}$$

If that enlarged box embeds, no two contributing cells are period equivalent and the same polynomial is the torus difference $V_i(F - i\eta) - V_i(F)$. Expand its exponential in characters:

$$e^{tV_i(F-i\eta)} = e^{tV_i(F)} \left(1 + \sum_\alpha b_\alpha(t)\varphi_{\beta_\alpha}(F)\right), \qquad \sum_\alpha |b_\alpha(t)| \le e^{tC(\eta)} - 1. \tag{147}$$

The coefficient norm is submultiplicative, so the exponential series proves absolute uniform convergence. All character supports lie in one fixed enlarged box, independent of i and t; sums of character exponents do not enlarge this support.

Set $P^\flat = \sum_a P_a$ in (143). Insert (147) with $\eta = du_a$ and divide by the positive denominator. Termwise integration is valid: at a fixed torus the functional on bounded bosonic factors multiplying P_a is continuous for their supremum norm, by the absolute integrability of its Berezin coefficients. We get

$$|\mu_i^t(P) - \mu_i^t(P^\flat)| \le (e^{Ct} - 1) \sum_a \sup_{\operatorname{supp}\beta \subset B'} |\mu_i^t(\varphi_\beta P_a)|, \tag{148}$$

where $C = \max_a C(du_a)$ and B' is a fixed box enlarged by h. The a priori bound (137) makes this $O(t)$ uniformly for all large i and $0 \le t \le 1$: each of the finitely many amplification factors is bounded once $\Delta_i \le 1$, and $e^{Ct} - 1 \le (e^C - 1)t$ on this interval.

The same estimate applies after transporting the identity. This needs care because D_i need not be invariant under a coordinate permutation. Reflect the equality on the reflected torus and then reflect it back, using functional covariance. The conjugations change a shift η to $-\theta^*\eta$. Indeed, if $\Phi^\theta(F) = \overline{\Phi(\theta^*\overline{F})}$, then $(\Phi^\theta(\,\cdot - i\eta))^\theta = \Phi(\,\cdot + i\theta^*\eta)$. Thus (143) becomes an exact identity for ΘP with shifts $-\theta^* du_a$ and polynomials ΘP_a on the original torus. Iterating reflections transports it by any g, with shifts $\det(\vec{g})(g^{-1})^* du_a$, where $\vec{g}$ is the linear part of g. Its support box is transported at the same time. Apply (147) and (137) on D_i itself to obtain the transported $O(t)$ estimate. No invariance of D_i or of its tilted array under g is used.

Proposition 8.2. For each $P \in \mathcal{P}$ and coordinate link plane c there is $Y_P \in \mathcal{P}_1(c)$, independent of the torus and tilt, such that, for every fixed lattice isometry g,

$$|\mu_i^t(P^g) - \mu_i^t((Y_P)^g)| \le C_{P,g}\, t \quad (0 \le t \le 1, i \ge i_0(P,g)). \tag{149}$$

At a plane normal to either other coordinate the corresponding reduction and its reflected estimate hold as well. ■

Proof. For $c = j_0 + \frac{1}{2}$, eliminate lowest planes until the lowest is at least j_0. Every step raises it by at least one, so this takes finitely many steps. Then eliminate highest planes until the highest is at most $j_0 + 1$. During this phase an eliminated plane $j \ge j_0 + 2$ can create links only as low as $j - 2 \ge j_0$, preserving the first reduction. The resulting polynomial reads just the two thick layers and their connecting links. Apply the upward recursions in thin layers $1, \dots, 5$ of the lower thick layer, followed by the downward recursions in thin layers $6, \dots, 2$ of the upper one. New fields remain at or above the lower layer six and at or below the upper layer one. The new type three phases are spatial phases in these two planes or the temporal phases between them. Thus the result Y_P is in $\mathcal{P}_1(c)$.

There are finitely many outer-plane steps, each with (148) and its transported version; the final ten thin-layer recursions are exact with the unchanged bosonic factor and contribute no error. Take one volume threshold for all their fixed support boxes and a priori bounds, and sum the finite $O(t)$ errors. This proves (149). For another direction first apply the construction to $P^{g_\mu^{-1}}$ and then transport its exact identities by the cyclic coordinate permutation g_μ. The general transported estimate, evaluated on D_i, proves the claim there and its reflected version. □

9. Comparison states

9.1. **Comparison states and translation invariance.** A comparison state for the chosen cell structure and O is a law ν_0 obtained by the ordered limits

$$\mu_{i_m}^{t_k} \longrightarrow \nu_{t_k} \quad (m \to \infty) \text{ for each } k, \qquad \nu_{t_k} \longrightarrow \nu_0 \quad (k \to \infty), \qquad t_k > 0, \quad t_k \to 0. \tag{150}$$

All convergences here are weak convergences of bosonic laws on Ω, and the volume subsequence is common to all strengths. Given a prescribed subsequence of volumes, a further subsequence produces such a state. To see this, take strengths $t_k = 1/k$, extract nested convergent volume subsequences by the tightness proved above, and choose their diagonal subsequence. The limiting laws ν_{t_k} are themselves uniformly tight: applying (120) to $\beta = \pm 1_\sigma$ and using $t_k \le 1$ bounds their exponential coordinate moments by a common constant. The same product-of-intervals argument gives a weakly convergent strength subsequence. This proves existence of (150) subordinate to any prescribed volume subsequence.

Every comparison state has Gaussian domination

$$\nu_0(e^{\langle \beta, F\rangle}) \le e^{\|\beta\|^2/2}, \qquad \beta \text{ of finite support.} \tag{151}$$

This follows from (120) by letting $t_k \to 0$ and truncating the exponential in the second weak limit. Wall invariance and wall reflection positivity also pass through both limits. For each local positive-side test, its support is contained in the positive half-period slab once i_m is large, so (99) applies. The continuity and L^2 extension described in section 6.2 give positivity for bounded Borel local tests. In particular ν_0 is invariant and reflection positive at $(x_1 - x_2, c_0)$.

The comparison with a prescribed cubic limit will be proved in proposition 10.6, using nearby divisible sides.

We now obtain the extra invariance required to use proposition 10.6 for observables at arbitrary positions. Fix a coordinate link reflection θ_c and a curvature character $P = \varphi_\beta$. Its polynomial representation is $e^{i\langle d^*\beta, \widehat{A}\rangle}$; on every sufficiently large torus this equals $e^{i\langle \beta, \widetilde{F}\rangle}$ on the local lift. Apply Proposition 8.2 to P and its reflection, and then Proposition 8.1 to its two-layer reduction. The finite errors are at most two $O(t)$ terms plus a finite sum of the terms in (132). Along the common volume subsequence in (150), take volume to infinity, then $t_k \to 0$ with the regularization fixed, and finally remove the regularization. Only the character expectations are passed to weak limits. The conclusion is

$$\nu_0(\Theta_c \varphi_\beta) = \overline{\nu_0(\varphi_\beta)}. \tag{152}$$

There is no need to extract limiting fermion expectations in this argument.

For a bosonic character $\Theta_c \varphi_\beta = \varphi_{-\theta_c^*\beta}$, while positivity of the probability law gives $\overline{\nu_0(\varphi_\beta)} = \nu_0(\varphi_{-\beta})$. Thus (152), with $-\beta$ in place of β, is invariance under the tensor pullback $F \mapsto \theta_c^* F$. Characters determine all finite plaquette marginals by uniqueness of their Fourier transforms [Pöt, Theorem 6.2]; these marginals determine the law because cylinder sets generate the Borel sigma algebra of Ω. Consequently every comparison state is invariant under all coordinate link reflections, in all three directions. The product of reflections at c and $c + 1$ is translation by $2e_\mu$, so

$$(\tau_v^*)_* \nu_0 = \nu_0, \qquad v \in 2\mathbb{Z}^3. \tag{153}$$

We need only these even translations. The original wall periods alone would not have permitted the next placement.

10. Comparison for arbitrary cubic sides

The comparison states constructed above require increasingly divisible periods. The original cubic sequence need not have this property. We compare its side n with a divisible side $m \le n$, using an empty slab to absorb the difference in each of the three folding directions. The empty slabs contribute partition functions without insertions. Their logarithms are controlled by a common pressure bound,

and their total contribution is measured by their volumes multiplied by the corresponding Hölder exponents. Propositions 10.4 and 10.6 establish the comparison for the original cubic laws used in theorem A.

Throughout this section the parameters retain their fixed values $q > 0$, $0 < \rho < 1$, $M \in \mathbb{R}$ and $w_i > 0$. Write $D_m = D(m/2, m, m)$ when m is even. The original torus is always $C(n)$; all tilted laws and comparison states in this section are formed on the auxiliary tori D_m.

10.1. **Unequal slabs.** Let $\ell(x) = n_\ell \cdot x$ be a mirror family and let $\Xi = (\Gamma_\parallel, \chi_\parallel)$ be a cross-section. Put

$$|\Xi| = [\mathbb{Z}^3 \cap \{\ell = 0\} : \Gamma_\parallel].$$

If $P > 0$ belongs to $|n_\ell|^2\mathbb{Z}$, set

$$u_P = \frac{P}{|n_\ell|^2} n_\ell, \qquad \mathbb{T}(\Xi; P) = (\Gamma_\parallel \oplus \mathbb{Z}u_P, \chi_P), \qquad \chi_P|_{\Gamma_\parallel} = \chi_\parallel, \quad \chi_P(u_P) = -1.$$

We use this notation only when the displayed period lattice has systole at least four. Then u_P is an integer vector, the torus is adapted to ℓ, and its volume is $P|\Xi|$. More generally, a torus with this cross-section and height period p has volume $p|\Xi|$. Indeed, $\ell(\mathbb{Z}^3) = \mathbb{Z}$, and the quotient is counted by its p height classes and the $|\Xi|$ classes in each height. The character does not affect this count.

Proposition 10.1 (Unequal slab folds)**.** Let $\mathbb{T} = (\Gamma, \chi)$ have cross-section Ξ in the family ℓ, and let $g \in \Gamma$ be a closing vector of minimal positive height $p = \ell(g)$. Choose an integer $k \geq 1$ and admissible positions

$$c_0 < c_1 < \cdots < c_{2k} = c_0 + p, \qquad h_j = c_j - c_{j-1}.$$

Assume

$$\min_{\gamma \in \Gamma_\parallel + \mathbb{Z}(2kh_j n_\ell/|n_\ell|^2),\ \gamma \neq 0} |\gamma|_\infty \geq 4, \qquad 1 \leq j \leq 2k. \tag{154}$$

Let G_j be a bounded scalar function of finitely many curvature plaquettes contained in the open slab $c_{j-1} < \ell < c_j$. For each j put $\mathbb{T}_j = \mathbb{T}(\Xi; 2kh_j)$ and form $\operatorname{fold}_\ell G_j$ by successive reflections in the planes $\ell = c_{j-1} + ih_j$. Then

$$Z_{\mathbb{T}_j}[\operatorname{fold}_\ell G_j] = \|\mathfrak{S}[G_j]\|_{2k}^{2k} \geq 0, \tag{155}$$

$$\left| Z_{\mathbb{T}} \left[\prod_{j=1}^{2k} G_j \right] \right| \leq \prod_{j=1}^{2k} Z_{\mathbb{T}_j}[\operatorname{fold}_\ell G_j]^{1/(2k)}, \tag{156}$$

$$|\mathbb{T}| = \sum_{j=1}^{2k} \frac{|\mathbb{T}_j|}{2k}. \tag{157}$$

In particular, the original torus need not admit a straightened torus of height p. ∎

Proof. Admissible positions form the group $\frac{1}{2}|n_\ell|^2\mathbb{Z}$. Thus h_j belongs to this group, $2kh_j \in |n_\ell|^2\mathbb{Z}$, and every plane used in the jth fold is admissible. Condition (154) therefore defines each $\mathbb{T}_j$. Its volume is $2kh_j|\Xi|$, so summing h_j proves (157).

The boundary spaces and the slab operator $\mathfrak{S}[G_j]$ depend on the cross-section and the two endpoints, not on the torus used to close a chain of slabs. Apply the reflected-pair identity (66) on $\mathbb{T}_j$, whose normal character is -1. Its closing translation cancels the translation of the repeated pairs, giving

$$Z_{\mathbb{T}_j}[\operatorname{fold}_\ell G_j] = \operatorname{Tr}(\mathfrak{S}[G_j]^* \mathfrak{S}[G_j])^k.$$

This proves (155).

On the original torus the exact trace (63) is

$$Z_{\mathbb{T}} \left[\prod_j G_j \right] = \operatorname{Tr} \left(V_g^{-1} U_{\mathrm{f}}^{(1+\chi(g))/2} \mathfrak{S}[G_{2k}] \cdots \mathfrak{S}[G_1] \right).$$

Its closing factor is unitary. All slab operators are Hilbert–Schmidt by Proposition 4.1, hence belong to the Schatten class of exponent $2k$. Apply (64) with every exponent equal to $2k$, and then use (155). This gives (156). The bounded scalar insertions are read on periodically extended curvatures throughout; they require no choice of a fermionic box or local lift. □

For example, when n is odd the first diagonal height period of $C(n)$ is n, whereas a straight normal closing vector would be $(n/2)e_-$. This is not an integer vector. In the preceding proposition each replacement normal period is $2kh_j$, which is even for integer diagonal slab heights. This is the step that permits odd original sides.

10.2. **An empty slab and its pressure cost.** The exceptional torus in an unequal fold may be long. We use a bound valid for every shape. Set

$$H_M = |M| + \frac{1+\rho}{2}\sum_{i=0}^{3} w_i, \qquad \bar{f} = 192\log H_M + 3\log(r_q C_q). \tag{158}$$

Both constants are finite, and $H_M > 0$.

Lemma 10.2. Every torus $\mathbb{T}$ used in the finite model satisfies

$$0 < Z_{\mathbb{T}}[1] \le e^{\bar{f}|\mathbb{T}|}.$$

Consequently $f_\infty \le f(a,b,c) \le \bar{f}$ for all integer $a,b,c \ge 4$. ■

Proof. The determinant bound is $\det \mathcal{D}_A^{\mathbb{T}} \le H_M^{192|\mathbb{T}|}$, and the Maxwell integral is at most $(r_q C_q)^{3|\mathbb{T}|}$. Together with the positive lower bound in (21), these give the first assertion. The second follows from the definition of pressure and Proposition 4.4. □

Use the diagonal cells (67), with heights $(\ell^1,\ell^2,\ell^3) = (x_1 - x_2, x_1 + x_2, x_3)$, admissible offsets o_a, and even width h. A finite family $\mathfrak{G} = (G_\kappa)_{\kappa\in\mathcal{K}}$ will mean bounded scalar curvature functions with G_κ supported in the open cell Q_κ. For a direction a and a layer j, write $\mathfrak{R}_a^{j,2k}\mathfrak{G}$ for the family obtained by retaining the factors with $\kappa_a = j$ and reflecting that layer into the $2k$ layers with indices $j,\dots,j+2k-1$. The other two indices are unchanged. Its product is exactly the slab fold of the original layer product. This follows because the unique wall-group element taking one layer to another acts on each of its cells by the corresponding cell reflection.

Lemma 10.3 (One empty slab)**.** Fix a direction a, a torus $\mathbb{T}$, and its cross-section Ξ and height period p for ℓ^a. Suppose $(2k-1)h < p$, and put

$$h^{\mathrm{exc}} = p - (2k-1)h, \qquad \mathbb{T}^{\mathrm{reg}} = \mathbb{T}(\Xi; 2kh), \qquad \mathbb{T}^{\mathrm{exc}} = \mathbb{T}(\Xi; 2kh^{\mathrm{exc}}).$$

Assume both replacement period lattices have systole at least four. If every index of $\mathfrak{G}$ satisfies $0 \le \kappa_a \le 2k-2$, then

$$\Big|Z_{\mathbb{T}}[\prod \mathfrak{G}]\Big| \le Z_{\mathbb{T}^{\mathrm{exc}}}[1]^{1/(2k)} \prod_{j=0}^{2k-2} Z_{\mathbb{T}^{\mathrm{reg}}}[\prod \mathfrak{R}_a^{j,2k}\mathfrak{G}]^{1/(2k)}, \tag{159}$$

$$|\mathbb{T}| = \frac{2k-1}{2k}|\mathbb{T}^{\mathrm{reg}}| + \frac{1}{2k}|\mathbb{T}^{\mathrm{exc}}|. \tag{160}$$

Every folded partition function on the right is nonnegative. ■

Proof. Cut at $o_a + jh$ for $0 \le j \le 2k-1$ and at $o_a + p$. The first $2k-1$ slabs have height h; the final slab has height $h^{\mathrm{exc}} > 0$ and contains no factor. The positions are admissible, since p and h are integers and o_a is admissible. Apply Proposition 10.1 to their layer products and to the constant 1 in the final slab. The regular folds are the products of $\mathfrak{R}_a^{j,2k}\mathfrak{G}$, while the empty slab folds to 1. Its three conclusions give the assertions. □

The empty slab has no local observable to transport. Thus its reflected copies may have a different width, while every regular factor remains in the original cell structure. Folding in direction a leaves the other two cell indices unchanged, so the same lemma can be applied successively in the remaining directions.

10.3. **The comparison inequality.** Fix a real local trigonometric polynomial O supported in a box $y+[-r,r]^3 \subset Q_0$, with $r \geq 1$. Choose the cell width even, $h \geq 18$, and $h > 8r+8$. The offsets are $o_1 = c_0, o_2 = c_+, o_3 = c_3$, with $c_0, c_+ \in \mathbb{Z}$ and $c_3 \in \mathbb{Z}+1/2$. For $4h \mid m$, let O_m^{fold} be the full reflected array on D_m, and let $\mu_{D_m}^t$ be the tilt (92). Its number of cells is $N = m^3/|Q|$, where $|Q| = h^3/2$.

Proposition 10.4. Suppose $4h \mid m$, $m \geq 4h$, and $n \geq m$ is an integer. Put

$$V^{\mathrm{reg}} = \frac{(m-h)^2(2m-h)}{2}, \qquad V^{\mathrm{exc}} = n^3 - V^{\mathrm{reg}},$$
$$\Delta'(n,m,h) = f(m/2,2m,m) - f_\infty + \frac{V^{\mathrm{exc}}}{V^{\mathrm{reg}}}(\bar{f} - f_\infty).$$

Then $\Delta'(n,m,h) \geq 0$ and, for every $t > 0$,

$$\mu_{C(n)}(O) \leq \mu_{D_m}^t(O) + \frac{|Q|\Delta'(n,m,h)}{t}. \tag{161}$$

There is no parity restriction on n. ∎

Proof. Set $s = n - m \geq 0$ and

$$k_1 = k_3 = \frac{m}{2h}, \qquad k_2 = \frac{m}{h}, \qquad \mathcal{K}^{\mathrm{reg}} = \{\kappa : 0 \leq \kappa_a \leq 2k_a - 2,\ a = 1,2,3\}.$$

These are integer ranges, with every $k_a \geq 2$. Write

$$R = \prod_{a=1}^{3}(2k_a - 1), \qquad N = 8k_1k_2k_3 = \frac{m^3}{|Q|}.$$

Then $R|Q| = V^{\mathrm{reg}}$, and $0 < V^{\mathrm{reg}} < m^3 \leq n^3$. Since $f(m/2,2m,m) \geq f_\infty$ and $\bar{f} \geq f_\infty$, the claimed sign of Δ' follows.

The array on the original torus. For $\kappa \in \mathbb{Z}^3$ let

$$v(\kappa) = \frac{h}{2}(\kappa_1 e_- + \kappa_2 e_+) + h\kappa_3 e_3, \qquad G_\kappa = e^{tO^{\tau_{v(\kappa)}}}.$$

The vector $v(\kappa)$ is integral because h is even, and its three height coordinates are $h\kappa$. Thus G_κ is supported in Q_κ. Define

$$\widetilde{O}_{\mathrm{reg}} = \sum_{\kappa \in \mathcal{K}^{\mathrm{reg}}} O^{\tau_{v(\kappa)}}, \qquad \mathfrak{G} = (G_\kappa)_{\kappa \in \mathcal{K}^{\mathrm{reg}}}.$$

Then $\prod \mathfrak{G} = e^{t\widetilde{O}_{\mathrm{reg}}}$. Jensen's inequality and translation invariance give

$$tR\mu_{C(n)}(O) \leq \log Z_{C(n)}[e^{t\widetilde{O}_{\mathrm{reg}}}] - \log Z_{C(n)}[1]. \tag{162}$$

The three replacement geometries. Apply Lemma 10.3 in the order (ℓ_-, ℓ_+, x_3). The regular tori are

$$\mathbb{T}_0 = C(n), \quad \mathbb{T}_1 = D'(m/2,n,n), \quad \mathbb{T}_2 = D(m/2,m,n), \quad \mathbb{T}_3 = D_m.$$

For the first step the cross-section of $C(n)$ is generated by ne_+, ne_3, with characters $+1, -1$, respectively. Its height period is n, with closing vector ne_1. The regular normal vector is $k_1 h e_- = (m/2)e_-$, and the exceptional slab has height $n - (2k_1 - 1)h = h + s$. Hence

$$\mathbb{T}_1^{\mathrm{exc}} = D'(k_1(h+s),n,n).$$

This calculation uses only that n is an integer.

In the second direction, $\mathbb{T}_1$ has cross-section generated by $(m/2)e_-, ne_3$, both with character -1. Its height period is $2n$, and the original closing vector ne_+ has character $+1$. The trace therefore contains the parity operator, which is unitary and is retained in Proposition 10.1. The regular normal vector is $k_2 h e_+ = me_+$, now with character -1. The empty slab has height $h + 2s$, giving

$$\mathbb{T}_2^{\mathrm{exc}} = D(m/2, k_2(h + 2s), n).$$

In the third direction, the cross-section of $\mathbb{T}_2$ is generated by $(m/2)e_-, me_+$, both with character -1, and the height period is n. The regular closing vector is $2k_3 h e_3 = me_3$. The empty slab has height $h + s$, so

$$\mathbb{T}_3^{\mathrm{exc}} = D(m/2, m, 2k_3(h + s)).$$

All these tori are admissible. For positive integers a, b, c, the lattice generated by ae_-, be_+, ce_3 has systole $\min(a, b, c)$: a vector has coordinates $(aA + bB, -aA + bB, cC)$, and the maximum of the first two absolute values is $|aA| + |bB|$. The stated minimum follows, and is attained by a generator. Here every displayed coefficient is at least $\min(m/2, 2(h + s), n) \geq 36$. This checks the regular and exceptional systoles at each application of Lemma 10.3.

The reflected arrays. The first fold can be applied because the first indices of $\mathfrak{G}$ lie in $[0, 2k_1 - 2]$. After retaining a first layer j_1 and reflecting it, the second and third index ranges are unchanged. The second fold may therefore be applied to each resulting family; afterwards the third range is still $[0, 2k_3 - 2]$, so the third fold applies as well.

For $j = (j_1, j_2, j_3) \in \mathcal{K}^{\mathrm{reg}}$, the final family is the full reflected array of G_j in the cells

$$j + \mathcal{K}_0, \qquad \mathcal{K}_0 = \{\kappa : 0 \leq \kappa_a \leq 2k_a - 1\}.$$

Indeed the three wall maps commute, and their product is the unique wall-group element carrying Q_j to the prescribed cell. The periods of D_m shift the cell indices by $(2k_1, 0, 0)$, $(0, 2k_2, 0)$ and $(0, 0, 2k_3)$, so $j + \mathcal{K}_0$ is a complete set of cell representatives. Changing those representatives does not change a scalar fold on periodic curvatures.

The full fold of G_j has the same partition function as the full fold of e^{tO}. Translation by $v(j)$ permutes the cell walls, conjugates their reflected maps, and preserves D_m. Thus it translates the full array of e^{tO} into that of G_j, and translation invariance of the finite functional gives

$$Z_{D_m}[\mathrm{Fold}_{D_m} G_j] = Z_{D_m}[e^{tO_m^{\mathrm{fold}}}].$$

The weighted exceptional volume. The three Hölder steps give R regular factors, each with exponent $1/N$, and the exceptional exponents

$$\vartheta_1 = \frac{1}{2k_1}, \qquad \vartheta_2 = \frac{2k_1 - 1}{4k_1 k_2}, \qquad \vartheta_3 = \frac{(2k_1 - 1)(2k_2 - 1)}{8k_1 k_2 k_3}.$$

All folded partition functions are nonnegative. Successive substitution in (159) is therefore valid and yields

$$Z_{C(n)}[e^{t\widetilde{O}_{\mathrm{reg}}}] \leq \prod_{a=1}^{3} Z_{\mathbb{T}_a^{\mathrm{exc}}}[1]^{\vartheta_a} Z_{D_m}[e^{tO_m^{\mathrm{fold}}}]^{R/N}. \tag{163}$$

The three volume identities (160) are

$$|\mathbb{T}_{a-1}| = \frac{|\mathbb{T}_a^{\mathrm{exc}}|}{2k_a} + \frac{2k_a - 1}{2k_a}|\mathbb{T}_a|, \qquad a = 1, 2, 3.$$

Substitute the third into the second, and then the result into the first. Since $|\mathbb{T}_3| = m^3$ and $Rm^3/N = R|Q|$, this gives

$$\sum_{a=1}^{3} \vartheta_a |\mathbb{T}_a^{\mathrm{exc}}| = n^3 - R|Q| = V^{\mathrm{exc}}. \tag{164}$$

Apply Lemma 10.2 separately to the exceptional factors in (163). The resulting bound is

$$Z_{C(n)}[e^{t\widetilde{O}_{\mathrm{reg}}}] \le e^{\bar{f}V^{\mathrm{exc}}} Z_{D_m}[e^{tO_m^{\mathrm{fold}}}]^{V^{\mathrm{reg}}/m^3}. \tag{165}$$

The weighted identity (164) is essential here. The exceptional volumes themselves may exceed n^3.

Normalization and convexity. All partition functions in (165) are positive. For the inserted ones this also follows directly from $|\widetilde{O}_{\mathrm{reg}}| \le R|O|_1$ and $|O_m^{\mathrm{fold}}| \le N|O|_1$. As in (96), convexity of the logarithm of the tilted partition function gives

$$\log Z_{D_m}[e^{tO_m^{\mathrm{fold}}}] \le \log Z_{D_m}[1] + tN\mu_{D_m}^t(O).$$

The pressure bounds give

$$\log Z_{C(n)}[1] \ge n^3 f_\infty, \qquad \log Z_{D_m}[1] \le m^3 f(m/2, 2m, m),$$

where the second inequality is (73) at the even side m. Combining these estimates with (162) and (165) yields

$$\begin{aligned} tR\mu_{C(n)}(O) \le tR\mu_{D_m}^t(O) &+ V^{\mathrm{reg}}(f(m/2,2m,m) - f_\infty) \\ &+ V^{\mathrm{exc}}(\bar{f} - f_\infty). \end{aligned}$$

Divide by $tR = tV^{\mathrm{reg}}/|Q|$ to obtain (161). □

10.4. **Divisible companion sides.** The comparison side must accommodate every fixed cell width used in the local estimates, including the coarser widths used for amplification. Rounding to multiples of a single fixed integer would not supply these widths. The following choice is eventually divisible by every positive integer and changes the side by less than its square root.

Lemma 10.5. Let $n_i \ge 4$ be integers with $n_i \to \infty$. There are positive integers m_i such that $m_i \le n_i$ eventually, $m_i/n_i \to 1$, and every positive integer divides m_i for all sufficiently large i. One choice, for $n_i \ge 64$, is

$$k(n) = \max\{k \in \mathbb{Z} : k \ge 1,\ 8k! \le \sqrt{n}\}, \qquad m_i = 8k(n_i)! \left\lfloor \frac{n_i}{8k(n_i)!} \right\rfloor.$$

The finitely many remaining terms may be set equal to 8. ■

Proof. For $n \ge 64$ the set defining $k(n)$ is nonempty and finite. The remainder after division by $8k(n)!$ satisfies

$$0 \le n - 8k(n)! \left\lfloor \frac{n}{8k(n)!} \right\rfloor < 8k(n)! \le \sqrt{n}.$$

Thus $n_i - \sqrt{n_i} < m_i \le n_i$ for all large i, which gives $m_i/n_i \to 1$ and $m_i \to \infty$. Given an integer $d \ge 1$, the inequality $n_i \ge 64(d!)^2$ holds eventually. It implies $k(n_i) \ge d$, so d divides $k(n_i)!$ and hence m_i. □

Proposition 10.6 (Comparison for arbitrary original sides)**.** Let $n_i \ge 4$ tend to infinity, and choose m_i as in Lemma 10.5. Fix the cell data and observable O of Proposition 10.4. Form the tilted laws $\mu_i^t = \mu_{D_{m_i}}^t$ and their ordered comparison states using these companion sides. Then

$$\Delta_i' = \Delta'(n_i, m_i, h) \longrightarrow 0.$$

If $\mu_{C(n_i)} \Rightarrow \nu$ on a prescribed subsequence and ν_0 is any comparison state subordinate to that subsequence, then

$$\nu(O) \le \nu_0(O). \tag{166}$$

■

Proof. For all sufficiently large i, one has $4h \mid m_i$, $m_i \geq 4h$ and $m_i \leq n_i$. Proposition 10.4 therefore applies. The pressure term $f(m_i/2, 2m_i, m_i) - f_\infty$ tends to zero by Proposition 4.4. Moreover

$$\frac{V_i^{\mathrm{reg}}}{n_i^3} = \left(\frac{m_i}{n_i}\right)^3 \left(1 - \frac{h}{m_i}\right)^2 \left(1 - \frac{h}{2m_i}\right) \longrightarrow 1. \tag{167}$$

Hence $V_i^{\mathrm{exc}}/V_i^{\mathrm{reg}} \to 0$, proving $\Delta_i' \to 0$.

All divisibility requirements used to construct comparison states hold for (m_i). Choose the common volume subsequence i_j and strengths $t_k \downarrow 0$ in (150), subordinate to the prescribed original subsequence. For fixed k, (161) gives

$$\mu_{C(n_{i_j})}(O) \leq \mu_{i_j}^{t_k}(O) + \frac{|Q|\Delta_{i_j}'}{t_k}.$$

The observable O is bounded and continuous. Letting $j \to \infty$ at fixed $t_k > 0$ therefore gives $\nu(O) \leq \nu_{t_k}(O)$. Now let $k \to \infty$ to obtain (166). The factor $1/t_k$ in the error is the reason for this order of limits. □

The law on the left of (166) remains the limit of the original cubic laws. The construction of ν_0 uses the companion volumes and retains the wall positivity and even translation invariance proved for ordered comparison states.

11. Reflection positivity of the original limit

11.1. **Convex separation and the nine mirror families.** Fix an integer c_0. Let $\mathcal{R}(c_0)$ be the set of laws on Ω satisfying Gaussian domination and both invariance and reflection positivity at $(x_1 - x_2, c_0)$. This is convex and weakly compact. Gaussian domination gives a common tail $\lambda(|F_\sigma| \geq R) \leq 2e^{-R^2/2}$ and hence uniform tightness. It is weakly closed by truncation of its exponential integrals. Mirror invariance and positivity against continuous local tests are also closed, and the approximation in section 5.3 extends the latter to the full open-side L^2 space. Each condition is affine or an affine inequality in the law, proving convexity.

We claim that, for every real local trigonometric polynomial O and every cubic limit ν along an arbitrary sequence of sides tending to infinity, some $\nu_0 \in \mathcal{R}(c_0)$ satisfies (168). The polynomial need not initially fit in a cell having its lower wall at c_0. Let $\operatorname{supp} O \subset y_0 + [-r, r]^3$, with $r \geq 1$, and choose an admissible even width $h > 8r + 8$. Choose an even integer a in

$$(c_0 + 2r - \ell_-(y_0),\ c_0 + h - 2r - \ell_-(y_0)).$$

The interval has length greater than two. For $v = ae_1$ and $y = y_0 + v$ its defining inequalities place the translated box between the two ℓ_- walls. Choose integer c_+ and half-integer c_3 so that

$$c_+ < \ell_+(y) - 2r < \ell_+(y) + 2r < c_+ + h, \qquad c_3 < y_3 - r < y_3 + r < c_3 + h.$$

The available intervals have length greater than one. The box therefore lies in the open diagonal cell, since $|\ell_\pm(x) - \ell_\pm(y)| \leq 2r$ on it. Form the tilted law with the translated polynomial O^{τ_v}.

Choose companion sides by lemma 10.5 and form a comparison state on those sides, subordinate to the original volume subsequence defining ν. It belongs to $\mathcal{R}(c_0)$, and proposition 10.6 gives $\nu(O^{\tau_v}) \leq \nu_0(O^{\tau_v})$. The original cubic laws are invariant under every lattice translation, and this invariance passes to ν. Equation (153) gives the same invariance under this particular v for ν_0. Hence

$$\nu(O) = \nu(O^{\tau_v}) \leq \nu_0(O^{\tau_v}) = \nu_0(O). \tag{168}$$

The comparison state may depend on O. Convex separation is what turns these individual inequalities into reflection positivity of the original law.

Lemma 11.1. Let $\mathcal{R}$ be a convex weakly compact set of laws on Ω. If for every real local trigonometric polynomial O some $\lambda \in \mathcal{R}$ satisfies $\nu(O) \leq \lambda(O)$, then $\nu \in \mathcal{R}$. ■

Proof. Give the real vector space of finite signed measures on Ω the weak topology defined by integration against these polynomials. It is locally convex and Hausdorff: if every such integral of a signed measure vanishes, its finite-coordinate marginals have zero Fourier transform, so its positive and negative parts agree on cylinder sets and hence on every Borel set. Its continuous linear functionals are precisely integration against the test space; a continuous linear functional for a topology defined by these seminorms depends on finitely many of them, and their finite real span is still that space. Since the tests are bounded and continuous, the inclusion from the usual weak topology of laws is continuous. Thus $\mathcal{R}$ is a compact convex subset of this locally convex Hausdorff vector space. It is nonempty by the hypothesis at $O = 0$.

If ν were outside $\mathcal{R}$, strict separation [Pet17, Theorem 6.2.9] of a compact convex set and a disjoint closed point would give a continuous linear functional with $\max_{\lambda\in\mathcal{R}} \lambda(O) < \nu(O)$ for some test O. Its maximum exists by compactness. This contradicts the assumed inequality for that O. □

Proof of Theorem A. The coordinate part of the theorem, including tightness, Gaussian domination, all lattice symmetries and coordinate reflection positivity, was proved in section 5.3 for arbitrary integer sides. Let ν be any limit along a prescribed original sequence. For the diagonal mirror $(x_1 - x_2, c_0)$, (168) holds for every real trigonometric polynomial. Lemma 11.1 therefore gives $\nu \in \mathcal{R}(c_0)$, proving its diagonal positivity. This is valid for every integer c_0.

A signed permutation transports this diagonal mirror and its positive half-space to any plane $x_\mu \pm x_\nu = c$. The corresponding reflection is the conjugate of the original one. Pull a positive-side test back by this isometry; invariance of ν identifies its reflected form with the already nonnegative one. This gives the six diagonal families and completes the proof. □

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